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Review article

Facets of hydro power and future trends in a Nordic Context☆

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ABSTRACT

Hydropower technologies bolster high penetration of variable renewable energies (VREs) in the net zero emissions scenarios. Nevertheless, there are various challenges to meeting the ambitious goal, such as stability, reliability, resiliency, security, lack of reactive power, voltage support and inertia, large-scale storage deployment and coordination, interconnectedness, demand-side response, higher thermal cycles with increased start/stops, and inadequate Levelized Cost of Energy (LCOE) for system-wise VRE integration and profitability. This survey conducts a bottom-up analysis to unveil the opportunities to utilize hydropower facilities and disentangle the nested problem for intertwining design features, control algorithms, operation, optimization approaches, incentives, services, and market mechanisms using a three-pillar framework perspective: grid owners, power producers, and machine designers.

The survey identified emerging trends in real-time and capacity markets, flexible power systems, and enhanced grid capabilities, including advanced voltage support and updated grid codes. These developments present significant opportunities for hydropower, such as achieving super-flexibility through hybridization, expanded reactive power capabilities, and advanced operational modes like a synchronous condenser and power *adequator* functionalities. These opportunities require novel design philosophies — including new winding, stator, and rotor configurations, optimized ventilation, and active cooling systems — to enhance performance under stressed grid and climate conditions. Finally, integrating climate and energy models for multi-basin optimization with finer spatial and temporal granularity enhances the planning accuracy for water management of hydropower while addressing environmental challenges.

The review delivers helpful prospective suggestions and tools that would serve researchers, power engineers, and stakeholders in making decisions about hydropower technologies and services in 2050 and beyond.

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1. Introduction

The most ambitious goal for combating climate change is limiting global warming to 1.5 °C goal, a target in which renewable energies will play an essential role in reducing CO_2 emissions, according to the Intergovernmental Panel on Climate Change (IPCC) (IPCC, 2021). By 2020, the new renewable-based capacity globally added to the grid was more than four times higher than other sources. Moreover, solar power and wind power represent the highest share of global annual renewable energy investment, i.e., 94 % by 2019. In one of the most ambitious projections by 2050, 73 % of the installed capacity and 63 % of all electricity generation will come from variable renewable energy (VRE) (IRENA, 2021). However, in more conservative estimates, like the International Energy Agency's (IEA's) stated policies scenario (Cozzi et al., 2022), the projected share of wind and solar is 40 %. Thus, there is high uncertainty and many obstacles to scaling up VRE.

1.1. Background

The world's Energy Trilemma (ET) proposes maintaining Environmental Sustainability through international climate change agreements; Energy Security, as nation's energy reliability and resilience; and Energy Equity as coherent and integral energy policies (Shirazi, 2022; Marti and Puertas, 2022; Khan et al., 2022). Nonetheless, the challenges in achieving a high share of VRE by 2050 are stability, real-time market, lack of reactive power and voltage support, and deployment and coordination of high-scale storage capabilities within a global energy mix (Munkhchuluun and Meegahapola, 2017; Flynn et al., 2017; Guerra et al., 2022; Spodniak et al., 2021; Xia et al., 2020; Heusinger et al., 2021; World Bank Group, 2016).

The main actors in power systems, shown in Fig. 1, can meet these challenges by achieving flexible power grids, i.e., extended power capabilities, enhanced controllability and observability, coordinated reactive power supply, enhanced voltage support during grid faults, reactive power market, reduced grid losses under regular operation, and real-time price for new services (Øyvang and Nøland, 2022). In this regard, hydropower technologies are a catalyst toward IPCC goals because they contribute to a more flexible and robust grid through high dispatchability and predictability for vast storage capabilities, supporting a high share of VREs (Skar et al., 2018; Øyvang and Nøland, 2022). Hence, achieving the ET.

1.2. Motivation

In 2020, hydropower supplied one-sixth of the global electricity generation, the third-largest source after coal and gas. Moreover, hydropower plants can ramp electricity generation up and down very rapidly compared with other power plants such as nuclear, coal, and natural gas (IEA, 2021). To reach the Net Zero by 2050 level, hydropower capacity expansion in the year 2030 must be 101.74% higher than the global forecast, i.e., 464 GW of new hydropower capacity installed by 2030 (IEA, 2021). Furthermore, the International Energy Agency (IEA) suggests that the world will need 2600 GW of hydropower capacity installed by mid-century to have a chance of keeping global

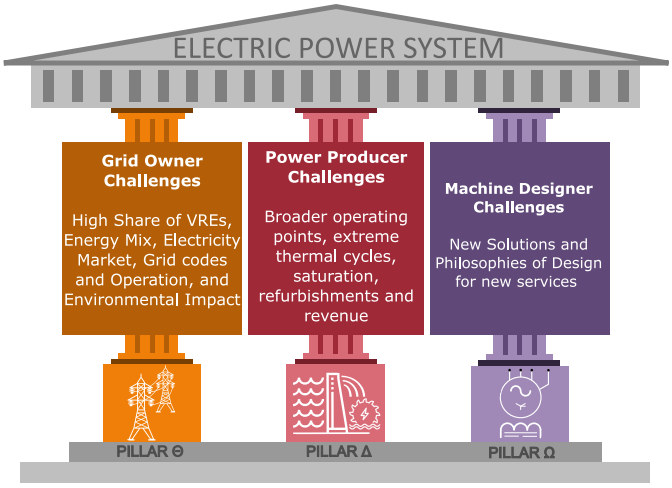


Fig. 1. Challenge-Pillars: The leading actors for operating an electric power system are shown. A heterarchical horizontal relational analysis is performed by assessing the grid owner $P\Theta$, power producer $P\Delta$, and machine designer $P\Omega$ challenges with focus on Hydropower.

temperature rising below 1.5°. That means building the same capacity in the next quarter century as the previous century (Association, 2021).

Power system stability can be classified as rotor angle, voltage, frequency, converter-driven, and resonance stability (Anon, 2021b). Large-scale hydropower plants contribute to the grid's robustness through the rotor and frequency stability due to inertia constant of 2∼4 s and active power control. Also, it contributes to voltage stability from reactive power control (Kundur et al., 1994). Long-term voltage stability is of utmost interest because the maximum power transfer and voltage support are limited when generators hit their field or armature current time-overload capability limits. Moreover, the study period of interest may extend to several minutes. Therefore, extended thermal availability is crucial to face long-term voltage stability, especially for a high share of VREs (Hatziargyriou et al., 2020). Furthermore, hydropower's operational flexibility and stability contributions create a robust, reliable, and resilient grid. Hence, its hybridization may enhance the optimal location of VREs by improving their inertial distribution and flexibility indices (Brahma and Senroy, 2020, 2021).

1.3. Scientific aim and research contribution

This paper presents the challenges and opportunities of hydropower technologies to enable high-share of VREs toward 2050 from a holistic, interdisciplinary, and heterarchical horizontal relational approach, as is shown in Fig. 1 and the sequential abstract of Fig. 2. This work investigates how hydropower technologies can be systematically improved to enable high shares of variable renewable energy (VREs) in power systems while ensuring grid stability, economic viability, and alignment with IPCC climate goals. The study focuses on three interdependent pillars of hydropower systems as subjects of the research:

$P\Omega$ **Hydrogenerator Design**: A chronological review of the hydrogenerators' Philosophy of Design (PoD) to uncover a paradigmatic historical outlook. Thus, it advises a range of opportunities for new PoDs while considering their techno-economic limitations, primarily for reactive power boosting.

$P\Delta$ **Power Producer Operation**: A power producers' viewpoint analysis of hydropower operation challenges uncovering the hydropower plant's impairments due to the abrupt wide range of operating points imposed by the erratic behavior of VREs and reactive power services.

$P\Theta$ **Grid Owner Integration**: A grid owners' standpoint analysis of hydropower operation challenges considering the constraints on a high share of VREs, energy mix optimization and coordination, electricity markets, power grid control, and environmental impact.

Briefly, this article brings a new look to the existing literature in the following areas:

(I) **Comprehensive Synthesis of Hydropower–VRE Integration**: Provides a comprehensive synthesis of challenges and opportunities for hydropower integration with VRE systems using a holistic approach as an intertwined three-pillar functional scheme for hydrogenerator design, hydropower producer, and grid owner.

- *Scientific Aim*: To unify technical, economic, and environmental challenges/opportunities into a holistic framework.
- *Extension of Existing Research*: Unlike prior fragmented studies, this work reveals the technical, economic and environmental interdependencies between the three pillars.

(II) **Critical research priorities for Hydropower in grids with high share of VREs:** Identifies research areas for the scientific community regarding hydropower technologies and VRE integration, including stochastic modeling approaches, multi-basin system optimization, hydropower hybridization strategies, adaptive balancing-oriented hydropower solutions, variable speed pumped storage technologies, improved market mechanisms to support hydropower's role and appreciate its value, harmonization of grid codes for Hydro-VRE integration, hydrogenerator thermal stress assessment, new design opportunities for refurbishment and new facilities, digital twin implementations, environmental and regulatory frameworks.

- *Scientific Aim:* To define actionable gaps in hydropower's role as a flexibility enabler for a high share of VREs in the grid.
- *Extension of Existing Research:* While earlier works address isolated topics, this study systematizes research priorities across the three-pillars, such as linking adaptive balancing solutions with environmental frameworks to meet IPCC targets.

(III) **Decision Support Framework for IPCC-Aligned Hydropower**: Develop a decision support framework to help stakeholders select appropriate technologies for addressing grid integration challenges in high VRE penetration scenarios while complying with the IPCC goals toward 2050.

- *Scientific Aim*: To bridge technical, economical, and environmental solutions and stakeholder decision-making.
- *Extension of Existing Research:* Existing tools focus on siloed decisions, e.g., turbine upgrades or market participation. This work codifies trade-offs between pillars, i.e., the economic challenges are linked to the hydropower services and the remuneration of grid owners for revenue models that value the flexibility of hydropower technologies; and the technical challenges are linked to incentives and energy policies to create new market mechanisms.

The remainder of the paper is divided into the following sections. Section 2 discusses the main characteristics of hydropower and states its importance in the global energy mix. Section 3 addresses a power system dominated by hydropower under stressed conditions imposed by VREs. Section 4 assesses the three-pillar scheme to uncover the opportunities and challenges for hydropower development. Section 5 presents a strategy with key elements to be present in hydropower modeling. Finally, Section 6 provides critical trends in power systems and hydropower inclusion before the paper is concluded in Section 7.

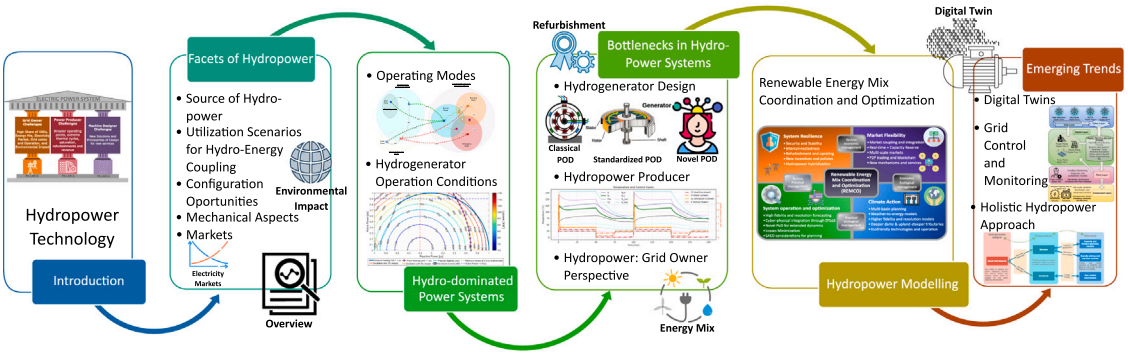


Fig. 2. Sequential abstract. Here, the review's main subjects are depicted in a linear fashion.

2. Methodology

This study adopts a three-pillar interdisciplinary framework to systematically analyze hydropower's role in enabling high shares of variable renewable energy (VRE) in power systems. The methodology integrates technical, economic, and environmental perspectives across the three pillars:

- The pillar of hydrogenerator design analyzes the machine-level challenges and opportunities such as: thermal/electromagnetic dynamics, refurbishment opportunities, new operating modes like synchronous condenser/power adequator, extended reactive power capabilities, hybrid excitation systems, super-flexible operation, etc.
- The pillar of power producers provides the operational challenges and opportunities such as: extended thermal cycling, revenue models, hybridization strategies with VREs or storage, impact of VRE variability on hydropower dispatch, etc.
- The pillar of grid owners gives the system-level challenges and opportunities such as: grid stability requirements, market mechanisms, environmental impacts and regulations, etc.

This study is structured to give a holistic view of hydropower's evolving role in enabling high shares of variable renewable energy (VRE) within power systems, aligning with IPCC climate goals. Beginning with foundational principles (Section 3), the article establishes hydropower's inherent technical and environmental characteristics, contextualizing its flexibility and constraints. Section 4 then diagnoses the challenges hydro-dominated systems face under increasing VRE penetration, such as operational stress from frequent starts/stops and voltage instability. Building on this, Section 5 adopts a three-pillar framework to holistically address these challenges: (1) hydrogenerator design innovations are historically reviewed to unveil novel technical prospects; (2) power producer strategies are analyzed to understand the challenges and opportunities of hydropower production within VRE-rich grids; and (3) grid owner perspectives are studied to provide a panoramic view of the challenges and opportunities on the grid side and markets. Section 6 progressively bridges theory to practice by synthesizing hydropower modeling strategies, such as digital twins and multi-basin optimization, to support climate-resilient planning. Finally, Section 7 culminates in a forward-looking synthesis, illustrating how the interplay of the three pillars — designer, producer, and grid owner — can unlock synergies to achieve a sustainable energy transition. This structured progression — from foundational analysis to systemic explanations — ensures a rigorous, interdisciplinary evaluation of hydropower's potential to underpin VRE-rich grids.

3. Facets of hydropower

Hydropower technologies were the first bulk generation that started the electrification of the world. Therefore, it is the most well-known technology and has achieved a high-level standardization after 140 years (Allerhand, 2020b), and it is institutionalized in power system engineering as an immutable system. Therefore, any action away from the hydropower standardized technologies and systems represents a risky task for designers and operators since it has worked fine over the years. However, the new paradigm in power systems with a high share of VREs has changed everything. Hence, core characteristics, such as flexibility, storage, costs, stability, capacity factors, and environmental challenges, must be assessed.

3.1. Source of hydropower

According to Association (2021), by 2020, the global hydropower installed was 1330 GW, and the hydropower generation was 4370 TWh. Therefore, the global capacity factor for hydropower was 0.375, compared to 0.394 in 2011 (WEC, 2013).

For the particular case of Norway, there are ten pumped hydro storage plants, according to Pitorac et al. (2020). The gross annual production is 2591 GWh, the pump consumption is 844 GWh, and the net annual production is 1761 GWh. The total capacity installed is 1369 MW. Hence, the capacity factor for pumped hydro storage is C_f = 14.7 %. In contrast, the conventional installed hydropower in Norway is 33 055 MW with an annual energy production of 136.4 TWh. Which is a capacity factor of 47.1 % (Anon, 2021c).

According to Nøland et al. (2022), hydropower has an average C_f =41.9 %, which is 68.95 %, 195.07 %, 219.85 %, and 23.23 % higher than Concentrated Solar Power, Solar PV, Solar PV rooftops, and wind onshore, respectively. However, it is 5.73 % smaller than wind offshore. Nonetheless, hydropower outperforms the spatial annual energy density (SAED) by far compared to other renewable energy sources, equal to 0.296 TWh/km^2, as shown in Fig. 3. Only surpassed by natural gas and nuclear, 3.280 TWh/km^2 and 6.703 TWh/km^2, respectively.

Bigger dams may improve water usage, however, the main issue is water consumption, based on the gross water evaporation from reservoirs, which changes as a function of its surface area. Studies suggest that evapotranspiration accounts on average 54 % of the gross water consumption. However, it is advised for a fair comparison with other energy types that the impact of the reservoir should be allocated among its multiple purposes, such as water supply, flood control, and navigation (Jin et al., 2019).

Furthermore, hydropeaking affects downstream rivers, i.e., daily or sub-daily rapid and marked discharge in streamflow fluctuations. The effects are geomorphological alterations and freshwater biota modifications, i.e., benthic macroinvertebrates, fishes, periphyton, and so forth (Salmaso et al., 2021). Hydropeaking has increased in the Nordic rivers over the last decade, especially for ROR hydropower plants constructed on rivers with small elevation heads. Introducing wind power projects may require a further increase in hydropower; hence, more hydropeaking (Ashraf et al., 2018). The cost related to head losses caused by rock fall in some Norwegian hydrological systems is found

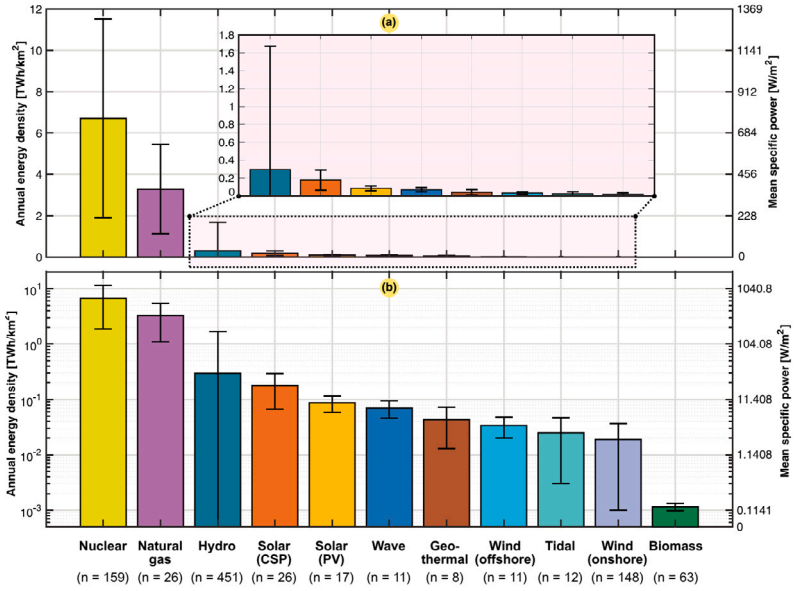


Fig. 3. Spatial Annual Energy Density (SAED) and Spatial Annual Power Density (SAPD) extracted from Nøland et al. (2022). Hydropower has the highest density after Nuclear and Natural Gas.

to have increased by 70 % (atveit et al., 2016). For instance, some mitigation actions can be implementing a minimum/maximum flow, slow ramping, and re-regulation of reservoirs, i.e., a second reservoir after the peaking facility that allows a flow similar to the natural regime, and then pumping back upstream when demand is low. Also, *offline* PSH whereby two created water bodies at different elevations are connected, producing power at peak demand and pumping back when low demand Smokorowski (2022).

Regarding the mitigation measures for the downside effects of hydropeaking discharge, Austria and Switzerland have established legal regulations. Therefore, to enhance regulations and implement them in more countries, the mitigation thresholds and targets can be defined on critical species, including particular features regarding season, a parameter that determines life-stage phases and diel variations, combined with the site-specific morphological characteristics (Moreira et al., 2019).

Environmentally friendly strategies for hydropower technologies have been addressed, such as oil-free turbines (polycrystalline diamond bearing–shaft assembly), demodulation reservoirs to reduce hydropeaking, environmentally enhanced turbines, local sustainable development, hydropower deployment using existing infrastructure, energy recovery from drinking water networks, fish-friendly very low head turbines (Quaranta et al., 2020, 2021, 2022).

Climate change will likely impact each bulk electric power system component, including increasing average and peak demand, increasing thermal generation and transmission capacity curtailment, decreasing summertime hydropower generation in some regions, decreasing wind and solar resources, and increasing air temperature globally, likely increasing peak electricity demand in summertime peaking systems (Craig et al., 2018).

Nearly all existing hydropower projects are river-based; they are susceptible to climate change phenomena that impact rivers, such as (1) depletion of the permanent glacier and ice storage with increasing average annual temperature; (2) depletion of seasonal snow storage with warmer winters; (3) intensification of extreme precipitation and variability; and (4) increased evaporation and water demand. The impact of these global phenomena at the local scale can only easily be guessed with a targeted study of the local context (Wasti et al., 2022). Moreover, according to Van Vliet et al. (van Vliet et al., 2016), it is expected a mean global increase of Gross Hydropower Potential (GHP) between 2.4-6.3 % by 2080s. However, in a region-wise overview, Asia and North America may increase GHP, and Australia and South America may decrease GHP for all scenarios. In the case of Europe and Africa, the best Greenhouse Gas (GHG) concentration scenario (RCP2.6) may present a slight increase in Europe and a possible decrease in Africa's GHP, and the worst scenario (RCP8.5) may show a reduction in Europe and an increase in Africa of GHP. Furthermore, the global mean cooling water discharge capacity (CWDC) is projected to decrease by 4.5-15 % by the 2080s. Therefore, the GHP and CWDC must be assessed region-wise to capture better the actual situation.

Paltán et al. (2021) estimate that about 65 % of current global hydropower installed capacity will be exposed to risks from recurrent high river flows. Up to 75 % of existing hydropower capacity in Europe, North America, Latin America and the Caribbean (LAC), and the Middle East and North Africa (MENA) are located in areas where droughts are projected to become longer and more intense. More frequent extreme flows may lead to shifts in sedimentation rates and the subsequent reduction of the available reservoir depth and volume (diminishing the hydropower generation potentials), structural damage to dams, which may result in dam failures, inundation of critical infrastructures such as power plants, inundation of downstream lands (as result of contingency floodwater releases), or general diminishing of life expectancies of hydropower projects. For instance, in Europe, maintaining a 1.5 °C scenario would avoid an additional 40 GW of installed capacity from being exposed to droughts, which are 50 % more prolonged and more intense. In contrast, up to 80 % of planned capacities in LAC, South Asia, Oceania, and Sub-Sahara would experience at least 10 % more intense or more prolonged drought in both scenarios.

The midterm hydrothermal coordination will be affected by decreased thermal seasonal capacity, increased seasonal demand, and increased or decreased seasonal hydropower generation (depending on the region). For instance, it might be increasingly concentrated during peak demand hours in areas with decreasing hydropower resources, requiring more generation from other plants during off-peak hours (Craig et al., 2018). Moreover, a review paper conducted in the West Africa region presents that hydropower plants must adjust their operation in response to extreme weather conditions. Also, dam construction may contribute to changes in land use and land cover and modify the surrounding climate and precipitation patterns, as well as the hydrological dynamics of the river. The construction of dams,

coupled with rising temperatures, may also favor the development of pest species such as water hyacinth, which is harmful to hydropower generation systems. The water hyacinth can disrupt water abstraction facilities and obstruct hydroelectric turbine intakes, irrigation channels, and watercourses, causing siltation, flooding, and high water loss through evapotranspiration (Obahoundje and Diedhiou, 2022).

Nonetheless, FPVs can mitigate and limit the Algae blooming, i.e., hyacinth, by reducing solar radiation and the algae proliferation mainly due to the combination of eutrophic water and photosynthesis (Rosa-Clot and Tina, 2020). For instance, FPV should ideally cover between 40 to 60 % of the dam's surface to limit Algae concentration without incurring revenue losses. The numbers may differ according to the shape and conditions of the water body (Haas et al., 2020).

Furthermore, Fan et al. (2022) found an asymmetry between constructed Global North and Global South dams. For the former, the dams are associated with an increased GDP. Meanwhile, the latter is associated with decreased GDP, urban land, and population in the Global South, especially Africa, which shows reduced greenness. The impact varied by the size of the dams. Large dams were significantly related to reduced GDP and greenness, whereas small and medium dams were coupled with lowered population and urban land.

There is still an extensive debate on whether reservoirs are a source or sink of GHG emissions as present (Lu et al., 2020). However, in a more recent study by Li and He (2022), the global GHG emissions from hydropower plants account for 1.0 % of global anthropogenic emissions. Nonetheless, the 44 % and 66 % of existing and future hydropower reservoirs were found to be incompatible with the 2030 Agenda goal. The high carbon intensity in most of these reservoirs was attributable to shallow water depth, eutrophication, or both. Therefore, shallow dams are not recommended for future hydropower construction to achieve the low-carbon goal.

Finally, decisions based on power density at individual proposed dams produce sub-optimal solutions compared to those examining power and emissions as separate objectives at a basin scale. Therefore, a dam siting study spanning multiple countries in the Amazon demonstrated the benefits of using such an approach. The authors concluded that GHG emissions could be avoided by sitting dams along steeper, upland tributaries with a high head rather than larger, lowland rivers (Almeida et al., 2019; Jager et al., 2022). Furthermore, to improve the resolution and accuracy of the hydrological model, it is critical to incorporate runoff generation, groundwater–surface water interaction, evaporation, snow processes, land use change, human withdrawal, water consumption, and regulation of flows from upstream powered and non-powered dams (Turner and Voisin, 2022).

The critical challenges for the long-term construction of new hydropower dams can be summarized as follows: regional variability due to climate change, hydropeaking, GHG emissions, transboundary coordination, and ecological disruption to riverine species. To achieve a successful long-term dam construction it is required an integrated, adaptive, and long-term strategy to balance electricity generation, biodiversity, climate resilience, socioeconomic equity, and technological innovation. Scholars suggest prioritizing basin-scale planning, fish passes and bypass systems, water quality, and sediment management, limiting hydropeaking, and providing habitat compensation (Twardek et al., 2022).

3.2. Utilization scenarios for hydro-energy coupling

Multi-energy coupling prediction approaches, i.e., wind–PV–Hydro, may improve system security, reliability, controllability, and flexibility and assist hydropower peak-regulating capacity management (Zhang et al., 2022b). Furthermore, significant storage capabilities of PSH may be exploited within the future energy mix. Consequently, combining the Energy Optimization-based dispatch algorithms and the broader reactive power capabilities of hydrogenerators (Øyvang and Nøland, 2022) may result in system optimization and system security strategies.

China is the primary growth point for the coordinated operation of hydropower and VREs (Liu and He, 2023; Liu et al., 2023). However, as the power market model differs totally from the one in continental Europe, which is the core market associated with this review paper, it is decided not to dig deeper into this topic as the size of the paper is vast. Similarly, the case of hydropower plants in Brazil is also very interesting to analyze due to its substantial technical complexity, which comes from a continental size and several river basins. Still, the power market design is more straightforward than the one in continental Europe. Nonetheless, the core focus of this paper lies in the continental Europe market model and thus has a considerable focus on the Nordics, where this market model is pinnacling. Especially as it is seen as nascent coupling between technics and economics to drive the future of optimization. Therefore, it is seen as the future for all continental European power markets and, thus, the focus point of this paper. Also, to limit the scope of the paper.

There is a myriad of research into coordination between VREs and Energy Storage Systems (ESS). Notwithstanding, the overall objective for VREs is to limit their active power to provide more reactive power for voltage control. In other words, change the power factor to provide voltage support (Rahman et al., 2021). Therefore, the percentage of VRE penetration must be optimized; otherwise, the social welfare and climate change goals will not meet each other. Thus, in Table 1 shows aspects to take into account for optimization and coordination in the energy mix.

A high share of VREs presents several challenges, including low inertia leading to frequency instability, high fault current resulting from short circuits, intensive uncertainties from varying wind speed and irradiance, and degraded power quality due to power electronic interfaced connections (Shafiul Alam et al., 2020). A crucial role in integrating VREs in planning and optimizing of the grid is hydropower technologies because it provides flexibility and facilitates the expansion of wind and solar. Nonetheless, climate change is a source of uncertainty in modeling (Miri et al., 2025; Wang et al., 2025). Ji et al. present long-term planning for hydro–PV–wind while addressing the multiple uncertainties with Markov Chains and Copula functions, achieving improvements up to 2.90 % compared to current algorithms (Ji et al., 2025). Moreover, to mitigate the effect of seasonality and climate change, transboundary and regional planning and optimization must be performed in conjunction with storage technologies in the energy mix (Tapia Carpio and Cardoso Guimarães, 2024; Gonzalez-Salazar and Roger Poganietz, 2022; Wood and Raade, 2024).

3.2.1. Wind power + hydro

Wind+Hydro integration may boost reactive power and voltage ride through support because hydropower chiefly collaborates into frequency and flexibility services (Milligan, 2018; Ahmed et al., 2020).

In a study made by Spodniak in Spodniak et al. (2021) for the Nordic case, it is found that wind forecast errors and hydro-deviations significantly and negatively impact the price spreads and have larger impacts on the day-ahead and intraday prices than on the regulating electricity market. Finally, it concluded that markets closer to real-time are becoming more critical due to increasing shares of wind power, and their role as reference prices relevant for price risk hedging, capacity markets, and other decision-making will probably increase in the future.

3.2.2. Solar power + hydro

The uncertainties introduced by solar PV impede effective participation in electricity markets, particularly short-term markets. Besides, the withdrawal of government incentives over time pushes the PV systems to compete with well-established, low-priced, fossil fuel-based generation technologies. Table 2 shows three technologies and approaches to increase PV penetration. Therefore, hydropower with FPV may fulfill PV participation and hydropower expansion by 2030 (IEA, 2021) by extending the actual reservoir installed capacity. Covering 1.2 % of

Table 1
Energy mix, optimization, and coordination.

Features	Optimization	Coordination
Storage (Sun et al., 2022; Ali et al., 2021; Thirugnanam et al., 2021; Li et al., 2018b; Xie et al., 2021; Xia et al., 2020; Zhou et al., 2021; Ali et al., 2022; Imcharoenkul and Chaitusaney, 2022)	- Multi-microgrid energy trading and storage optimization. - Short-term optimization for bulk generation. - Peer-to-peer multi-microgrid system framework for energy trading may reduce 36 % the payoff value of the multi-objective function. - Minimum system costs and maximum generation revenue. For instance, the minimal total system cost in Thailand is achieved at 70 % VRE penetration.	- Voltage control for active distribution networks with mobile ESS. - Multi-time scale multi-energy coordination - Coordinated advanced control to provide voltage support for multi-microgrid energy management systems.
Wind	- Share of Wind vs. hydro - Share of Wind vs. solar	- Coordination between forecast of different resources to reduce prices.
Solar	- Share of PVs vs. hydro - Share of PVs vs. wind	- Coordination with ESS to reduce uncertainties and intermittencies.
Hydrothermal	- Improve optimization algorithms with better formulations	- Coordination in scenarios when thermal units decrease, and more active use of water is needed.
PSH	- Size of PSH against mix of energy	- Fast regulation services

Table 2
Technologies to achieve higher PV penetration.

Technologies	Description	Challenges
Storage (Shafiullah et al., 2022)	Subsidies and mechanisms to support Energy-Storage Systems	- Cost reduction - Power electronics
Forecasting (Yang et al., 2022)	Holistic approaches between power engineers and atmospheric scientists, i.e., energy meteorology.	- Satellite-based irradiance - Aerial solar forecasting - Numerical methods for multi-day-ahead forecasting. - Flexible reserve and firm generation coordination - Ready-to-use data and software packages. - Interdisciplinary Work
Distributed Generation (Cousse, 2021; Yin et al., 2021; Huxley et al., 2022)	- Distributed solar rooftop PVs. - Distributed PV farms at a local level.	- Popular acceptance toward solar PV in communities. - Uncertainty-based policy making and markets. - Unique solar bidding strategies. - Coordination between up and downstream markets. - Problematic measurement for System Operators

US reservoirs with floating PV produces more energy than installed hydropower (Perez et al., 2018), for 1 % cover of Africa's reservoirs may achieve 29 TWp (Gonzalez Sanchez et al., 2021). The global potential for FPV is between 4.2 TW to 7.6 TW, with annual generation ranging from 4251 TWh to 10 616 TWh annual generation (Kakoulaki et al., 2023), where the cheapest LCOE for India is 46 USD/MWh (Vidović et al., 2023).

The estimated global potential for large-scale hydro–PV–Wind hybrid systems (HS) is 1699 GW, generating 4348 TWh/y, which could meet up to 12 % of global electricity demand in 2040. Nowadays, Asia-Pacific, Central and South America (CSA), and Africa account for two-thirds of the total potential. In the future, HS may provide up to 54 % and 33 % in CSA and Africa. The fluctuation ratios for HS can be up to 99 % lower than of those independent systems, and the increase of average utilization efficiency of transmission networks as high as 72 % (Wang et al., 2021).

3.2.3. Multi-sector coupling: Europe case

A recent high Spatio-temporal multi-sector coupling study by Pickering et al. (2022), made in Calliope (Pfenninger and Pickering, 2018) with 441 technically feasible and cost-effective options for energy-self-sufficient, carbon-neutral Europe, shows that firm generation is not strictly mandatory. Five scenarios are presented in Table 3. Here, it shows the intertwined trade-off between storage and transmission expansion and the hydropower generation share among different scenarios. The zero storage and maximum transmission expansion scenario provides the highest hydropower share. However, further development and research must be done to achieve optimal solutions.

3.3. Configuration opportunities

Hydropower technologies may need help with the different electricity production peaks and loads for large-scale multi-grid and multi-basin systems to balance the VREs. Thus, it entails a complex optimization for inter-regional systems (Cheng et al., 2012). Spatiotemporal coordination of multi-energy-hybridized sources for a single basin may increase the energy balance (Hu et al., 2023). Similarly, hybridization may extend a hydropower plant's life by 5 %, an average damage reduction of 26 %, with a reduction of operational costs (Bhatti et al., 2023), and improve its flexibility with energy storage systems (Gerini et al., 2021b).

Multipurpose small hydropower plants are being investigated to uncover the hidden flexible potential of hydropower technologies. For instance, the *SmallFLEX* project, a run-of-river (ROR) power plant at the Rhone River in Switzerland, presents a low environmental impact with intelligent use of peak energy and ancillary services as a business case (Münch-Alligné et al., 2021). Another project is the DuoTurbo counter-rotating hydro turbine for energy recovery in drinking water networks with a technology readiness level (TRL) of 7 (Biner et al., 2021). Optimization of ROR hydropower plants has effectively increased the provision of frequency containment reserve (Gerini et al., 2021a).

3.3.1. Pumped-storage hydro

In the case of Pumped-Storage Hydropower (PSH), it can provide long-term energy storage duration from hours to months with a discharge rate of 10 h (Luo et al., 2015). Also, PSH may supply fast frequency reserves to support the grid in terms of flexibility (Bontemps et al., 2020). Variable Speed Hydrogenerators (VSHs) can provide virtual inertia (VI) with grid-connected converters for fast frequency reserve. Usually, VI control utilizes energy stored in batteries, capacitors, or rotating masses. In that sense, VSHs with VI control may use kinetic energy stored in the turbine and generator rotating masses. Moreover, by having a small or large reservoir and controlling water flow, the power converters can quickly change the power injected or

Table 3
Hydropower Penetration Analysis of most crucial Spatially-Explicit Practically Optimal Results (SPOREs) for Europe.

SPORE	Description	Comments
345 ZTI (Base case)	- An energy-wise penetration of 21.77 %, 55.89 %, and 13.19 % for PV, onshore and offshore wind, respectively. Key aspects: 3.66 % curtailment, High EV flexibility, and up to 4.62 TWh of storage.	- Linearize model - Small firm generation is still needed due to curtailment. - Hydropower penetration equal to 3.09 %.
323 MTE 73 TWh Imports	Change in the energy-wise penetration: 6.27 % PV reduction, 3.27 % onshore wind increase, and 5.70 % offshore wind increase The main difference is the storage capacity: - Storage decreases to 1.21 TWh, around a 76 % reduction.	- Intertwined trade-off between transmission and storage expansion for optimize high penetration of VREs -Hydropower penetration equal to 2.86 %.
ZSE 39 results	Zero storage expansion does not necessary entails maximum transmission expansion.	-mTE Hydro Share 2.99 % -MTE Hydro Share 3.44 %
438 MCS	MCS does not indicate zero transmission expansion. - Reduced offshore wind energy to near zero.	- Hydropower penetration equal to 3.04 %.

[A] ZTI: Zero Transmission Investment
[B] MTE: Maximum Transmission Expansion
[C] ZSE: Zero Storage Expansion
[D] mTE: minimum transmission expansion
[E] MCS: Maximum Capacity Storage

withdrawn from the grid by controlling the turbine's rotational speed. Thus improving the stability of a large grid with advanced control algorithms (Reigstad and Uhlen, 2021).

PSH provides flexibility as a valuable technology to cope with the increasing share of VREs. The reservoir storage capacity may be managed at a system level, reducing the system generation costs and maximizing social welfare. Hence, PSH can pump at the lowest prices or excess of VREs and generate at higher prices or low generation of VREs (Huang et al., 2022; Wang et al., 2020).

The PSH flexibility against the wind, solar, and mixed energy scenarios will vary seasonally, according to the region of interest's load, solar production, wind production, and water flow. For instance, a study by Van Meerwijk in van Meerwijk et al. (2016) states that PSH plants in Switzerland are more useful in wind-dominated scenarios.

It is essential to address the energy mix toward climate mitigation goals with a transdisciplinary approach. For instance, Craig et al. (2022) propose an energy-system-tailored climate dataset for historical and future meteorological conditions and energy system models that can effectively leverage those datasets. That is, align climate modeling Spatio-temporal resolutions to energy system modeling and add climate-related uncertainty quantification analysis to energy system modeling. Furthermore, it is crucial to carefully choose the modeling conversion from weather to energy; the capacity factors affecting the CAPEX and OPEX for VREs, transport, and storage technologies; and the potential implications of choosing a specific meteorological dataset for energy system analysis (Kies et al., 2021).

Most existing fixed-speed PSH units typically operate above 60 % of their rated power output to avoid *rough zone*, i.e., vibrations and increased cavitation (material pitting) due to reduced water flow. However, designers are now able to make turbines that have an extended operating range from 0 % to 100 % with little to no rough zone. Nonetheless, fixed-speed cannot provide regulation services in pumping mode because it always consumes the same amount of power from the grid. Therefore, VSH technology may be considered for offering regulation services during pumping mode. The two main types of VSH are doubly fed induction machine (DFIM) with a frequency converter of a fraction of the rated power of the DFIM unit and converted-fed synchronous machine (CFSM), which uses a full-size frequency converter (Koritarov et al., 2022).

The most common method of converting an existing hydropower plant into a PSH is constructing an upper reservoir and using the existing hydropower reservoir as the lower reservoir, where a new PSH powerhouse would be located. In other cases, an existing hydropower plant can be converted to a pump-back PSH plant by replacing its turbines with reversible pump turbines or adding separate pumps that pump the water downstream to the upstream reservoir. Nonetheless, only some hydropower plants can be converted into PSH plants, i.e., little storage capacity (Koritarov et al., 2022).

New fixed-speed PSHs have faster response, i.e., fast ramp rates and short mode change times, and broader operating ranges, i.e., lower minimum load and broader operating head range. VSH has advantages (see Table 7), but higher capital investment cost than fixed-speed units is a significant drawback.

Lastly, for off-grid systems, PSH may offer economic advantages over Battery Energy Storage Systems (BESS) (Guruprasad et al., 2023).

3.3.2. *Refurbishment opportunities*

The energy mix transition toward a high share of VREs will be a regional and local challenge, where the unique scenarios require holistic approaches beyond evaluating the components. Therefore, the current fleet of hydropower plants must be evaluated to achieve superflexibility by assessing their contextual particularities through refurbishment opportunities.

Hydropower plant refurbishment can be uprated with less capital and a shorter gestation period while maintaining the existing plant fleet, avoiding decommissioning, and minimizing forced and planned outages. This reduces maintenance costs, energy production losses, and revenue reductions while improving plant efficiency. The refurbishment indicators can be the condition of the component, age, operating conditions, trouble spots, time of breakdowns, and maintenance charges, whereas, for uprating, the indicating parameters can be the time of license, period of overflow, and efficiency (Rahi and Chandel, 2015). Table 4 shows some considerations for assessing the refurbishment of a hydrogenerator.

The Hydrogenerator's refurbishment of the rotor may be necessary for enhanced reactive power requirements. Table 5 shows three options for rotor refurbishment (Electric Machinery Committee of IEEE, 2010).

For conventional power grids, the field windings of hydrogenerators are not subjected to as high electrical stresses as stator windings. It has been found that for uprates of 50 %, the increase in excitation current requirement is in the order of only 10 % (Znidarich, 2013). Moreover, when uprating, in most cases, it can be assumed that the rotor is not the limiting factor (Blecken, 1997). However, the rotor winding is a critical limiting component for extended reactive power capabilities due to its thermal dynamics and the insulation class (Øyvang and Nøland, 2022; Tome-Robles et al., 2023).

In the case of ventilation systems, the air-forced cooling method can be improved by using adaptive or load-dependent systems or optimization of the rotor fan to increase the airflow rate. For instance, it has been found that with an improvement in the rotor fan geometry, the cooling airflow can be increased up to 19 % (Behravesh et al., 2021). Similarly, an upgrade to water or evaporative cooling systems can be achieved to improve the machine's cooling further.

Table 4
Basic considerations for refurbishment of a hydrogenerator.

Features	Considerations for refurbishment
Operation (Electric Machinery Committee of IEEE, 2010)	Define whether base loading, load following, load cycling, peaking, pump storage, and switching between generating and motoring.
Mechanical (Lehoczky, 1988; Moore, 2001)	- Stator buckling due to incompatibility of new and old components. - Increased power output requires thorough stress analysis to requalify existing parts, such as shaft, spider, keys, keyways, and anchor bolts to the foundation.
Insulation Materials (Znidarich, 2008a)	- Vacuum Pressure Impregnation (VPI) requires large initial capital expenditure and is more suited for large-scale manufacturing. - Resin rich system is better suited for refurbishments and rewinds - Poorly annealed copper may increase Joule losses by 2 %.
Armature Winding (Akiror et al., 2016)	- An analysis of harmonics and radial forces must be done to assess the need of rewinding. - Armature rewinding may considered a new well-designed winding to reduce stator losses.

Table 5
Rotor winding refurbishment according to IEEE Std 1665–2009 (Electric Machinery Committee of IEEE, 2010).

Task	Description
Reinsulating with original conductors	- Cheapest option - Needs cleaning, straightening, and re-annealing - Could be more expensive than expected
Rewind with new conductors	- Minimizes risks - Improves conductor's lifetime - Quickest option
New winding design	- Prefer when reliability issues - For uprating the output - Suitable for reactive power boosting

Table 6
Hybridization of Hydropower plants to enhanced flexibility.

Type	Advantages	Challenges
Hydropower with Storage (Shrivastwa et al., 2020; Banerjee et al., 2021; Kougias et al., 2019; Kim et al., 2019)	Mitigates flexibility issues and provides ancillary services: - Synthetic inertia response - Frequency restoration reserve - Voltage regulation/reactive power support - Black Start - Broader capability for load frequency control signals - Coordination with run-of-river hydropower technologies	- Limited penstock's inertia and gates opening: Response time, available volume, and overloading capacity - Lack of technology maturity: Supercapacitors, High-speed flywheels (TRL 4), Thermal Storage, Fuel cells, and Compressed air
Hydropower with VREs (Piancó et al., 2022; Haas et al., 2020; Gadzanku et al., 2022; Lee et al., 2020; Rosa-Clot and Tina, 2020; Gorjian et al., 2021; Cazzaniga et al., 2019)	Floating PVs on hydropower's reservoir optimizes water use while maximizing with the zero marginal costs of PVs. - Reduces PV curtailment - Reduce dependency to other non-renewable energies. - Up to 50 % increase generation - Increase of capacity factor - Increase in hydrogenerator efficiency - Estimated global potential up to 7.6 TW and 10.6 PWh.	The main challenge for hybrid hydro-VREs system is cost. Especially, for FPVs in hydropower's reservoirs. - Large FPVs increases curtailment during solar peak hours. - There is a trade-off between ecology, health, costs, transmission, and curtailments. - More highly granular spatial and temporal data is needed to conduct accurate analysis. - Results depends on water's body shape and conditions. - Cooling systems to increase production.

Table 7
Variable speed pumped-storage hydrogenerator advantages (Koritarov et al., 2022).

Feature	Advantages
Regulation	- 70–100 % rated capacity at pumping mode.
Efficiency	- High efficiency at partial and minimum load operation
Operation	- Narrower or non rough zone - Lower technical minimum load, 20 %. - Less wear and tear
Grid support	- Flexible voltage support (decouple control) - Better dynamic response characteristics

3.3.3. Hybridization of hydropower plants

The paradigm shift in power systems toward high VRE penetration demands flexible assets, such as hydropower facilities. While PSH can help address the challenges posed by VREs, some techno-economic challenges may slow its implementation.

Providing hybrid hydropower plants may deal better with techno-economical challenges if proven and well-known technologies are used. Table 6 shows the two types of hybridization for hydropower facilities to enhance flexibility and achieve grid reliability and resiliency.

For instance, in a study made by Gonzalez Sanchez et al. (2021), it is found that in the African context, Floating Photovoltaic (FPV) systems covering 1 % to 10 %, of 146 considered reservoirs, can increase the annual electricity production by 50 % to 500 %. In a similar study for EU, Kakoulaki et al. (2023) estimate that 42.31 TWh is the potential for FPV installation in 292 hydropower reservoirs with a 2.3 % surface coverage (same installed capacity as hydropower). That is a 68.35 % increase in annual electricity production. Moreover, in the Continental United States (Spencer et al., 2019), 786 TWh per year (9.6 % of

electricity production in the US by 2016) can be produced by covering 27 % of 24 419 eligible water bodies in the US.

In the case of energy storage, it may extend hydropower's life by reducing wear and tear (Bhatti et al., 2023, 2021; Casarin et al., 2023), i.e., the guide vanes and runner blades mileage is reduced (Kadam et al., 2023), increase the revenue and reduce the hydropeaking (Chalishazar et al., 2022), fast-response ancillary services and support for demands beyond optimally efficient setpoints (Bellgraph et al., 2021).

3.4. Mechanical aspect — hydropower turbine runners

A crucial aspect of hydropower production is turbine runner efficiency. A myriad of work has been done to optimize the design with increased efficiency, chiefly focusing on the geometry optimization using CFD simulations (Masood et al., 2021; Rahi and Chandel, 2015; Aggidis et al., 2023a,b; Mrope et al., 2021; Tiwari et al., 2020). The recent challenge of pump and turbine modes with a wide range of operations is being addressed with CFD simulations as well (Zanetti et al., 2023). Similarly, the minimization of erosion and cavitation has been achieved using CFD (Lama et al., 2022; Kumar et al., 2024). Furthermore, efforts have been made to implement environmentally enhanced turbines, especially on the Alden and Minimum Gap Runner turbines, to reduce fish injury and self-aerating and self/free-lubricant turbines to improve water quality (Quaranta et al., 2021). Furthermore, hydrostatic pressure machines for ultra-low hydraulic heads and high-efficiency low-head Mariucci turbines are further addressed (Quaranta et al., 2020).

3.5. Markets

A substantial challenge for future power grids (FPG) is allocating economic resources to R&D, optimization, reliability, and resilience—especially incentives for baseload units to become more flexible in the long term with a range of extra capabilities, e.g., rapidly and frequently change output to ensure reliability (del Granado et al., 2023). Moreover, the FPG trend is toward a finer spatiotemporal granularity for forecasting, planning, operation, and electricity markets. The finer spatial granularity of prices allows for efficiently managing grid congestion. Similarly, the finer temporal granularity of market operation, i.e., real-time markets, enacts corrections between VREs and demand. Lastly, active and flexible demand-side help integrate VREs (Bichler et al., 2022).

In that sense, the use of Flow-based market coupling (FBMC) may present higher precision, better resource utilization (finer spatial granularity), increased total welfare, and smaller prices for a high share of Wind Power (Cui et al., 2021; Bo et al., 2020). For instance, FBMC allows the transport of 1.5 TWh of electricity through the Nordic grid, especially from the cheap hydro production in the northern areas. South Norway has a lot of flexible hydro production, resulting in flatter bid curves on the supply side (ENTSOe: Nordic TSOs, 2023).

Other approaches include creating a stochastic programming improvement, such as by Zakeri et al. in Zakeri et al. (2019), where revenue adequacy, expected cost recovery, and incentive compatibility are established. However, future work is still needed for an effective multistage model with all the stochastic variables, e.g., intermittency and transmission capabilities.

The framework for developing electricity markets is with the Locational Marginal Price (LMP). Nonetheless, the greatest challenge is an ultra-high penetration of VREs, where the costs increase for the grid resiliency and reliability, e.g., capacity investment for power adequacy, reliability against instability, advanced monitoring and control, and flexibility capacity. Moreover, the power supply is primarily driven by capacity reserve markets and balancing resource procurement, as the short-term variable costs for VREs are near zero (Tan et al., 2022). In that sense, hydropower plays a role in future investment for power adequacy and storage capacity.

3.5.1. Scarcity pricing, capacity and forward electricity markets

The merit order pushes conventional power plants backward because of VRE's near-zero marginal costs. However, in the dark doldrums, i.e., scarcity periods, real-time reserve capacity is required to ensure the system's reliability. Therefore, conventional power plants' business models may shift from baseload suppliers to real-time reserve capacity, i.e., capacity mechanisms in centralized markets. Likewise, scarcity pricing in energy-only (decentralized) markets may lead to new power plant business models to ensure system reliability (Bichler et al., 2022). Moreover, studies show that centralized designs are more efficient in the short run for thermal-dominated markets and with significant network congestion (Ahlqvist et al., 2022). In that sense, hydropower offers real-time reserve capacity as a dispatchable resource, i.e., PSH and large reservoirs, improving system flexibility; it may become a crucial actor for price formation in future zero-carbon systems and exert market power under certain market conditions (Zhou et al., 2022).

In the case of an inefficient market design and associated energy and ancillary service, prices are insufficient to recover the investment and operational costs of generating resources; this is known as the missing money problem (MMP). There are two main approaches to overcoming the MMP. First, forward capacity markets, which provide additional payments in advance to power producers who invest in a future needed capacity for the system; and second, for energy-only markets, the use of higher offer caps to allow higher spot prices and price adders, e.g., scarcity pricing, for energy to incentivize new investments and existing power producers (Liu et al., 2018). Table 8 presents some challenges and opportunities for the two *diametrical* market mechanisms to cope with the MMP.

Although scarcity pricing and Real-Time Reserve Capacity Markets (RT-RCM) operate as diametrical market mechanisms, their compatibility is a false dichotomy. The precedence of achieving scarcity pricing and then introducing RT-RCM may matter. Therefore, a mixed mechanism of an accurate Operating Reserve Demand Curve (ORDC) and RT-RCM may account for short- and long-term price signals for an efficient market and investment for capacity reserves and grid resiliency (Bajo-Buenestado, 2021; Papavasiliou, 2020; Papavasiliou and Bertrand, 2021).

There may need more than market-based solutions such as scarcity pricing and forward capacity markets for short- and long-term markets to ensure system reliability and resilience. Additional studies must be conducted to understand better the interdependency, interoperability, and interchangeability between frequency control schemes beyond new investment for capacity and control services (Prakash et al., 2022). For instance, regulatory intervention for mandatory forward contracting and capacity markets may resolve market incompleteness problems to ensure resiliency (Mays et al., 2022).

Furthermore, balancing services are present after markets have closed (gate closure), in which a TSO acts to ensure that demand is equal to supply, in and near real-time. Hydropower offers short-term scheduling as a dispatchable generation, ramping abilities, and inertia to the system, thus offering ancillary services such as flexible ramping products and frequency response (Aubin et al., 2023).

A case example of achieving decarbonization and system security is China. The market-based measures are the primary constraint toward improving flexibility, such as (1) increasing power producers' flexibility through refurbishments and uprating existing power plants; (2) promoting flexible power grids by constructing an efficient meshed power grid with improved transmission interconnection capacity; (3) encouraging demand flexibility, through demand-response measures for load shifting over time and space; and (4) providing additional flexibility from energy storage technologies (Li et al., 2022).

Table 8
Challenges and opportunities against missing money problem for electricity markets (Thomaßen et al., 2022; Hogan, 2022; Bajo-Buenestado, 2021; Papavasiliou, 2020; Papavasiliou and Bertrand, 2021; Duggan, 2020; Ahlqvist et al., 2022; Liu et al., 2018; Subramanyam and Zhang, 2020).

Market design	Challenges	Opportunities
Centralized Capacity Market	- Do not consider intraday clearing. Thus, it does not account for real-time operation. - Overprocurement of capacity, resulting in the exacerbation of the MMP.	- New generator technologies - Better control-optimization systems - Introduce intra-day clearing - Financial incentives to operate any time.
Decentralized Energy Market	- Relies on short-term markets that depends on random and highly unpredictable VREs. - Example: high share of Wind Power in ERCOT may reduce ORDC prices to zero. - Undermines the valuation of reserve capacity and the back-propagation of price signals to forward reserve markets for future investment. - Improved ORDC mechanism, e.g., adding a correction in dispatch with price-lift adders. - Accurate loss of load probability for ORDC.	- Scarcity pricing to reflect real-time price. - ORDC: A real-time price adder for reserves, which can be use for investment in power adequacy and system resilience. - Demand participation, storage investment, and carbon pricing. - Flow-based zonal pricing - Increase the spatiotemporal granularity of zonal pricing and the number of zones.

3.5.2. Local market mechanisms

Since the power grid is growing fast with more participants, it boosts the development of peer-to-peer (P2P) market mechanisms (Zahraoui et al., 2023). However, the computational burden is the main drawback of achieving real-time P2P electricity markets. A proposal can be an asynchronous online mechanism, where it can freely trade with a neighboring agent without waiting for the idle or inactive one (Guo et al., 2021).

Real-time Local Marginal Pricing (LMP) is critical for energy market participants to develop optimal bidding strategies and maximize their profit. However, the increasing integration of VREs leads to high uncertainties, which increases price volatility in electricity markets and makes LMP prediction difficult for market participants. For instance, a convolutional neural network may predict the real-time LMP (Zhang and Wu, 2022) or energy storage systems to secure real-time balancing by applying a data-driven methodology (Toubeau et al., 2021).

Furthermore, an active distribution network will play an essential role in the flexibility of the entire power system, with mechanisms that enhance TSOs and DSOs market interaction. For instance, Jiang et al. (2022) presented a prospective bi-level market-clearing solution that considers the distributed generation and clearing process of the energy market, the flexibility market, and the distribution market. It shows that it is possible to reduce costs. However, further development is needed to account for the uncertainty. Sanjab et al. (2022) proposed a cooperative game approach to analyze the cooperation potential between the DSOs and TSOs, showing the necessity for investment to improve power exchange capabilities among the different grids. Similarly, the power system expansion under uncertainty is crucial to achieving the system's reliability. Thus, Mishra et al. (2019) proposes a multi-agent system approach for optimal microgrid expansion.

Hydropower may be suitable for forming and operating resilient microgrids (Poudel et al., 2023) if capability metrics are considered, such as microgrid formation, reactive power, inertial response, ramping capability, and storage capacity (Gautam et al., 2023). Hence, hydropower may play a crucial role in local market mechanisms.

3.5.3. Market flexibility and storage capacity

Flexibility can be characterized in four dimensions: (1) time as response, duration/availability, and ramping characteristics. (2) spatiality at a local scale with reactive power provision and large distances through active power availability. (3) Resource types include demand, supply, grid, or storage. (4) Risk is defined by its availability of multivariate probability distribution for variables such as response, duration, available power, and ramping (del Granado et al., 2023). The analysis done by Bistline in Bistline (2019) for coal- and natural-gas-fired power plants suggests that achieving minimum load operation is most valuable when policies rearrange the merit order, and mid-merit units are considered. Units are more flexible than other comparable generators. Moreover, these extra ancillary services revenues may increase the value of enhanced operational flexibility of the generator. Therefore, it further opens the question of extended capability for super-flexible hydrogenerators, especially for reactive power markets. Some reactive power markets manages the following: Great Britain's average cost is £3.27/Mvarh (ESO), and in the United States of America, NYISO is US$ 2,747.61/Mvar, and ISO-NE is US$ 1,188.33/Mvar (Anaya and Pollitt, 2020).

Redesigning short-term market mechanisms may allow for the efficient and comprehensive integration of intermittent technologies to reward flexibility and accurate forecasting while enhancing liquidity and transparency. Furthermore, the minimization of market distortion may include explicit and implicit technology subsidies, Capacity Remunerated Mechanisms, mandated long-term contracts, and Regulated Asset Based models (Batlle et al., 2021).

Some of the consequences of an inflexible power grid may be difficulty balancing demand and supply, significant renewable energy curtailments, area balance violations, negative market prices, and price volatility (Cochran et al., 2014). Furthermore, some metrics and cost-wise improvements are shown in Table 9 to measure flexibility.

Storage technologies are required to increase the penetration of VREs into the power grids. A study by Haas et al. in Haas et al. (2022) found that the perspectives of electricity storage systems from an economic viewpoint highly depend on the storage's operation time, the nature of the overall system, availability of other flexibility options, and sector coupling. In which PSH is the cheapest one.

The main challenges for investing in PSH include the need for a better-designed capacity market and the uncertainty of future electricity markets, particularly given the large-scale deployment of VREs with near-zero marginal operating costs. For instance, PSH technologies offer non-remunerated system services, such as inertia, reduction of VRE curtailment, reduced cycling and ramping of other units in the system, improved power system stability, increased reliability and resilience of grid operation, and reduced transmission congestion and transmission deferral benefits (Koritarov et al., 2022). The flexibility enhancement capacity can be further improved with the fast ramping product (FRP) that PSHs can provide; however, accurate, very short-term net-load forecasting and multi-variate net-load uncertainty estimation need attention. Moreover, the absence of structured pricing mechanisms for FRP procurement may restrict the participation (Sreekumar et al., 2022).

A study by Naversen et al. (2022) using a simplified Northern European System where Norwegian hydropower is coupled to Continental Europe with offshore wind power shows that hydropower can provide 73.5 % of balancing energy for base case scenario, allowing thermal units to cover peak load. However, when the wind power is

Table 9
Flexibility measures and improvements.

Cost-wise interventions	Markets		
	Day-ahead	Intraday	Balancing
System Operation Optimization	× □		
Flexible generation with services	× □		
Storage massive deployment	× △		
Transmission expansion and reinforcement	× △		
Markets mechanisms improvements	× △	△	
Active Demand Response		□ △	
Flexible Generation Adjustments		□	
Enhanced Forecasting Techniques		△	
Fast-ramping Generation			□
Enhanced System Operation Protocols			□
Real-time balancing reserves			△
Emergency Balancing Reserves			△
Metrics	Percentage of installed capacity relative to peak demand by type (×) Maximum change of power balance given an initial state (□) Expected percentage of incidents in a period when cannot cope with net load changes (△)		

increased by 50 %, the hydropower reaches its limits mainly because of transmission capacity, forcing thermal units to increase the real-time balancing energy. Therefore, Norwegian hydropower is a balancing asset for the European green transition.

There are generally three schools of thought regarding the future electricity market structure. First, existing market designs will function as well as today or with minor incremental changes. Second, substantial changes are required for markets to function properly, given the future resource mix. Third, markets should be eliminated or minimized in favor of a return to central planning, vertical integration, and cost-of-service pricing (Ela et al., 2021).

The most significant challenges to achieving the interventions mentioned above on a planetary scale are power system operations, institutional factors, policies, asymmetric economies, market design, and accurate transdisciplinary climate-energy-system models. The overall recommendation is broader and finer spatio-temporal granularity for design, planning, operation, and markets (Mohandes et al., 2019).

3.5.4. Hydropower costs

The Levelized Cost of Energy (LCOE) for hydropower by 2022 is 61 USD/MWh (IRENA, 2022). In the case of PSH, the power capital cost average is 3722 USD/kW, the operating and maintenance cost is around 18 USD/kW/year, and the Variable O&M is 0.51 USD/MWh (NREL, 2022). The estimated Levelized Cost of Storage (LCOS) may vary between different technologies; however, for a TRL of 8-9, the Open-pit mine PSH technology has a cost of 193 USD/MWh according to the U.S. Department of Energy. PSH is the energy storage option with the lowest LCOS value with long-duration storage (Koritarov et al., 2022). Fig. 4 present the LCOE for different technologies.

It is important to remark that LCOE has three shortcomings: (1) it only covers energy services, thus is inadequate for other services such as frequency regulation; (2) it does not account for time or locational variations in the system's need for energy, e.g., comparing a peaking unit with baseload one does not reveal much more; (3) LCOE reveals nothing about the system value of the provided technology's services. Therefore, LCOE fails in capturing a technology's total economic value to the system, referred to as *system value*. This concept can be defined as the suite of services a particular technology can provide within the system interaction, e.g., energy, capacity, and ancillary services. In other words, the system value is sensitive to the conditions of the system (Mai et al., 2021; Mowers and Mai, 2021). Furthermore, the system's flexibility may be crucial in achieving cost-effectiveness for high VREs penetration. Which flexibility can be provided by: interconnections, storage, demand-side response, and fast ramping generation (Chow-Kambitsch et al., 2018). For instance, inflexible systems may incur up to four times the integration costs for VREs compared to very flexible systems, especially for high penetration of VREs doubling the costs per MWh (Heptonstall and Gross, 2020). Finally, flexible resources can substantially reduce the cost of the necessary network reinforcements, e.g., storage and demand-side response (Vivid Economics, 2019).

Recently, a proposed metric named Profitable-LCOE (PLCOE) by Mowers et al. (2023), as is shown in (1), described that the PLCOE increases when the value factor VF_x declines. The value factor reflects the system's spatiotemporal and requirement price distributions and how well they correspond to a technology's location, generation profile, and mix of services it provides. For instance, it is found that the VREs' value factors decline as a function of their market shares and other technologies. Specifically, wind and solar value factors have steeper declines when the system has higher shares of other VREs. On the other hand, the gas combined cycle can have an increasing value factor with increasing VRE market shares. Therefore, without an appropriate increase of the value factor, it may lead to an increase in PLCOE for wind and solar at higher penetration of VREs. In that sense, strengthening hydropower services may improve its PLCOE (Jåstad et al., 2022; Nagel et al., 2023).

$$PLCOE = \frac{LCOE_x}{VF_x} \tag{1}$$

4. Hydro-dominated power systems

The high share of VREs motivates hydropower plants to use their *standardized* synchronous generators for broader operating modes that entail higher cycling, lower utilization, and increasing relative generation costs (Bistline, 2019). Moreover, the hydrogenerators will operate in two main plant modes: (1) synchronous condenser and (2) power *adequator* modes. These two plant modes may require additional reactive power and voltage support; consequently, new operation points of the hydrogenerator over the capability diagram, primarily, an extended overexcited region. However, the extended overexcited part of the capability diagram produces thermal and electromagnetic challenges at short- and long-term schemes.

Studies show the probability of occurrence of operation throughout the year within the capability diagram of the hydrogenerators deviating from the designed nominal points. In the Brazilian case, Bortoni et al. (2021) show that this hydropower plant mainly absorbs reactive power and has many occurrences of synchronous condenser mode (zero active power) to absorb reactive power. Meanwhile, the case for Norway, by cyiza Karekezi et al. (2022), depicts that the hydropower plant is used around the nominal active power with little contribution to absorbing or generating reactive power. The successful case of Tasmania with 90 % of inverter-based resources (IBRs) proves the shift of hydrogenerators to operate in synchronous condenser mode with the fast transition to generation mode and variable governor tuning for boost mode (Halley et al., 2022).

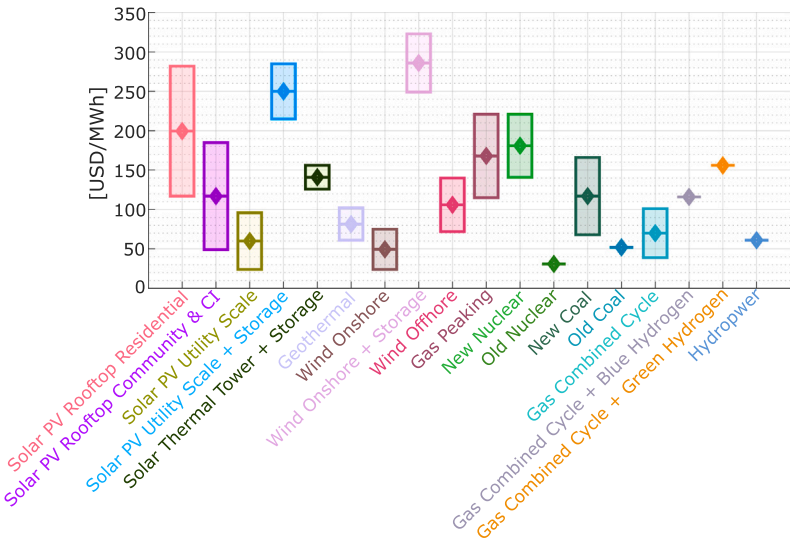


Fig. 4. Levelized Cost Of Energy Comparison (IRENA, 2022; LAZARD, 2023). Here, solar thermal tower + storage is taken from LAZARD (2021). The old coal and nuclear depicts the mean cost of a depreciated assets including decommissioning.

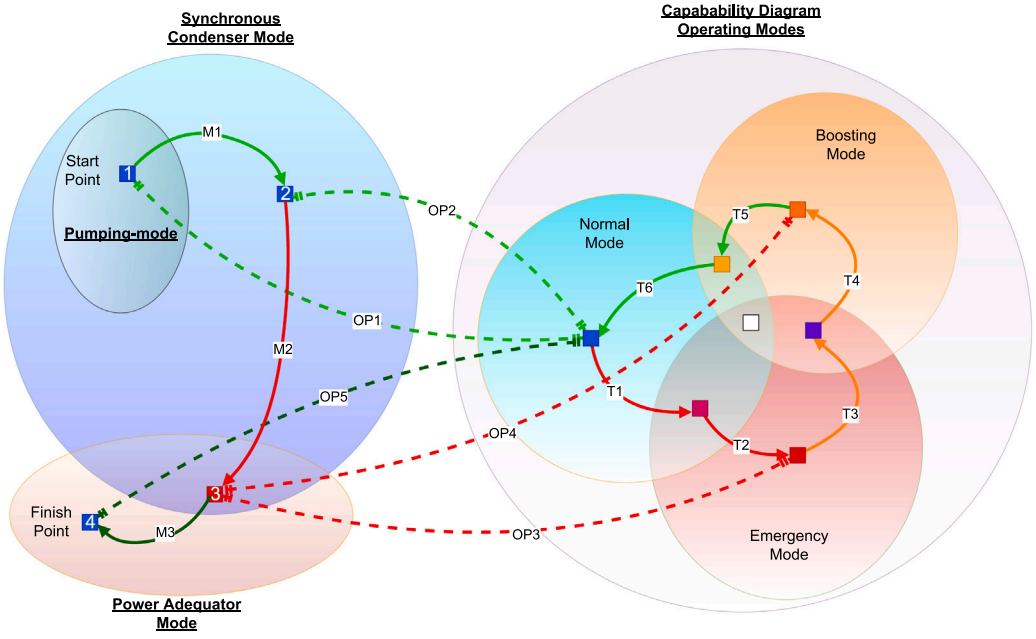


Fig. 5. Super-flexible Hydropower Plant Modes and Operating Modes Scenario. M1, M2, and M3 represent the transition between the plant mode operation. OP1, OP2, OP3, OP4, and OP5 link the plant and hydrogenerator modes. Finally, T1, T2, T3, T4, T5, and T6 are the transitions between the operating modes. After each boosting mode, a recovery mode is implemented to return to nominal temperatures. Here, *Power Adequator Mode* is a new type of operation mode to face power adequacy challenges during *dunkelflaute* periods.

Current projects do not consider synchronous condenser capabilities with inertia and system restart in any revenue modeling for hydropower redevelopment (HydroTasmania, 2023). However, the Australian Energy Market Commission has proposed an efficient inertia provision spot market for Inertia Ancillary Services (HydroTasmania, 2023). The main argument is that dynamic forecasting and allocating inertia in real-time could reduce over-procurement and increase transparency, as well as incentivize innovation in inertia technologies.

Therefore, a new concept of a super-flexible hydrogenerator is presented in Fig. 5, where it needs to constantly account for extended thermal, electromagnetic, and mechanical dynamics. The synchronous condenser and power *adequator* are power plant modes, while normal, boosting, and emergency modes are the hydrogenerator modes. Where the normal mode is to operate inside the capability diagram, whereas boosting and emergency mode depend on the dynamic thermal limits, e.g., 120 °C and 155 °C, respectively. It is essential to point out that the power *adequator* mode is set to offer all the services to ensure system reliability and resiliency; hence, fault-ride through, fast frequency response, and other features to maintain stability during scarcity periods are included.

For instance, an operating scenario could be the one presented in Fig. 5: (I) the start point operating in synchronous condenser plant mode 1 and providing regulation services through a pumping mode, both within the normal hydrogenerator mode (OP1); (II) stop the pumping mode and change to another point of operation (M1), e.g., more or less reactive power within synchronous condenser mode 2 and

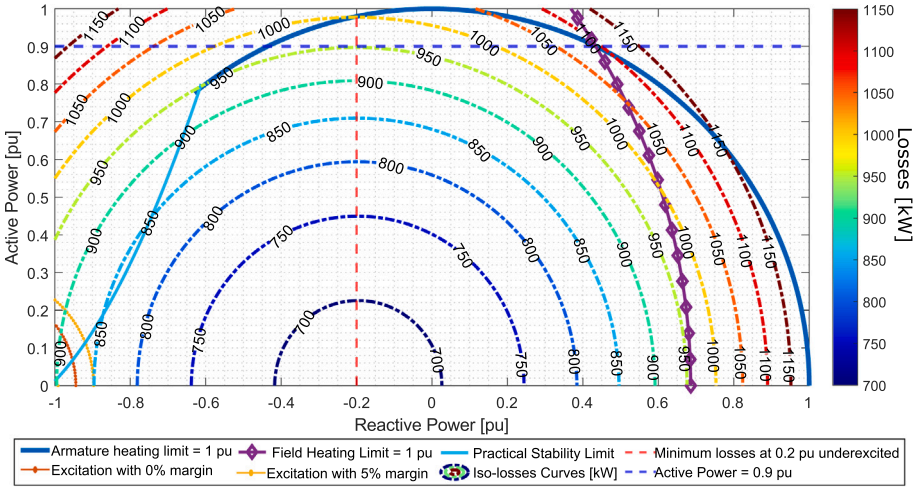


Fig. 6. Iso-losses map over the capability diagram from cyiza Karekezi et al. (2022), Karekezi et al. (2022).

normal operating mode (OP2); (III) engage the power adequator plant mode 3 with reactive power enhanced capabilities (M2), and transition (T1→T2) from normal mode to emergency mode (OP3) in the hydrogenerator capability diagram; (IV) transition (T3→T4) from the emergency mode to boost mode (OP4); and (V) a transition (T5→T6), to normal mode after the windows of opportunities are exceeded according to the dynamic thermal limits, reaching a new steady-state for the power adequator plant mode (M3) without reactive power boosting 4 and a normal hydrogenerator mode (OP5). Moreover, it is vital to notice that the synchronous condenser plant mode can operate in an emergency and boost hydrogenerator modes. At this stage, performing a recovery mode via thermal estimation and control is essential to restore a safe temperature and prepare for any new contingency.

From a power producer perspective, the challenges for hydropower operation are presented in Table 27. These barriers must be addressed with the previously discussed new operating modes characterized. Establishing synergies between the designer and power producer is crucial to elucidate the new services that enable a high share of VREs while maintaining a robust, secure, and resilient power system.

Finally, since one of the boundaries of the lifetime of a rotational electrical machine is thermal dynamics, it is of utmost importance to set a more comprehensive understanding of the underlying physics involved in enhancing the life cycle of the super-flexible hydrogenerator under extreme conditions. Bistline (Bistline, 2019) suggests strategic modifications such as proactive inspections, training programs, and forced cooling for existing fleets. Furthermore, the thermal degradation may be reduced by controlling each component's different thermal expansion and gradients. For instance, Fig. 21 shows a Dual-MPC advance control work example. Here, the temperature rate of change in Emergency, Boosting, and Recovery mode can be further adjusted to prevent wear and tear of the insulation. Hence, it is a paramount Hydrogenerator's PoD to strengthen its adaptability using active cooling with advanced control to predict future operating regimes and pre-heat/pre-cold the machine accordingly to reduce the tear and wear while keeping safe temperature values and rate of change.

4.1. Hydrogenerator operation conditions

Hydrogenerators, as complex multi-phase rotating inductors, use reactive power. According to Holzer et al. (2020), the minimum total loss of a synchronous generator can be achieved around a $\cos\phi = 0.975$ under-excited. The per unit reactive power absorption is calculated with (2) for nominal power output.

$$Q = P\sqrt{\frac{1}{p_f^2} - 1} = 0.9\sqrt{\frac{1}{0.975^2} - 1} = 0.2051 \text{ pu} \tag{2}$$

The iso-losses curves are centered around 0.2 pu of reactive power under-excited (Karekezi et al., 2022), as is shown in Fig. 6. Hence, under-excited operation (UEO) presents the minimum loss. Over history, the UEO of synchronous generators has been used to reduce the line-charging of transmission lines during light load periods and provide voltage regulation (Adams and McClure, 1948; Kekela and Firestone, 1964). However, during the last mid-century, synchronous machines with UEO showed steady-state instability due to the old voltage regulators and not a proper definition for limiting the UEO (Phillips and Rubenstein, 1956). Moreover, synchronous generators have an inherent limitation concerning end iron heating because the leading power factor increases the resultant leakage flux. Thus, the heating in the end iron in the machine increases (Estcourt et al., 1953; Zhang et al., 2022a). Moreover, Table 10 shows the different armature reactions, where capacitive load increases the magnetizing effect as is shown in Figs. 7 and 8.

UEO increases currents circulating through the stator winding and a voltage drop across its terminals, reducing the impedance seen at the machine terminals, which may trigger a relay with a positive offset mho characteristic. This relay is set according to the stability and heating limits of the synchronous machine (Coelho et al., 2015; Morais et al., 2016; Baracho et al., 2018; Park et al., 2018).

Therefore, an efficiency-based optimization controller in a hydrogenerator will present challenges in two directions: First, the power producers will face end-core region wear and tear (Tian et al., 2019; Jichao et al., 2019, 2020). Second, the Grid's Owner will need to coordinate the lack of reactive power for voltage regulation and, therefore, the system-wide optimization of the grid. An open research area is to calculate the magnetic energy absorbed in UEO to design HESMs and size the volume of the permanent magnets (Al-Adsani and Beik, 2022; Mörée and Leijon, 2022) to improve the reactive capability, as is shown in Fig. 9 done by Ammar et al. (2012).

In the present scenario, a critical research question is whether extended reactive power markets will be sufficient incentives to modernize super-flexible hydropower plants to support voltage stability and regulation.

Finally, regarding the high share of VREs, hydropower generation is diminished because of the natural functioning of the market. Moreover, the minimum generation level for hydropower can be understood as a *Flexible Service*, which can be used to integrate VREs into the power grid. The primary constraints to minimum hydropower generation may be hydrological conditions and affect up and down the river, such as agricultural draws downstream, species or river systems

Table 10
Armature Reaction Effects regarding Load Type.

Load Type	Effect
Resistive	Rotor magnetic field ∠90° regarding armature magnetic field reaction. - Crossmagnetizing effect
Inductive	Rotor magnetic field and armature reaction in a opposite direction. - Demagnetizing effect
Capacitive	Rotor magnetic field and armature reaction in the same direction. - Magnetizing effect

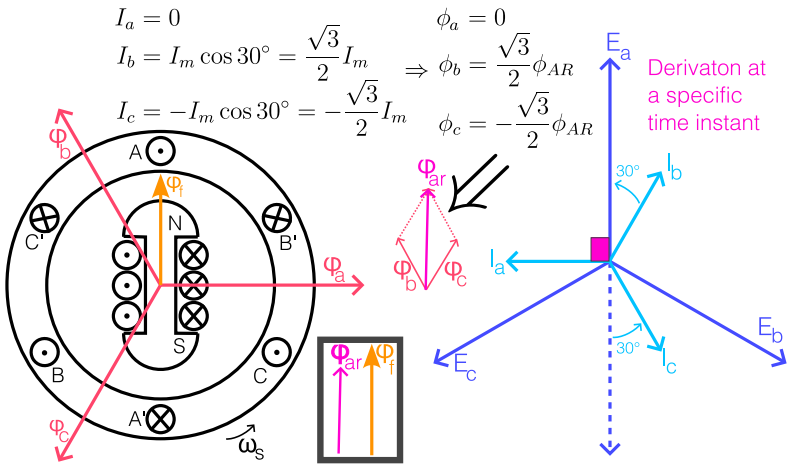


Fig. 7. Magnetizing (capacitive) effect at a particular instant of time. Field flux ϕ_f and armature reaction flux ϕ_{ar} are parallel. The direction of ϕ_a, ϕ_b, and ϕ_c are obtained from the armature winding configuration.

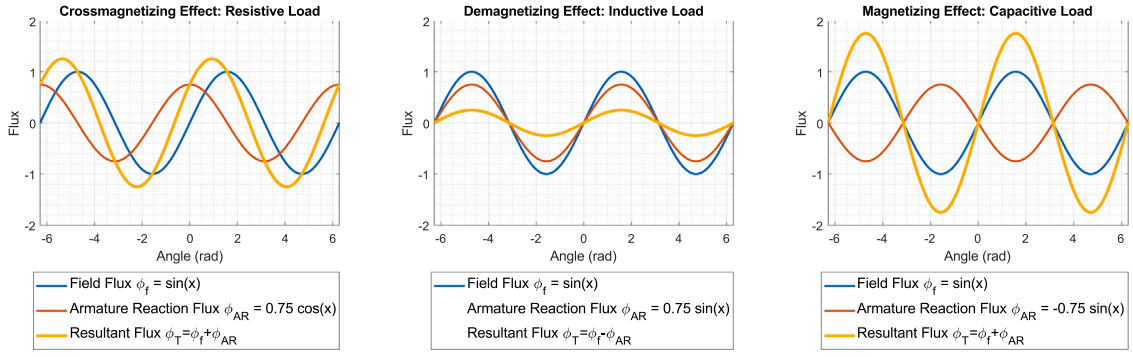


Fig. 8. Resultant Flux regarding Load Type.

downstream, recreational constraints, and stakeholder and international agreements—furthermore, water runoff demand for a significant hydropower generation, and potential VREs curtailment (Denholm et al., 2018).

4.1.1. Insulation degradation

The hydrogenerator stator winding deals with different degradation mechanisms, as is shown in Table 11. Furthermore, to prolong the winding life, it is possible to swap leads after some years of operation (Znidarich, 2008a).

The insulation failures represent 56 % of total hydrogenerator failures, as is shown in Fig. 10. Moreover, hydropower intermittency operations are rising due to the high penetration of wind and solar power and the phasing out of thermal and nuclear power plants. These frequent starts/stops will increase the electro-thermo-mechanical stresses, whereas nowadays, thermal cycling represents only 7 % of the insulation failures. Nonetheless, the lifetime insulation may be higher at 160 °C than 20 °C due to the higher flexibility of binder resin and in part to a post-curing effect (Brutsch et al., 2008).

5. Bottlenecks in hydropower proliferation

5.1. Hydrogenerator's design

This section chronologically addresses the hydrogenerator philosophy of design (PoD) to present a panoramic view. The chronological PoD assessment is divided into three subsections: the classical, a period from the first machines to the evolution of modern standardized generators; the transitioning period, which stands for significant standardization processes; nowadays, the standard PoDs for hydrogenerators. Moreover, it is crucial to notice that the transition periods from different aspects of the hydrogenerator's PoD change depending on the advancements of technology, e.g., the transition from mechanical to analog circuits and then to digital occurred around the '40s and '90s, respectively, in contrast to better vents design improving the cooling of a machine at the beginning of the 1900s. Therefore, various components of the hydrogenerator were standardized and developed at different moments -diachronically. The analysis from classical to modern PODs

Table 11
Insulation degradation mechanisms.

Degradation mechanism	Causes	Prevention
Thermal Heat	- I^2R losses	- Better design - Optimized operation - Good cooling system
Electrical Partial Discharges	High voltage air breakdown: - Insulation voids - Corona effect - End windings surface - End winding coil-to-coil	- Good manufacturing - Ensure same potential (good contact) - Special voltage grading conductive layer in the end-winding - Ensure minimum end-winding spacing
Mechanical Forces	- Mechanical stress over coils due to magnetic rotating wave - Coil MMF interaction vibrations - Thermal expansion shear stress - Magnetostriction core stress	- Good winding design - Optimized operation to prevent large thermal expansion
Environmental Contamination	Moisture, dirt, and chemicals.	- Proper maintenance

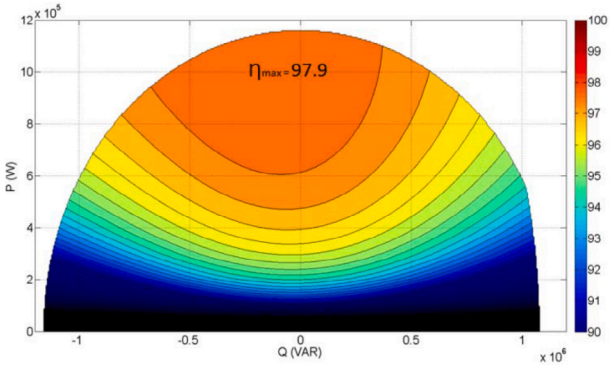


Fig. 9. Improved capability diagram of a Hybrid Excited Synchronous Generator with ratio of hybridization (RH) in per unit of 0.8. RH is the contribution per unit of the permanent magnets in the value of the useful flux.
Source: Extracted from Ammar et al. (2012).

is summarized in Fig. 11 to limit the length of the paper. Here are the citations used (Allerhand, 2017, 2020a; Barns and Bowness, 1922; Bezdek, 2020; Blalock and Woodworth, 2008; Blecken, 1997; Cordeiro et al., 2014; Digital Excitation Applications Task Force of the Excitation Systems Subcommittee, 1997; Halacsy, 1971; Emery and Williams, 2011; Evans and Bergvall, 1924; Evans, 1981; Anon, 2016; Smith, 2013; Fard, 2007; Field, 1905; Floyd and Dunbar, 1931; Forster and Klataske, 1981; Furfari, 2000; Gbegbe et al., 2017; Gu and Lin, 2003; Harder, 1999; Head and Rao, 1967; Heller, 1896; Islam et al., 2021; Kamiev et al., 2009; Kinnunen et al., 2006; Kirkpatrick, 1992; Lamme, 1916; Li et al., 2018a; Li, 2019; Yanping et al., 2012; Lyon, 1921, 1922; Lyles and Smith, 1988; Lyons, 1930; Maurer, 2019; Maurer and Noland, 2022; McCarty and Hart, 1922; McClure et al., 1946; Michell, 1912; Neidhöfer, 2011; Nøland, 2017; Noland et al., 2019; Nuzzo et al., 2017; Owen, 1997; Park and Bancker, 1929; Electric Machinery Committee of IEEE, 2010; Ridley and GEC Large Machines Limited UK, 1988; Roebel, 1912; Royal Society and Royal Society, 1964; Steinmetz, 1894; Sugimoto, 2019; Summerhayes, 1920; Summers, 1927; Tessarolo et al., 2012; Traxler-Samek et al., 2010a; Traxler-Samek and Lecker, 2020; Voith, 2023; Wagner, 1931; Waters, 1903; Znidarich, 2008b,c,a, 2009, 2013). Afterward, a prospective and techno-economic assessment gives an insight into different technologies that are not standardized to provide new services and hybrid solutions. Finally, a summary for each component is depicted in Table 22.

5.1.1. Prospective and techno-economic analysis for hydrogenerators future PODs

The future scenarios are based on several approaches for the iron core, windings, excitation, cooling, and power electronics components. This section is not intended to give an unequivocal POD for future scenarios. Instead, its main endeavor is to systematize scientific knowledge

Table 12
High-Phase Order (HPO) machines possibilities.

Features	Possibility
Conductors (Tessarolo, 2010)	Reduced power per-phase - Use coil windings to avoid Roebel Bar - Increase the filling factor (> 0.6).
Power electronics (Huang et al., 2008)	- Reduce the per-phase inverter rating - Use most competitive drives (2-level topology).
Non-sinusoidal waveform (Huang et al., 2008, 1998; Al-Adsani and Beik, 2022; Vu et al., 2021b,a; Kalaj et al., 2022)	- Performance gain with harmonic injection - Reduced high order spatial harmonics. - Optional distributed or short-pitched windings - Use of concentrated or full-pitch windings. - Better performance with rectangular waveforms.
End-winding reduction (Farshadnia, 2018)	Fractional Slot Concentrated Windings reduces end-windings. - Lesser copper losses, and compact designs. - Improve heat transfer between copper and iron - Higher reactance to limit short-circuit currents.

for new POD possibilities. Based on the prospective analysis, a techno-economical assessment is presented. Moreover, it is advised to perform additional research to understand the challenges and the groundings for a general framework to comply with the IPCC goals. Additionally, other important remarks are addressed in the section of Hydropower Operation: Power Producer and Grid Owners Perspectives.

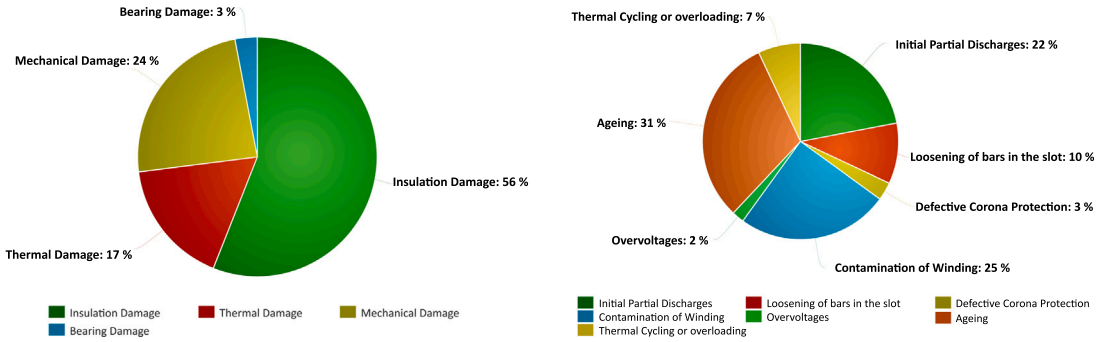
5.1.2. Stator, armature, winding layout, and conductors analysis

In terms of winding layouts, using high-phase order technology reduces the power for each phase. Therefore, according to the PoD, the current or the voltage can be reduce to comply with the design requirements. Moreover, multi-phase machines with prime phase number $m \geq 5$ have many advantages, such as lower torque pulsation, high power density, and better fault tolerance (Abdel-Khalik et al., 2019a). In Table 12, the possibilities for High-phase order machines are shown.

The sizing equation constants K_i and K_p shown in (3) from Huang et al. (1998) can be used to validate that a rectangular waveform will achieve higher power densities. K_i and K_p are the current and electrical power waveform factors, respectively.

$$K_p = \frac{1}{T}\int_0^T \frac{e(t)\times i(t)}{E_{pk}\times I_{pk}}dt \qquad K_i = \frac{I_{pk}}{I_{rms}} = \sqrt{\frac{1}{T}\int_0^T \left(\frac{i(t)}{I_{pk}}\right)^2 dt} \quad (3)$$

Furthermore, the harmonic order on multiphase machines depends on the number of fictitious machines (FMs). The total number of FMs is $(m-1)/2$, with m being an odd number of phases (Vu et al., 2021b). Once the number of FMs is calculated, the associated harmonics are expressed in Table 13.



(a) Hydrogenerator Failures (b) Insulation failures

Fig. 10. Hydrogenerator Failures Statistics (Alagoz et al., 2021; Brutsch et al., 2008).

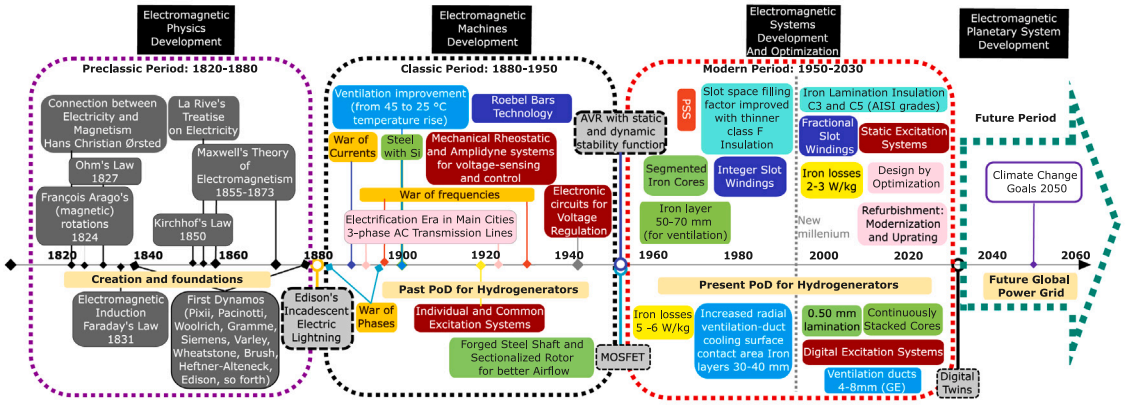


Fig. 11. Timeline of Development of Electromagnetics from Physics to Machines and Systems.

Table 13
Fictious machines, reference frames, and associated harmonics for m-phase machine (only odd harmonics). Extracted from Vu et al. (2021b).

Fictitious machine	Reference frame	Associated harmonics[a]
1*st* fictitious machine (FM-1)	$\alpha_1 - \beta_1$	$mk \pm 1$
2*nd* fictitious machine (FM-2)	$\alpha_2 - \beta_2$	$mk \pm 2$
⋮	⋮	⋮
l−th fictitious machine (FM-l)	$\alpha_l - \beta_l$	$mk \pm l$
Zero-sequence machine (ZM)	z	mk

[a] m is the number of phases, l the FM order, and k the associated harmonic order, with $(mk \pm l) \wedge mk \in \mathbb{N}_o$ (Odd harmonics).

The extension of the 2-reaction theory (Park's transformation) to the multiphase machine has been done since 1937 to nowadays, showing the reduction in the torque fluctuations in 1969 (Ku, 1937; Ward and Harer, 1969; Frikha et al., 2023). Nonetheless, the physical meaning for enhanced performance is argued to be related to "saturation" harmonics, resulting in a flattened MMF traveling at the same speed and direction as the fundamental (Kindl et al., 2020). Recent studies have shown that multiphase machines add more degrees of freedom in control, i.e., harmonic current injection control systems may reduce the losses and improve the overall performance of the machines (Kindl et al., 2020; Spas et al., 2020; Abdel-Khalik et al., 2011; Yang et al., 2021; Duran et al., 2017; Huang et al., 2008; Kong et al., 2016, 2017). Typically, multiphase machines are modeled using the vector space decomposition (VSD) approach provided by the generalized Clarke's transformation (Duran et al., 2017; Kindl et al., 2020; Abdel-Khalik et al., 2011; Spas et al., 2020; Yang et al., 2021; Kalaj et al., 2022). The multiphase winding design could be improved for the harmonics if a 4-layer or Multi-Layer FSCW is considered to perform a short-pitched winding design by further reducing the overall rotor losses, unbalanced magnetic pull, and torque ripple (Tessarolo, 2018; Tessarolo et al., 2019, 2021).

A critical remark is the stator frame, standardized for 3-phase machines, meaning that the number of slots is a multiple of 3. Therefore, implementing multi-phase winding layouts, especially prime-phase order (symmetrical), becomes a real challenge. However, Abdel-Khalik et al. in Abdel-Khalik et al. (2019a,b) presented an optimal selection of Slot/Pole combinations to implement High-Prime-Phase Order ($m > 3$) machine topologies within Standard Three-Phase Stator frames. Nonetheless, with a stronger trend toward multi-phase machines, stator frame industrialization should be constructed without any complication for any number of phases.

Moreover, multi-phase machines are already in use for traction, wind energy, and ship propulsion system (Yaramasu and Wu, 2017; Duran et al., 2017). Recently, a 24-phase (8 × 3-phase) 4-MW class generator demonstrator for Future Hybrid-Electric Aircraft has been tested for a power density of 17.3 kW/kg (Trentin et al., 2021; Anon, 2021a). Furthermore, some multi-phase topologies may allow Symmetrical/Asymmetrical winding reconfiguration as is shown in Slunjski et al. (2020), giving more degrees of freedom for control systems. The unique advantages of high-phase order machines are shown in Table 14 (Duran et al., 2017).

The multi-phase windings can be divided into symmetrical for a prime number of phases, and asymmetrical for a non-prime number of phases (Levi, 2008). Moreover, for ship propulsion, 15-phase machines have been used with General Electric motors of the destroyer Type 45 and Queen Elizabeth Class (QEC) aircraft carrier (Duran et al., 2017; Pisani, 2019). Furthermore, some investigations into thermal behavior in post-fault operation have been carried out for 15-phase machine, where it has been confirmed that after one phase open-circuit fault occurs, the maximum temperature does not exceed the

Table 14
High-phase order machines advantages.

Feature	Advantages
Faults	High tolerance and smooth post-fault operation without additional hardware
Current	Current density reduction for same voltage and power rating
Torque	Lower ripple
Voltage	Many phase voltage combinations
Harmonics	- Power density improvement by current harmonic injection - Suitable for concentrated windings

Table 15
Multiphase power system challenges.

Multiphase systems	Challenges	Opportunities
Generation n-phase to 3-phase (Tabrez et al., 2022; Djebli et al., 2019a; Tabrez et al., 2021; Djebli et al., 2019b; Tabrez et al., 2019; Schroeder and Ponick, 2018; Shen et al., 2022a; Althi et al., 2023; Shukla et al., 2023)	Step-up transformer for a prime number of phases.	- Optimization of number of turns, windings, and core geometries to achieved balanced voltage, current, losses, and thermal distribution. - Multiphase transformer are more economical, reliable, efficient, and have lesser THD than PE systems.
Transmission n-phase to 3-phase (Krizauskas et al., 1997; Landers et al., 1998; Bortnik, 1998; Stewart and Hudson, 1988; Fan et al., 2011; Yan et al., 2012; Pierre, 2019; Koley et al., 2015; Eboule et al., 2022; Shen et al., 2022b)	- Break-even distance for 3-phase multi-circuit systems and new multiphase systems -Protection schemes: Fault identification and online monitoring and control -Roll and fully transposed	- Reduced right-of-way (ROW) - Compact, smaller, symmetric and more pleasing structures - Better voltage stability performance - Keep a better stability margin - Require fewer var reserves. - Reduced reactive power losses - Increased line loadability - No need for a complete transposition
Mixed grids n-phase to n-phase	Standardization: - Generation - Transmission - Distribution	- Loss reduction (Ribeiro Miranda et al., 2020)

windings insulation temperature and can continue to operate (Kuang et al., 2019).

Nowadays, hydrogenerators are standardized as 3-phase machines with fractional slots windings for $q > 1$. However, it is well-known that multiphase machines offer several advantages over 3-phase machines, such as lower torque pulsation, high power density, higher efficiency, and better fault tolerance. Nonetheless, they present techno-economical challenges for multiphase power systems, such as Interconnection, Protection Schemes, and security, i.e., Stability, Inertia, and Reliability. These challenges have been addressed for PE-interfaced machines, where n-phase to three-phase is decoupled electromagnetically. Hence, the grid does not see the n-phase generator. Nonetheless, non-PE-interfaced (physical inertia) units for multiphase power systems can be assessed in three ways, as is shown in Table 15.

Since all the research done in multiphase power systems conveys that achieving lower prices for specific cases is possible, there is still a need for a significant technology gap in implementing these systems. Therefore, it can be advised to strengthen the research projects into multiphase power systems within the System Optimization and Reliability framework, especially bulk hydropower services and flexibility.

In the case of the stator, it is strictly linked to the winding layout, size, and cooling of the machine. Commonly, 3-phase stators would have a multiple of 3 number of slots. However, odd-phase number winding layouts require new stators. Furthermore, the most natural extension of the stator number of phases is to have a multiple of 3.

Moreover, in terms of materials, grain-oriented (GO) teeth offer the advantage of loss reduction, i.e., less heating to cool down if the stator is appropriately constructed. However, grain-oriented steel stators' construction is more expensive than non-oriented ones due to their complexity. Moreover, hydrogenerators are expected to operate in a wide-range area of the capability diagram for future scenarios, going from low saturation to high saturation regions. Hence, this magnetic flux density long-cycle may create problems related to magnetostriction because of the anisotropy in its *B-H* characteristics (Somkun et al., 2009) because the magnetic flux angle changes according to the load and power factor angle.

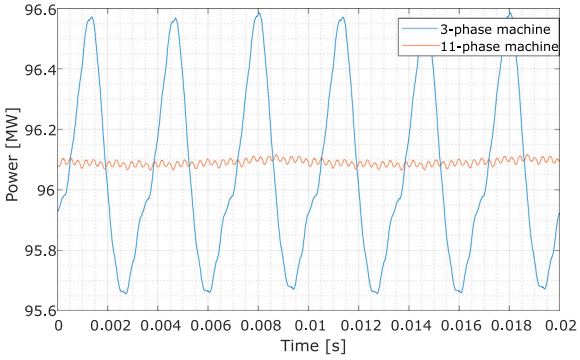


Fig. 12. Power comparison between a 3-phase and 11-phase machine with the same topology. It shows how the multiphase can reduce the torque pulsations while keeping the same output power.

Since hydrogenerators may be a salient-pole rotor topology, the resulting direction of the (DC) flux densities in the poles is forced by the field coils to stay in the radial direction. Therefore, using GO materials in the poles opens a new window of opportunities to improve the POD for hydrogenerators.

In multiphase hydrogenerators, according to the POD, the voltage or current can be further decreased, which might allow transitioning from expensive Roebel Bars to other cheaper conductors. For instance, a multiphase winding layout from the machine in Øyvang (2018) exhibits the current density decreases with a proper POD, with a torque pulsation reduction of 94.51 % while the line voltage is kept constant at 11 kV as shown in Fig. 12. The original machine is presented in Fig. 13, a 3-phase winding layout with a base winding of 2 poles and

Table 16
Segregated Losses Comparison and Yearly Loss Cost Savings. Here the difference in the rotor loss is because the three-phase machine has a field current of 1055.2 A, and for the 11-phase is 1065 A. The electricity price used is 0.1066 USD/kWh. The yearly loss costs is calculated for a uniform nominal operating regime.

Machine	Stator loss	Rotor loss	Total loss	Yearly loss costs
3-phase	387.26 kW	516.13 kW	903.39 kW	843 600.04 USD
11-phase	319.10 kW	525.76 kW	844.84 kW	788 902.11 USD
Difference	−16.3038 %	1.865 81 %	−6.481 14 %	−6.481 14 %
Yearly Loss Cost Savings				54 674.93 USD

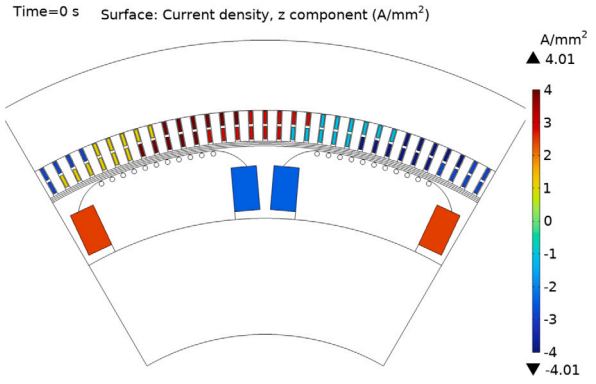


Fig. 13. 3-phase machine Current Density Distribution.

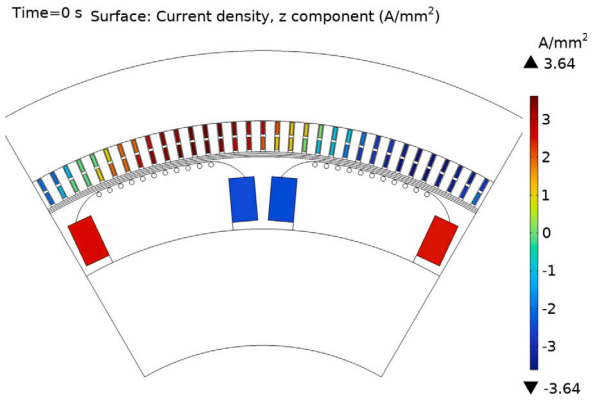


Fig. 14. 11-phase machine Current Density Distribution.

33 slots, connected with three circuits in parallel and 2 in series. The multiphase POD is depicted in 14, an 11-phase winding layout with all the circuits in series for the same base winding. Moreover, 11 phases are handpicked because the base winding layout is 33 slots with two poles, and the highest prime number of phases to achieve the optimal winding layout is 11 (Tome-Robles et al., 2022).

The physics behind this is a better magnetic flux density distribution along the airgap; thus, a smaller current density is needed to produce the same power. Furthermore, more than a reduction of 9.2 % in current density may be needed to justify a transition from Roebel Bars to simpler coil winding configurations. Instead, it may lead to higher power densities. For instance, a multiphase winding layout may reduce iron losses (Tome-Robles et al., 2022), providing a broader thermal capacity and extra savings for loss reduction as is shown in Table 16 from the previously worked case, only with the new winding layout. Additional considerations for connection may be needed, either through PE or physical connections, as pointed out in Table 15.

It is vital to notice that both winding layouts comparison are not Fractional Slot Concentrated Windings (FSCW). Nonetheless, to achieve FSCW, the base winding layout would need to be changed, which fails to accomplish a one-to-one comparison. However, FSCW where $q < 1$ has many advantages over other PODs, mainly reduction of end-windings. Therefore, achieving FSCW with Roebel Bar will depend on the mechanical constraints of bending, magnetic forces, lamination, and heat dissipation in end-windings.

Thinner insulation is needed to achieve a high fill factor for the conductors. Thus, increasing the power density of the machine. Moreover, higher-class insulation may lead to greater current density, i.e., power density. However, since new and better conductors are a trend in power systems, the main drawback is the economics of using them in hydrogenerator. Therefore, the feasibility of implementation depends mainly upon new market mechanisms and services for the remuneration of extra capabilities of the hydrogenerator.

Finally, further research into multiphase windings POD may lead to a higher degree of freedom for improvement, such as parallel and series circuits to change voltage or current, reducing insulation thickness and preventing PDs.

5.1.3. Rotor and excitation systems analysis

Static excitation systems are preferred whenever a fast dynamic response is required. However, it may produce brush wear and possible sparking. Brushless exciters overcome these challenges at the cost of worse dynamic performance, limited measurement and protection to the field winding, and an installation of an auxiliary rotating machine, i.e., an exciter, on the main generator shaft (Noland et al., 2019). Nonetheless, satisfactory results were tested in a brushless high-speed de-excitation system for a 15 MVA hydrogenerator (Rebollo et al., 2017), and in a high-speed response brushless excitation system (HSRBES) with a numerical Average-Value Model (AVM) for 47 MVA hydrogenerator (Noland et al., 2020a). Furthermore, brushless exciters have already been used commercially for large-hydrogenerators (Voith, 2013).

Additional possibilities for excitation systems is to provide field current with specific harmonics to the DC field current to minimize either voltage harmonics or magnetic forces (Evestedt et al., 2021). For example, a multiphase machine with third harmonics injection improves stator core utilization (Yao et al., 2021). Furthermore, a 32-phase brushless hybrid excited synchronous machine (HESM), i.e., Permanent Magnets + Wound Field, has been proposed for a 9-phase generator in Anon (2019), Beik and Al-Adsani (2022). HESM hydrogenerators have been proposed (Kamiev et al., 2016), where the efficiency and reactive capability can be improved (Ammar et al., 2012; Ployard et al., 2014, 2016).

Finally, nonlinear models and controllers may be required for excitation systems, e.g., neural networks, fuzzy control, and adaptive control, to maintain the power system stability during transition state, such as synchronization, short-circuit, curtailment of load, and variability due to intermittent power generation, e.g., wind and solar (Jerkovic et al., 2018).

5.1.4. Iron core analysis

A high silicon (6.5 %) steel sheet (10JNEX900) has achieved 0.50 W/kg at 50 Hz and 1.0 T. Specifically, for cores driven by a PWM inverter, the power losses are reduced compared to grain-oriented 3 % Si steel due to a good performance at high order harmonics (Namikawa et al., 1998; Takashi and Hironori, 2012). Moreover, tests have been performed with Permendur in switched reluctance motors, achieving

Table 17
Rotor and excitation systems challenges.

Features	Challenges	Opportunities
Insulation	Extended Thermal limits for broader operating points	Class H insulation
Materials	Rotor iron core heating losses	Grain-Oriented poles in the radial direction. - Avoid magnetostriction due to anisotropy of BH characteristics
Excitation	- Reactive power support - Voltage stability - Joule heating losses	Hybrid excitation with Laminated Permanent Magnets
Control	- Grid code static envelopes - Lack of coordination and optimization system-wise for a high share of VREs. - Grid stability for a high share of VREs	Grid code dynamic envelopes: - Nonlinear and Adaptive control models - Model-free Neural networks and Machine learning - Weather-based control
Exciter	Brush wear and sparking	Brushless exciter: - Harmonic injection - Thermal measurement

higher efficiency. This material is capable of high AC field operation due to high magnetic flux density saturation (Hasegawa et al., 2010, 2011; Nakajima, 2019).

The use of grain-oriented (GO) magnetic materials for hydrogenerator stator cores cannot be justified (Znidarich, 2008b). However, some reports propose using grain-oriented magnetic material in the teeth for small motors. Since the tooth flux density direction is relatively uniform in the radial direction regardless of the position of the rotor. In Son and Lim (2021), shows by partially using the GO on the stator teeth, the improvement in torque and efficiency is 9.89 % and 1.7 %, respectively. In Pei et al. (2017), a small motor was constructed to test the GO material in teeth, resulting in an output increment of 4.3 % and the efficiency increases by 2.7 %.

Furthermore, Soda et al. (2020) shows a loss reduction if the rolling direction is considered in segmented cores. Further research must be done to include the inclination angle of the flux lines due to the machine's loading level, i.e., the Maxwell Stress Tensor Angle between radial and tangential magnetic flux density in the airgap.

For the hydrogenerator's rotor and excitation systems, the techno-economical matters can be divided into voltage stability support, which entails reactive power contribution and implies losses and heating minimization; rotor angle and frequency (RAF) stability support, which involves damper windings and the exciter's fast dynamic response. The challenges are shown in Table 17.

HESM may manage voltage and RAF stability with brushless exciters. The drawbacks of HESM are the technical issues retaining PMs in large rotor poles due to mechanical forces and speed response, de-excitation, field forcing requirements, and starting performance for brushless exciters (Noland et al., 2019). However, fiber-reinforced sleeve rotors for high-performance PM motors have become a fascinating technique to research on Wang et al. (2022). From the economic perspective, the cost of HESM adoption must come with increased revenue for the extra reactive power services for voltage stability and fast frequency support in a high share of the VREs scenario.

Moreover, rotor optimization can lead to mechanical and economic improvements, as is pointed out in Vukšić et al. (2016), mainly as a cost-effective method to improve performance. Hence, advanced optimization methods must be done for ventilation and loss reduction. Lastly, the hydrogenerator's operation under extreme intermittent VRE scenarios may be reasonable with adequate temperature measurement, loss reduction, and optimal ventilation.

The iron core rationale is solely relevant to adopt new and better materials and lamination.

For the former, it is an expensive solution; nonetheless, if the losses and heating are worth minimizing for a wide range of operation points while obtaining revenue for these extra services, it is advised to research further. Otherwise, special iron core materials are exclusive for unique applications such as aviation and high-performance vehicles.

For the latter, the major drawback is the insulation thickness of laminations. For instance, if the lamination is a JNEX-Core with 0.10 mm thickness and 0.5 W/kg losses at 50 Hz and 1 T, the stacking factor may be as low as 90 %; thus, reducing the power density of the machine. Therefore, it may be advised to use thinner lamination for loss and heating minimization if the extra costs are compensated with services to the grid. Alternatively, the opportunities are reduced to use thinner lamination in the iron core's end-section to further reduce the losses without threatening the costs.

Table 18
Peltier Modules possibilities.

Challenges	Possibility
Ambient heat	Maintain a reduced temperature
Intermittent load	Reduce the exceeding temperature
Thermal cycling	Reduce the thermal gradients
Moisture ingress	Warm up the core and windings when low load conditions
Predictive Control	Anticipate peak load and reduce the core temperature

5.1.5. Cooling systems and ventilation analysis

Cooling system for hydrogenerators are vital for extreme condition operation. The most common system is cooling by forced air. In contrast, liquid, e.g., water, is the most efficient cooling system and more challenging to implement (Pyrhönen et al., 2008). Moreover, a new evaporative cooling system stretches to have the advantages of air and liquid (Ruan et al., 2007).

An alternative effort may be Thermo-Electric Coolers (TEC), i.e., solid-state heat pumps, a new technology that can improve the heat flow from the core and casing to the surrounding air, lowering the temperature gradient. For instance, a TEC installed instead of a cooling fan in a 370 W induction motor could remove up to 11 % of the heat, using Peltier Module TEC1-12706 of 57 W heat removal at 50 °C on the hot side (Toren and Mollahasanoglu, 2021).

Furthermore, in Lucas et al. (2022a), three TEC setups are used: (1) heat sink, (2) heat pipes, and (3) water-cooled heat sink. The results show a reduction in the slot temperatures by 14.9 %, 12 %, and 18.96 %, respectively. The temperature reduction allows an increased insulation life cycle or output operation without exceeding insulation thermal ratings. The possibility of removing heat from electric machine cores is shown in Table 18.

Thermoelectric modules can have cooling power densities of 3.89 kW/kg, i.e., the TEC1-12730HTS Peltier Module with 350 W of max cooling power at 30 A, 14 V and η_{max} =83.3 %, which price is US 24.19 with a power density of 91.05 kW/m^2 and size of 62 × 62 mm. However, the heat transfer coefficient may vary due to dissipating conditions in the hot side, i.e., exterior cooling system (Nasri et al., 2017; Liao et al., 2018; Agarwal and Kong, 2020). Tables 19 and 20 compare different mediums of heat transfer coefficients and cooling systems performance. Here, TEC modules may have a low cooling ratio; however, they have the advantage of actively controlling the temperature (Lucas et al., 2023a), i.e., heat Lucas et al. (2023b, 2022b) or cool (Lucas et al., 2022a; Morikawa and Katsura, 2022, 2023), when needed. Thus, an advanced control model may be implemented to increase efficiency and energy and reduce wear and tear. Therefore, the iron core may act as the heat source or sink accordingly.

For instance, the 103 MVA hydrogenerator described in Øyvang et al. (2019), cyiza Karekezi et al. (2022), has 211.92 kW of stator iron

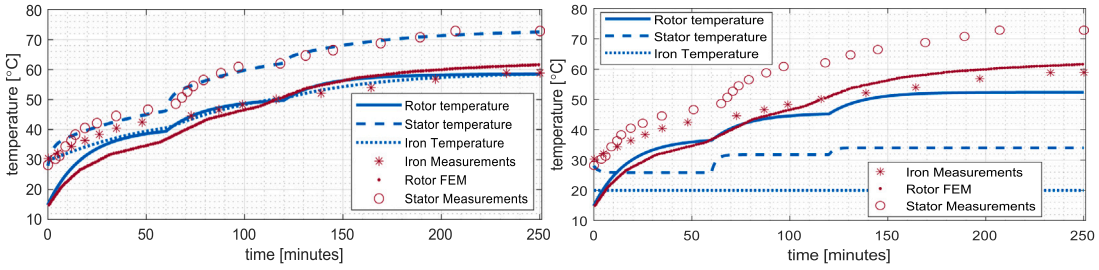
Table 19
Cooling efficiency and Temperature Rise comparison of different cooling systems.

Cooling type	Cooling fluid temperature	Highest temperature	Temperature rise	Cooling efficiency[e]	Cooling ratio[f]	System description
Air-Cooled (Øyvang, 2018; Øyvang et al., 2019)	23.44 °C of air	94.63 °C of rotor 82.5 °C of slots	71.19 K 59.06 K	0.595	1.0818	Counterflow heat exchanger
Water-Cooled (Yuan et al., 2013; Milic et al., 2013)	58.43 °C of water[a] 25°C of water[b]	83.33 °C of rotor[a] 53.2°C of slots	24.91 K 28.2 K	0.6[b]	2.6629	Heat Exchanger[b]
Evaporative[c] (Lin et al., 2007, 2015; Feihui and Lin, 2021; Yuan et al., 2013; Wang and Ruan, 2022; Xie and Ruan, 2013)	25 °C of water[b] 25.09 °C of water	69.1 °C of rotor 62.2°C of slots	44.1 K 37.11 K	0.60–0.63	3.1343	Condenser
Air-TEC-Cooled[d] (Lucas et al., 2022a)	25 °C of air[b]	58.71 °C of slots	33.71 K	0.636	0.6546	14 simple heat sink modules
Water-TEC-Cooled[d] (Lucas et al., 2022a)	11.1 °C of water	55.92 °C of slots	44.82 K	0.636	0.3666	10 water-cooled heat sink modules

[a] Temperature calculated at 250 min from the heating gradient value obtained by a first-order polynomial curve fitting.
[b] Assumptions. For the case of Air-TEC-Cooled, the value corresponds to the ambient temperature.
[c] Evaporative cooling presents better temperature distribution in the stator bars compared to water cooled.
[d] Both cooling systems are tested in the same machine.
[e] The cooling efficiency is calculated by Q/P. Here, Q is the cooling capacity and P is the total power used.
[f] The cooling ratio is found from $P_{loss}/P_{cooling}$. In the case of the TEC modules, is tested in a 545 W motor, assuming an efficiency of 78.8 %.

Table 20
Heat transfer coefficient comparison.

Medium	h[W/m^{2}K]	Citation
Air-cooled Hydrogenerator Airgap Air-cooled Hydrogenerator Rotor End Windings	145 83.6	Øyvang et al. (2019)
Air-cooled Hydrogenerator Field Windings	75–150	Li et al. (2020), Dang et al. (2018)
Water-Hydrogen^2-Cooled Turbogenerator End Windings	533.9–1149	Han et al. (2014)
Over-all calculation of evaporative condenser	284–318	Liu et al. (2009)
Evaporative Cooling End Windings	1240–2500	Li et al. (2009)
Evaporative Cooling Hydrogenerator Field Windings	up to 10,000	Wang and Ruan (2022)
Evaporative Cooling Hydrogenerator Stator Busbar	74–142	Dong and Ruan (2019b,a)
Evaporative Cooling Motor Stator Busbar Air-Cooled Motor Stator Busbar	73–83 54–57	(Dong et al., 2021)
Nanofluid-Based Cooling Stator End Winding	817.6	Xiong et al. (2019)
Air, free Air, forced Liquids, forced	2.5–25 10–500 100–15,000	(Kosky et al., 2013)
TEC Module	20–1500	Nasri et al. (2017), Liao et al. (2018), Agarwal and Kong (2020)



(a) Heat-Run Test without Peltier Module (b) Heat-Run Test with Peltier Module ($T_{Fe} = 20°C$)

Fig. 15. Thermal dynamics for a heat-run test of a hydrogenerator from Øyvang et al. (2019) with a DC active control of the Peltier Modules to fix iron core temperature at ($T_{Fe} = 20$ °C). Here, the iron core acts as a heat sink.

core losses. Therefore, with 673 modules is possible to remove up to 235.55 kW of thermal losses, that is 11 % of the total surface area of the stator, i.e., 2.59 m^2 out of 23.5 m^2, with a total cost of US 16,279.87. The total consumption of the Peltier Modules will be around 318.18 kW, representing a 0.003 432 % reduction in the efficiency of the generator. In other words, the Peltier Modules without any exterior cooling system have an efficiency of 74.03 %. In Fig. 15, an example where the iron core temperature is assumed fixed to 20 °C with active control of the TEC modules shows a substantial temperature reduction in the stator copper and a minor reduction in the rotor copper. Moreover, if 90 % surface area is meant to be used for cooling purposes, it will correspond to 5502 modules, giving a total cooling power of 1.92 MW with a cost of US 133,093.38. Therefore, further research must consider all the economics and thermal dynamics.

Furthermore, nanoparticles may enhance thermal properties by dispersing them into conventional heat transfer fluids, e.g., water, oil, ethylene, propylene, and glycol, to obtain the resulting nanofluid or create a ferrofluid to enhance magnetic and thermal properties in the airgap of a machine. However, the most effortless manner is to substitute a conventional cooling fluid with a nanofluid to enhance heat transfer capacity and decrease the number of cooling channels in the slots. Thus, the current-carrying ability of the machine increases, which in turn increases the power production (Vaschetto et al., 2019).

In order to improve the electrical, thermal, and mechanical properties of the conductors, polymer nanocomposites can be used. For

Table 21
Cooling systems and ventilation challenges.

Features	Challenges	Opportunities
Ventilation ducts	Narrower ducts require higher pressure	- Ferrofluids to improve heat transfer capabilities - Tiny TEC modules
Cooling System (Yuan et al., 2013; Milic et al., 2013) (Liang and Mudawar, 2019; Bakthavatchalam et al., 2020; Svoboda and Blecken, 2018)	- Air passive cooling - Water cooling have problems with clogging, plugging, corrosion, or leaking. Also, their compact POD reduces the inertia. - Further worldwide adoption of evaporative cooling systems. *Nanofluids challenges: erosion, sedimentation, clogging, corrosion, high cost, production difficulties, and stability for long-term use.	- Adaptive and fault-dependent cooling: (1) change in fluid flow (2) change in airflow (3) change in both - Nanofluids for water-cooled generator improves thermal properties, and reduces the water channels. - Well-known Evaporative Cooling to improve stator bar performance.
Iron core thermal control (Griffith et al., 2000)	No thermal control. There is little research regarding TEC modules for large machines	Active Thermal Control with TEC modules to reduce wear and tear.

example, Micro- or nano-fillers can be added to the epoxy impregnation resins, and the thermal conductivity would be improved, resulting in temperature reduction in the stator winding. Thus, the machine efficiency, insulation lifetime, breakdown voltage, thermal conductivity, and current density may be improved (Hemmati et al., 2019). Moreover, in Nguyen et al. (2021), has been demonstrated that using 2D nano-structured winding insulation exhibits high thermal conductivity of $>0.8\,\mathrm{W/(m\cdot K)}$, high breakdown strength of $>40\,\mathrm{kV/mm}$, low dielectric constant of less than 5.5, and low dielectric loss factor of less than 2.5 % at 155 °C.

Finally, research on thermo-mechanical insulation aging of evaporative cooled stator bars have shown better anti-aging abilities than air-cooled insulation (Chen and Ruan, 2015).

Improving ventilation performance in a hydrogenerator is an optimization problem. Nonetheless, further PODs for cooling systems and ventilation, alternative, cutting-edge technologies are mentioned in Table 21.

In the end, despite overcoming some challenges by designers, the Power Producers prefer more conventional generators. Then, motivating POD changes for better cooling of hydrogenerators depends upon the remuneration of novel market services to ensure the grid's reliability and resilience (Eriksson, 2007).

5.1.6. PE-interfaced hydrogenerator analysis

Variable Speed Hydrogenerators (VSHs) are used for pumped-storage hydropower (PSH) plants, mainly to shift excess energy to peak loads periods (Schmidt et al., 2011). This type of technology uses power electronics (PE) to connect to the grid, which may provide complete flexibility of active and reactive power. VSH may allow for higher power production to be achieved at higher speeds (Fraile-Ardanuy et al., 2006). Hence, a broader efficiency optimization can be performed through speed variations for every loading and hydraulic condition (Nobile et al., 2018) by coupling the turbine's hill chart and the hydrogenerator efficiency map. In Gao et al. (2021), a 12.5 % increase is achieved for a Small VSH Plant.

VSH can be designed to have smaller volumes, lower material costs, and higher efficiencies than their conventional counterpart. For instance, a unity power factor generator does not need an extra size for reactive power contribution. As a result, a 10 % volume reduction, 0.22 % efficiency increase, and 15.4 % cost reduction can be achieved with a Unity Power Factor POD (Holzer et al., 2020). Moreover, increasing the synchronous reactance may lead to a total machine cost decrease by 7.6 % as achieved in Hestengen (2016). Another POD is to change the electrical frequency of operation while maintaining the mechanical frequency, which means not optimizing the turbine for higher rotational speeds but increasing the number of poles to increase the electrical frequency. The pole increase means narrow poles, then less magnetic flux, which means reduced stator yoke, representing less machine volume. The stator and rotor back iron length changes concerning the number of poles as follows.

$$h_y = \frac{B_g}{B_y} \cdot \frac{\pi \cdot D_{ag}}{2 \cdot k_i \cdot p} \tag{4}$$

where h_y is the yoke back iron length, B_g is the average flux density of the airgap, B_y is the design flux density of the yoke, i.e., rotor or stator, k_i is the iron stacking factor, p is the number of poles, and D_{ag} is the diameter of the airgap.

However, with a fixed mechanical rotational speed, the same linear current density, airgap flux density, and identical inner machine volumes, the higher efficiencies and lower costs are found to be around an electrical frequency of 50-60 Hz. Furthermore, for an under-excited operation of $\cos\phi = 0.98$, a cost reduction of 18 % is achieved due to less field winding power losses; hence, the minimum of total losses trade-off is reached, and the efficiency is enhanced reducing the operation costs (Holzer et al., 2020).

The HydroWIRES initiative from the U.S. Department of Energy has identified some potential technologies for PSH, such as submersible pump-turbines and motor-generators, geomechanical PSH, open-pit mine PSH, and hybrid PSH technologies (Koritarov et al., 2022).

For a scenario with power electronics (PE)-interfaced hydrogenerators (PEIH), where synthetic and physical inertia are vital, the number of phases is constrained to the number of parallel and series converter structures after a techno-economical size-dependent optimization. Therefore, the PEIH is decoupled from the grid, meaning the power system can be DC or the standardized 3-phase AC power system. Nonetheless, the major drawback of this scenario is the assumption of good enough synthetic inertia and coordination of PEIH to prevent high-frequency resonance. Thus, the technical challenges for PEIH focus on grid resiliency. On the contrary, the economic challenges are naturally tackled with new market mechanisms and services, such as fast frequency and storage reserves for PSHs.

Full-size PEIH opens a window of opportunities for VSH in PSH. The significant advantage of this technology is the broader range of degrees of freedom for optimization from an electromechanical point of view. Nonetheless, the main preeminent inconvenience is the size of the power electronic system, which is bilateral, i.e., technical because of the power rating and series/parallel structure, and economical because of the costs for large tailored systems. From an economic standpoint, it is incited to expand the electricity markets nowadays through in-depth research of the value of water for coordination and optimization against wind and solar variability for future power grid scenarios. If the revenue and incentives for VSH are insufficient to cope with new PODs, fixed-speed PSH may become a standard.

Moreover, the converter can perform harmonic injection control to the generator to optimize the operation. Techno-economics is related extensively to power electronics since the control system algorithms are well understood.

For the case of damper windings, it has been found that even PEIH kept them in the POD. However, the electromagnetic dynamics between the grid and hydrogenerator are decoupled. Therefore, further research must be done on the techno-economical constraints of whether quitting damper windings is the optimal solution.

5.1.7. Generator's design characterization

Since the synchronous generators are used for many applications besides hydropower technologies, they have broader options for new design features. However, the design must be constrained to techno-economical, power producer, and grid owner requirements.

For instance, in the case of thermal design, more compact generators reduce the machine's thermal inertia and the thermal time constant. Hence, the time window for enhanced reactive power is tethered. Moreover, in the slot, the stator's bars transfer the heat through the stator's core and the airgap. Since there is a *thermal distance* of heat transfer from the copper passing through the insulation and reaching the airgap or external air outside the stator's core, the temperature gradient seen by the insulation is considerable. Hence, this temperature gradient and the different thermal expansion coefficients make a relative motion and friction between copper and insulation and between insulation and stator core. This friction wear on the insulation (Gu and Lin, 2003). Therefore, it is of utmost importance to keep minor temperature variations to reduce the tear and wear of the insulation.

In that sense, an approach for prospective POD may be the monitoring and control of thermal inertia, time constants, and temperature gradients. Similarly, to provide ultra fast response and motor-generator operation PE-interfaced synchronous machines can be further studied. Moreover, research on which operating points within the capability diagram can be optimal for PE-interfaces PSH.

Finally, a hydrogenerator may have various novel PODs for optimization and general improvement. However, creating a new POD is driven by how the market will set the rewards and remuneration for the different services that hydropower plants can provide to improve the grid's reliability and resilience. In Table 22, a chronological overview summarizes past, present, and future PODs. The latter consists of different stages of technology readiness; however, further development will depend on market mechanisms, energy policies, and incentives, as pointed out in the discussion section.

5.2. Hydropower operation: Power producer perspective

This section addresses the impairs on the hydrogenerator experience by the dramatically wide range of operating points imposed by a high share of VREs. Moreover, the nature of the electricity market structure may reduce the active power of hydropower generation; however, it might require synchronous condenser mode operation for extreme conditions. Therefore, an extended capability of operation points may be needed.

This broader range of operating points introduces new extended thermal and electromagnetic dynamics. Hence, new refurbishment opportunities, Digital Twins, and a more comprehensive hydrogenerator operation regime are assessed to provide a hydrogenerator characterization toward 2050.

It is essential to notice that other components from the hydropower plant, e.g., transformers and power electronics, are not assessed. Nonetheless, dynamic thermal models for power transformers can be found in Paulhiac and Desquiens (2022), Chen et al. (2022), Susa and Lehtonen (2006), and for power electronics in Probst et al. (2013), Vydrova et al. (2021), Tang and Thiringer (2023).

5.2.1. Extended thermal dynamics

For hydrogenerators, especially for a wider reactive power capabilities during short- and long-term conditions, the losses must be accurately determined to establish the machine's efficiency and the proper requirement for the cooling system capacity to prevent temperature rise beyond insulation threshold. The losses depend on the machine *load* and *power factor*. Tables 23 and 24 show the core segregated losses and the loss components, respectively.

Understanding the rotor–stator airgap region's flow dynamics and heat transfer coefficient (h) is vital for mitigating losses and improving overall machine thermal performance (Zeaiter, 2020). However, it is very complex to compute and to get good accuracy (Dang et al., 2018). Using Reynold-averaged Navier–Stokes (RANS) equations, mixing plane interface model for rotor–stator coupling, and the Shear Stress Transport $k-\omega$ model, it is found that the rotor–stator interface location has a significant impact on the air ventilation and thermal performance prediction in the airgap. In addition, the aerodynamics, e.g., counter-vortex and up-and down-stream regions, significantly impact the pole face's convective heat transfer coefficient profiles. Moreover, the choice of turbulence models plays a crucial role in temperature prediction on the pole face (Dang et al., 2020). Nonetheless, Conjugate Heat Transfer Computational Fluid Dynamics (CFD) can be used to optimize the ventilation for synchronous machines (Bersch et al., 2021).

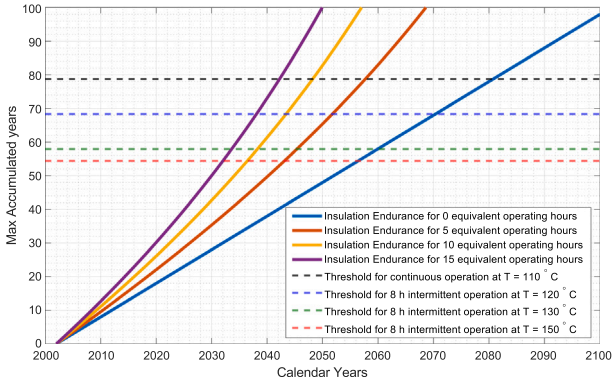


Fig. 16. Insulation Class F Endurance for 300 starts/stops with 2.8079 % of annual growth. The temperature threshold is computed based on (Znidarich, 2008a). Here, different equivalent operating hours for each start/stop are presented. 2002 is used as the initial year. The eight-hour intermittency means that the insulation reaches the maximum temperature for eight hours per day and then is returned to the continuous operation at 110 °C.

Therefore, a deeper understanding of hydrogenerator thermal dynamics ensures the grid's operational security, the machine's lifetime, and revenue outcomes. With the new hydrogenerator's PODs opportunities, a 3D FEM model enables identifying and optimizing the thermo-electromagnetic and fluid behavior during extreme conditions. For instance, a sensitivity analysis over different rotor structures depicts the hotpots and temperature changes according to the geometry and fluid dynamics by CFD (Jichao et al., 2021a). Moreover, the heat-transfer coefficient distribution of the rotor excitation winding is uneven along the axial direction, and the highest temperature of the rotor excitation winding appears on the leeward side (Jichao et al., 2021b). In other words, proper geometrics optimization and thermal modeling may improve thermal management, e.g., reduce the thermal gradient, for hydrogenerators.

Furthermore, surrogate modeling via reduced order thermal modeling may offer near-real-time with sufficient accuracy for extended transient thermal dynamics for different components and locations in electrical machines (Boscaglia et al., 2021).

Additionally, studies done by Carounagarane et al. in Carunaiselvane et al. (2018), Carounagarane et al. (2020) show that hydropower facilities can operate at 120 % rated conditions when water's excess. Nonetheless, it comes at the cost of reducing ten years of a lifetime due to thermal aging. Thus, further analysis must consider outcome revenue, operation and maintenance, and future refurbishments. Table 25 shows various features to unveil extended thermal dynamics.

In the case of Multiphase Distributed Winding Machines under Normal and Open-Phase Fault conditions, Boglietti et al. in Boglietti et al. (2020), state that for a particular quadruple three-phase induction machine, the current may increase by 11.5 %, 40 %, and 95 % when one set, two sets, and three sets are turned OFF. Here, a mutual heat exchange phenomenon is observed among different windings sets. Therefore, the State-of-the-Art (SoTA) of Lumped-Parameter Thermal Networks

Table 22
Philosophy of Design (POD) - Chronological overview.

Feature	Past (1880–1950)	Current (1950–2030)	Prospective: Concept-Based[a]	Citations
Stator, Armature Winding Layout, And Conductors	- Integer slot windings ($q \in \mathbb{Z}$) - War of 1-, 2- & 3-phase machines - High number of slots to fit the armature coils: Deeper and narrower slots - Higher leakage reactance and copper losses - Frequency: $8\frac{1}{3}$, $16\frac{2}{3}$, 25, 30, $33\frac{1}{3}$, 38.6, 40, $41\frac{2}{3}$, 42, $45\frac{1}{3}$, 46, $47\frac{1}{2}$, 50, 53, 60, 62.5, $65\frac{1}{3}$, 66, 67, $83\frac{1}{3}$, 100, 125, $133\frac{1}{3}$, and 140 Hz - Class A & B Insulation: 10 mm thickness; low fill factor 0.3 (1990); loosen coils that produce, corona and slot discharge. - Early-stage design: solid conductors - Transitioning design: Laminated, stranded, or interweave conductors - Standardized design: Roebel Bar (1920s)	- Fractional slots windings ($q = \frac{z}{b} > 1$) - 3-phase machine (as standard) - Reduced number of slots - Frequency: 50 or 60 Hz - Roebel bar transposition optimization - Pre-formed single-turn bars - Under-Roebling optimization - Insulation: 2.5 mm thickness; better fill factor 0.41 (1990); Class F (155 °C). - 40 kV/mm voltage strength	- Multiphase windings (multiple of 3) - Fractional Slot Concentrated Winding (FSCW) ($q = \frac{z}{b} < 1$) - Phases calculated according to parallel and series converter structure optimization - Harmonic Injection - Reduction of End-Windings (FSCW) - Possibility of Coil Windings instead of Roebel Bar. (Increased fill factor) - Reduction of Thickness <1 mm - High fill factor > 0.6 - Class H (180 °C) Insulation - Insulation with improved nanocomposites	Owen (1997), Furfari (2000), Blalock and Woodworth (2008), Neidhöfer (2011) Steinmetz (1894), Field (1905), Harder (1999), Roebel (1912), Lyon (1921, 1922), Summers (1927), Lyons (1930) Znidarich (2009), Blecken (1997), Znidarich (2013), Evans (1981), Forster and Klataske (1981), Ridley and GEC Large Machines Limited UK (1988), Lyles and Smith (1988) Traxler-Samek and Lecker (2020), Znidarich (2008a) Fard (2007) Bezdek (2020), Emery and Williams (2011) Schmidt et al. (2011), Fraile-Ardanuy et al. (2006), Nobile et al. (2018), Gao et al. (2021), Holzer et al. (2020) Hemmati et al. (2019), Nguyen et al. (2021) Tessarolo (2010) Huang et al. (2008), Al-Adsani and Beik (2022) Farshadnia (2018) Duran et al. (2017), Kindl et al. (2020), Abdel-Khalik et al. (2011), Spas et al. (2020), Yang et al. (2021), Tessarolo (2018), Tessarolo et al. (2019, 2021), Kong et al. (2016, 2017), Abdel-Khalik et al. (2019a,b), Yaramasu and Wu (2017), Trentin et al. (2021), Anon (2021a), Slunjski et al. (2020), Levi (2008) Kuang et al. (2019), Son and Lim (2021) Pei et al. (2017)
Rotor and Excitation System	- Forged steel and sectionalized rotors - Electromechanical excitation controls - Amplidyne regulators - Power factor regulators - Rotating DC Exciter machinery - Common exciters connected in parallel - Individual exciters for large hydropower plants - No consensus for high or low resistance damper windings (amortisseurs)	- Shaft-and-spider rotor configuration - Integrally forged shaft (to comply with standards) - Fully insulated laminations for rotor poles - Excitation controls digitalization - 25 different excitation models standardized (IEEE Std 421.5-2016: DC, AC, and ST types) - Static and Individual exciters most common - Damper resistance: computational optimization	- Hybrid excited rotors (PM+Windings) - Multiphase Excitation System - Brushless Exciters - Class H (180°C) insulation	Summerhayes (1920), Head and Rao (1967), McClure et al. (1946), Li (2019), Michell (1912), Barns and Bowness (1922), Evans and Bergvall (1924), Wagner (1931), Park and Bancker (1929) Blecken (1997) Znidarich (2008c), Electric Machinery Committee of IEEE (2010), Kirkpatrick (1992), Kinnunen et al. (2006), Kamiev et al. (2009), Li et al. (2018a), Nuzzo et al. (2017), Islam et al. (2021) Digital Excitation Applications Task Force of the Excitation Systems Subcommittee (1997), Anon (2016), Nøland (2017) Noland et al. (2019), Rebollo et al. (2017), Voith (2013), Noland et al. (2020a), Evestedt et al. (2021), Yao et al. (2021), Anon (2019) Yao et al. (2021), Anon (2019)
Iron Core	- Losses at 1.5 T and 50 Hz between 5 W/kg to 6 W/kg - Lamination ∼ 0.43 mm (17-mil) - Segmented cores - Core layers between 50–70mm.	- Losses at 1.5 T and 50 Hz between 2.05 W/kg to 3.0 W/kg - Lamination down to ∼ 0.35 mm. However, too small laminations may increase the effective machine's length if the insulation between laminations is not further decreased. - Continuously stacked core - Core layers between 30–40mm.	Better iron core materials: Permendur, 10JNEX900, Metaglas, PB Permalloy, Grain-oriented config. - Losses at 1.0 T and 50 Hz between 0.056 W/kg to /SI0.5W/kg - Stator grain-oriented teeth - Rotor grain-oriented poles - Thinner insulation and laminations	Waters (1903), Lamme (1916) Sugimoto (2019) Blecken (1997), Znidarich (2013) Znidarich (2008b,c) Namikawa et al. (1998), Takashi and Hironori (2012), Hasegawa et al. (2010, 2011), Nakajima (2019)
Cooling System & Ventilation	- Poor ventilation - Badly designed vents - 10 mm-wide ventilation ducts - Exhaust fans - Curved fan blades	- Good ventilation - Well-designed vents - 6 mm to 8 mm ventilation ducts - Air-forced cooling (up to 700 MW) - Optimized fan blades and auxiliary fans - Load-dependent cooling (water control)	- Multi-objective optimization design - New opportunities for Thermoelectric (TEC) cooling - Ferrofluids applied on vents. - Adaptive and fault-dependent cooling - Nanofluids (Counterflow Heat-Exchanger) - Evaporative Cooling Systems	Waters (1903) Floyd and Dunbar (1931), McCarty and Hart (1922) Blecken (1997), Znidarich (2013) Znidarich (2008b), Gu and Lin (2003) Ruan et al. (2007), Toren and Mollahasanoglu (2021), Lucas et al. (2022a) Vaschetto et al. (2019)
PE-Interfaced*	–	–	- Variable Speed Machines - Electromechanical POD optimization - No need for damper windings - Multiphase armature windings (any number) - FSCW	Koritarov et al. (2022) Schmidt et al. (2011), Fraile-Ardanuy et al. (2006), Nobile et al. (2018), Gao et al. (2021), Holzer et al. (2020), Hestengen (2016) Huang et al. (2008), Duran et al. (2017) Kindl et al. (2020), Spas et al. (2020) Yang et al. (2021), Kong et al. (2016) Kong et al. (2017)

[a] Early stage concepts based upon the research trends, IPCC goals, IEA requirements and SysOpt project goals.

Table 23
Segregated core losses (Znidarich, 2008c).

Core losses	Description
Open-Circuit or Load-dependent	- Stator back iron, teeth, end-region, and interlaminar losses. - Rotor pole surface losses
Short-circuit or Stray load (AC) current-dependent	- Eddy currents induced in core, frame, and windings. - Short-circuit pole face loss and third harmonic stator tooth loss.

(LPTN) for High-Phase Order (HPO) machinery may have reached a technology readiness for implementation in commercial machines.

Regarding hydropower's operation, the high share of VREs imposes an increasing number of start-stops and load changes that may reduce the equipment's life. A single start-stop is estimated to add an *equivalent operating time* between 15 h (Savin et al., 2021) and 5 h (EPRI, 2017). Fig. 16 depicts a hydrogenerator with 300 starts/stops (Signore, 2017) and an annual growth of 2.8079 % (Neupane et al., 2021). Thus, for regular operation with increasing starts/stops, the insulation is expected to last between 40 and 56 years. In contrast, for intermittent operation of 8 h per day at $T = 120$ °C and then back to normal operation, the insulation may last between 36 and 50 years. In the case of $T = 130$ °C, the insulation may last between 32 and 44 years. Lastly, for $T = 150$ °C, the insulation may last between 30 and 42 years.

In a steady state, the machine's final operating temperature depends on the performance of its cooling system. However, the I^2R losses increase as the cube of dimensions, whereas the surface area available for heat dissipation increases as the square. Hence, the bigger the

Table 24
Loss components (Traxler-Samek et al., 2010b; cyiza Karekezi et al., 2022).

Iron losses	Winding losses	Mechanical losses
P_c: Iron losses in the stator core (teeth and yoke) including the effect of harmonics and rotating fields	P_a: Stator (armature) winding copper I^2R losses	P_{br}: Friction losses of rotating parts (bearings)
P_{ps}: Eddy current losses on the pole shoe surface due to tooth-ripple pulsation	P_r: Rotor winding copper I^2R losses	P_{air}: Air friction losses of the forced cooling airflow
P_{cp}: Eddy current losses in the stator clamping plates	P_{br}: Rotor brush losses	P_p: Pressure generation losses (e.g., fan)
P_{cf}: Eddy current losses in the stator clamping fingers	P_{ex}: Excitation system losses	
P_{lam}: Eddy current losses in the stator core end laminations	P_l: Eddy current losses in the stator winding	
P_{ag}: Eddy current losses in external metallic air guides	P_{end}: Circulating current losses in stator winding	
	P_d: Damper winding losses	

Table 25
Extended thermal dynamics to ensure grid's reliability and resiliency.

Feature	Description	Challenges
Overexcitation Limiter (OEL) (Lomei et al., 2017)	Increase the operating time - Thermal-Energy-based calculation	- Real-time control and monitoring - No adaptive cooling
Lumped-Parameter Thermal Network (Armando et al., 2021; Boglietti et al., 2016, 2018, 2019)	Better prediction after 120 s during thermal transients with calibrated 2nd-order LPTN for 5 h heat-run test.	- Parameter estimation - Hotspots prediction - First-Order LPTN
Reactive Power Boosting (Øyvang et al., 2019, 2020)	Available thermal capabilities for boosting in a 15 min time window. - Field current may rise more than three times. - Armature current may rise more than two times. -Rotor winding is the bottleneck of available thermal capability.	- Proper incentives require - Test in real hydrogenerators - New Thermal Phenomena - Thermal cycling and expansion

machine, the more difficult it is to cool down. In the end, the insulation deterioration depends on time and temperature. For Class F insulation, the thermal aging threshold is between 120 °C-130 °C depending on the epoxy formulation and curing temperature, in which the typical thermal endurance is 100 000 hours (Znidarich, 2008a).

However, some research suggests no severe insulation deterioration due to start-stop thermal cycling; the aging varies with the length of the core, operation pattern, and type of cooling. Hence, it is not easy to estimate the aging accurately due to start-stops thermal cycling, and different measurement techniques offer various interpretations (Brügger, 2010; Istad et al., 2011; Savin et al., 2020). Furthermore, load cycles with high thermal gradients increase the insulation failures (Morin and Bartnikas, 2012). Therefore, thermal gradient minimization control systems are needed to enhance hydrogenerators' reactive power capabilities during boost mode (Noland et al., 2020b).

Therefore, as previously discussed, Peltier Modules (PEM) can be used easily, as is shown in Ponikvar (2022), to maintain a reference temperature, reduce the temperature gradient during thermal transient operations (Stoop et al., 2022) or be used as a heat pump for open cryogenic systems (Ivanov et al., 2022). Furthermore, PEMs have been assessed for battery cooling systems for Electric Aircraft, thermoelectric Generators for hypersonic vehicles (Gou et al., 2022), cooling and heating in the rear air-handling unit (AHU) for Sport Utility Vehicle (SUV) (Lee and Hwang, 2022). For these reasons, the research into PEMs is of utmost interest to control temperature and achieve higher efficiencies. Thus, it opens the research for the benefits of the hydrogenerator's thermal capabilities to achieve higher resiliencies in extreme operation conditions.

Moreover, it is vital to have fast and accurate models to compute the temperature of the hydrogenerator online. For example, Aleikish et al. in Aleikish et al. (2021) presented a hybrid model of a heat exchanger and a thermal model of an air-cooled synchronous generator with computing times of 1.5 µs and 2.5 ms, respectively. Moreover, since all the data is not always available, some guests based on experience can be done. Thus, a parameter estimation can be made to the thermal models, as in Pandey et al. (2020), Phuc et al. (2021).

Online monitoring and control demand accurate sensors, especially in the rotor. Thus, a wireless sensor seems the most acceptable option since infrared cannot accurately measure the rotor temperature due to the weak infrared radiation (Zhao et al., 2021), whereas a smaller module measures only temperature at a low sampling rate (Llano et al., 2017).

Ultimately, the hydrogenerator's thermal management is of utmost importance to cope with a broader range of operating scenarios where extended reactive power capabilities may be needed.

5.2.2. Extended electromagnetic dynamics

Extended electromagnetic dynamics considerations are essential during super-saturated operation regimes (SSOR). For instance, when the Short-Circuit Ratio (SCR) is enhanced, the Short-Circuit Torque (SCT) is also enlarged, which means that at SSOR, the synchronous machine can improve the SCR of the grid. Furthermore, the saturated reactance and SCR can be computed with (5), according to Kundur et al. (1994), cyiza Karekezi et al. (2022).

$$\begin{aligned} I_f^{occ} &= k \cdot (E_g + C_m \cdot {E_g}^m) & I_f^{air} &= E_g / b_v \\ K_d &= I_f^{air} / I_f^{occ} & X_d^{sat} &= K_d \cdot X_d^{uns} \\ SCR &= 1 / X_d^{sat} \end{aligned} \tag{5}$$

where I_f^{occ}, I_f^{air}, E_g, b_v, K_d, X_d^{uns}, X_d^{sat}, and SCR are Open-Circuit Characteristic field current, airgap line field current, induced voltage, airgap line's slope, saturation factor, unsaturated reactance, saturated reactance, and short-circuit ratio, respectively. Therefore, using the formulation in cyiza Karekezi et al. (2022), different saturation factors are derived according to the operation points. A case study for the hydrogenerator in Øyvang et al. (2019) is presented in Fig. 17, in which two operation points are depicted: the nominal operation point where $K_d = 0.6120$, and a handpicked operation point at 4 times higher

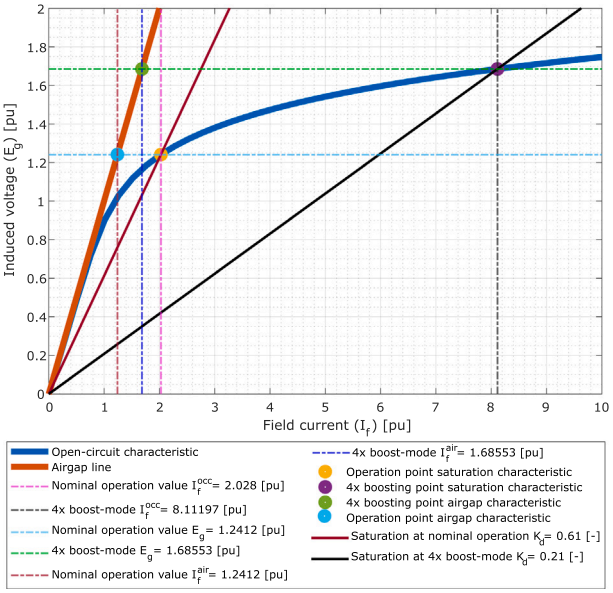


Fig. 17. No-load characteristic: airgap line and Open-circuit characteristics (cyiza Karekezi et al., 2022). The base field current and line voltage are 525.15 A and 11 kV.

reactive power, in which $K_d^{4\times} = 0.2078$. Thus, it is easy to see how the saturation factor decreases. From the values obtained in Fig. 17, and with X_d^{uns} =1.087 pu and leakage armature reactance X_l =0.08 pu, the saturated reactances and SCRs are calculated in (6) and (7). The results show an increase of 194 % SCR for a field current four times the nominal.

$$\begin{aligned} X_d^{sat} &= K_d \cdot (X_d^{uns} - X_l) + X_l \\ &= 0.6963 \text{ pu} \end{aligned} \tag{6}$$

$$\begin{aligned} X_d^{4\times sat} &= K_d^{4\times} \cdot (X_d^{uns} - X_l) + X_l = 0.2367 \text{ pu} \\ SCR &= \frac{1}{X_d^{sat}} = \frac{1}{0.6963} = 1.4361 \text{ pu} \\ SCR^{4\times} &= \frac{1}{X_d^{4\times sat}} = \frac{1}{0.2367} = 4.2239 \text{ pu} \end{aligned} \tag{7}$$

The short-circuit torque in synchronous machines is strongly intertwined with the stored magnetic energy (Penney, 1929). Neglecting losses, the stored energy in a ferromagnetic core inductor with current I_o equals the work required to establish the current through the inductor.

$$W = \frac{1}{2} \int \mathbf{H} \cdot \mathbf{B} \, dV = \int_0^{I_o} L_d(I) I dI \tag{8}$$

The dependency of the magnetic energy on the inductance and current is shown in (8). Nonetheless, the inductance in salient-pole synchronous machines changes over the rotor angle and saturation of the ferromagnetic core. Moreover, physical insights can be mentioned from the classical approach of the Poynting theorem (Calamaro et al., 2015) shown in (9).

$$-\oiint_{\partial V} \mathbf{E} \times \mathbf{H} \cdot d\mathbf{A} = \int_V \mathbf{J} \cdot \mathbf{E} \, dV + \frac{\partial}{\partial t} \int_V \left(\frac{\mathbf{H} \cdot \mathbf{B}}{2} + \frac{\mathbf{E} \cdot \mathbf{D}}{2} \right) dV \tag{9}$$

The left-hand side of (9) is the total power transferred out of the volume V, and the right side is divided into the first term known as *Active Power* or *Joule losses*, as Castilla pointed out in Castilla et al. (2009); and the second term representing the energy stored in the magnetic and electric fields, known as *Reactive Power* or non-active (Montoya et al., 2022). The Poynting vector is further worked with geometric algebra by Montoya et al. (2022) to obtain the geometric power as the sum of active power (P) and non-active power ($\mathbf{M_n}$), as shown in Eq. (10), where k and l are the harmonics, assuming both signals have the same harmonic content.

$$\mathbf{M} = \mathbf{ui} = \sum_{k=1}^{2n} u_k i_k + \sum_{\substack{k,l \\ k<l}}^{2n} (u_k i_l - u_l i_k) \sigma_{kl} = \mathrm{P} + \mathbf{M_n} \tag{10}$$

An electromagnetic wave contains energy stored in the electric and magnetic fields (van der Sluis, 2001). Since a synchronous generator is a sophisticated rotating multi-phase inductance, the electric field energy is negligible, i.e., capacitive energy, and the magnetic field energy is related to the machine's reactive (or non-active) power. Hence, when the magnetic field energy is increased during reactive power boosting, the total power of the machine is increased. Therefore, for an Ideal Machine (Penney, 1929) with negligible resistance, constant inductances, and mutual inductance between the field and armature circuits varies as a cosine, the change in stored magnetic energy can be written as (11), for which a single-phase short-circuit torque is ∼12 pu, as a worst-case scenario. In that sense, transforming magnetic energy to kinetic energy in a boost-mode operation may increase the short-circuit torque.

$$\tau = \frac{dW}{d\alpha} \tag{11}$$

The hydrogenerator study case's maximum short circuit torque is 22 885 kNm, more than 11 times larger than the rated torque. Here, an approximation of the sub-transient electromagnetic torque is computed by multiplying the sub-transient reactance by the same saturation factor as the steady-state reactance, as is shown in Eq. (12).

$$X_d'' = X_l + \frac{K_d}{\frac{1}{X_{ad}} + \frac{1}{X_D} + \frac{1}{X_f}} = X_l + K_d \cdot X_{eq}'' \tag{12}$$

where X_d'', X_l, X_D, X_f, K_d, and X_{eq}'' are the subtransient reactance, leakage reactance, dampers reactance, field reactance, saturation factor, and equivalent unsaturated subtransient reactance in d-axis. For the worked example, the unsaturated subtransient reactance is X_d'' =0.15 pu. Then, the equivalent unsaturated reactance is X_{eq}'' =0.07 pu.

The sub-transient electromagnetic torque is defined by Machowski in Machowski et al. (2020) as follows:

$$\begin{aligned} \tau_\omega(t) &= \frac{1}{X_d''} \cdot g_3(t) \cdot e^{-t/T_a} \sin \omega t \\ g_3(t) &= X_d'' \left[\left(\frac{1}{X_d''} - \frac{1}{X_d'} \right) e^{-t/T_d''} + \left(\frac{1}{X_d'} - \frac{1}{X_d} \right) e^{-t/T_d'} + \frac{1}{X_d} \right] \end{aligned} \tag{13}$$

From Eq. (13), the short-circuit torque peak per-unit value can be obtained for four times boost mode operation.

$$\tau_{sc}^{pk} = \frac{1}{X_d''^{4\times}} = \frac{1}{X_l + K_d^{4\times} \cdot X_{eq}''} = 10.57 \text{pu} \tag{14}$$

The short-circuit torque for 4x boost mode is ∼10.6 times higher than the rated torque value, below the maximum short circuit torque given by the manufacturer, i.e., τ_{sc}^{max} =11.63 pu. The maximum saturation factor for the working example is equal to $K_d^{max} \approx 0.086$, with a $SCR \approx 6$ pu. Here, the field current must be $I_f = 23.2711$ pu, i.e., I_f^{op} = 11.4749 pu of the operating field current, as is shown in Fig. 18. A remark is that the q-axis saturation and cross-saturation effects are not considered.

Ten boosting reactive power iterations are computed for a deeper dive using cyiza Karekezi et al. (2022) field current model. The short-circuit torque is calculated using (13) and (15), and with X_d' =0.24 pu. The saturation factor and operating field and armature currents for the last iteration before the short-circuit are K_d =0.1666, I_f^{op} =5.2197 pu and I_a =4.4508 pu, respectively. Fig. 19 shows the short-circuit torque results in which the highest is τ_{sc} =10.63 pu.

$$\begin{aligned} \tau_R(t) &= \left[\frac{1}{X_d''} \cdot g_3(t) \right]^2 \cdot R_s \\ \tau_{total} &= \tau_\omega(t) + \tau_R(t) \end{aligned} \tag{15}$$

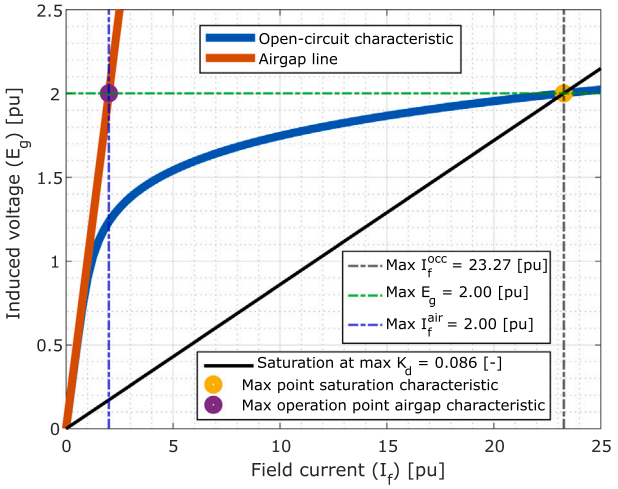


Fig. 18. Maximum saturation $K_d \approx 0.086$. No-load characteristic: airgap line and Open-circuit characteristics (cyiza Karekezi et al., 2022). The base field current and line voltage are 525.15 A and 11 kV.

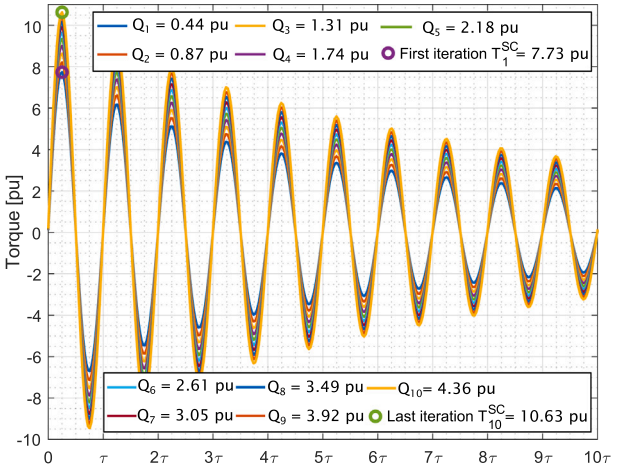


Fig. 19. Short circuit torque for ten different reactive power boosting. The graphs depicts the torque per unit with respect to the electric period.

Moreover, from a simplistic mechanical perspective for the rotor cylinder, the yield strength for steel is around σ_{yield} =300 MPa. Moreover, the shear stress of the rotor cylinder is as follows

$$\sigma = \frac{r\tau}{J} = \frac{r}{\frac{\pi}{2}(r^4 - r_i^4)} \cdot \tau \tag{16}$$

Where the term multiplying the torque is r/J =0.2938 m^{-2}, thus, the necessary torque to surpass the yield strength of steel is τ =1.0211 GNm, i.e., more than 500 times the rated torque. Hence, operating in the super-saturated region does not compromise the entire mechanical limits of the machine. Nonetheless, further research most be done to consider all the mechanical components.

This approximation sets up a scientific gap regarding the generator's super-saturated region when operated at boosting mode. Furthermore, it shows how the saturation level can affect the short-circuit torque, i.e., the mechanical dynamics. Therefore, detailed saturation models with cross-saturated must be used to achieve superior accuracy, such as the one proposed by Levi & Levi in Levi and Levi (2000). Nonetheless, the study was performed at nominal operation with a step of 5% increase of the stator terminal voltage for transient analysis. Thus, a super-saturated operation regime is not considered prior to the transient. Finally, for multiphase machines, considering interplane cross-saturation due to dq_1-axis (fundamental) and dq_n-axis (harmonics) must be done (Jecmenica et al., 2019).

Furthermore, a 2D FEM model is used to validate a short-circuit scenario during boosting mode. The field current is calculated according to cyiza Karekezi et al. (2022), assuming an active power 10.6 times the nominal value. Hence, the short-circuit field and armature currents are computed equal to $I_{f_{sc}}$ =12.4406 pu, and $I_{a_{sc}}$ =9.8133 pu, respectively. Moreover, the angle δ is adjusted in the FEM model equal to δ =0.9275 rad to obtain a handpicked reactive power equal to Q_{boost} =2.3 pu. In Fig. 20(a), the magnetic flux density is depicted up to 4.57 T in the poles, and ~2 T in the iron yoke. Moreover, in Fig. 20(b) the active power (torque) is P_{sc} =981.8887 MW or τ_{sc} =9.5329 pu, Finally, the saturation factor is approximated from (17) equal to $K_{d_{sc}}$ =0.0792, where the internal voltage is computed as open-circuit voltage E_a =1.9354 pu and the terminal voltage is V_a =1.4076 pu. Here, the saturation of the iron is higher than the computed analytically. Hence, it is more difficult to get a higher short-circuit torque due to the saturation of the iron core.

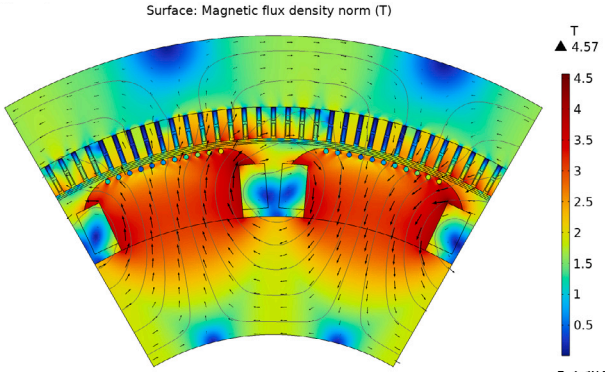
$$X_{d_{SC}} \approx \frac{\sqrt{E_a^2 + V_a^2 - 2 \cdot E_a^2 \cdot V_a^2 \cdot \cos\delta}}{I_a}$$
$$K_d \approx \frac{X_{d_{SC}} - X_l}{X_d - X_l} \tag{17}$$

Additionally, other approaches for reactive power optimization could be done. For example, optimizing the electromagnetic sub-transient parameters to support voltage and short-circuit capacity and enhancing the system's operation can cope with VRE variability by achieving a broader operation for synchronous generators with real inertia (Ma et al., 2021). Here, the minimization of X_d'' and T_d' enlarges the dynamic reactive power during the sub-transient process and achieves rapid response to the regulation of its excitation system. Then, the maximization of T_d'' extends the sub-transient process for larger dynamic reactive power output. Nonetheless, the reactance X_d is maintained to have the same steady-state performance, such as the short-circuit ratio (SCR).

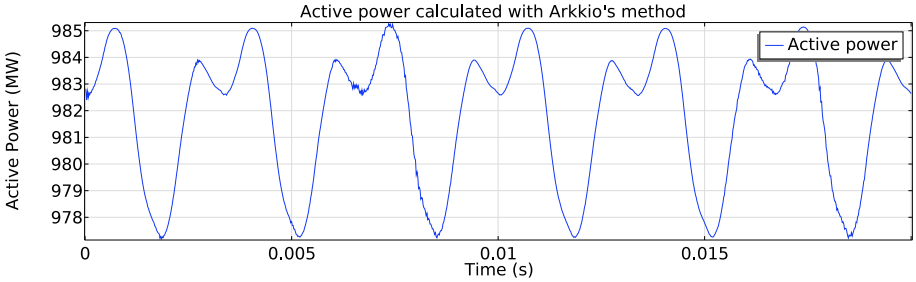
Large synchronous condensers (SCs) role during a severe voltage plunge failure is to supply sufficient excitation for a large amount of reactive power in a short time and provide emergency reactive voltage support for the systems, which helps to recover power and system voltage quickly and prevent voltage collapse. Since a smaller X_d'' is required, the optimization is performed by optimizing the stator slot shape, rotor slot shape, and structure. Furthermore, the T_d' is improved by optimizing airgap length, rotor slot type, the number of rotor turns, and stator diameter (Chen et al., 2019). The optimization outcomes are summarized in Table 26. Moreover, the large SCs use conductive rotor slot wedges, which are welded with silver-plated end rings to form a damper cage with conductivity to extend the sub-transient process (Ma et al., 2019). Furthermore, research into distributed SCs and superconducting SCs has been reported in Zhi-peng et al. (2022), Duan et al. (2022), respectively. Both emphasize the reduction of the steady-state, transient, and sub-transient reactances for improved reactive power dynamics.

5.3. Hydropower operation: Grid owner perspective

The core bulk power systems are suffering a revolution, where grid owners comply to collaborate in the Global Power System Transformation (G-PST) Consortium effort, which focuses on 59 research questions organized into six Research Programmes: Inverter Design, Stability Tools & Methods, Control Room of the Future, Planning, Restoration & Black Start, and Services (O'Malley et al., 2021). The G-PST consortium survey states that the standards' highest priority is energy storage, renewable energy solar, distributed energy resources, and microgrids.



(a) Magnetic flux density 2D Electromagnetic FEM Calculations.



(b) Active power behaviour in time.

Fig. 20. Simulation at Short-circuit condition in Finite Element Method.

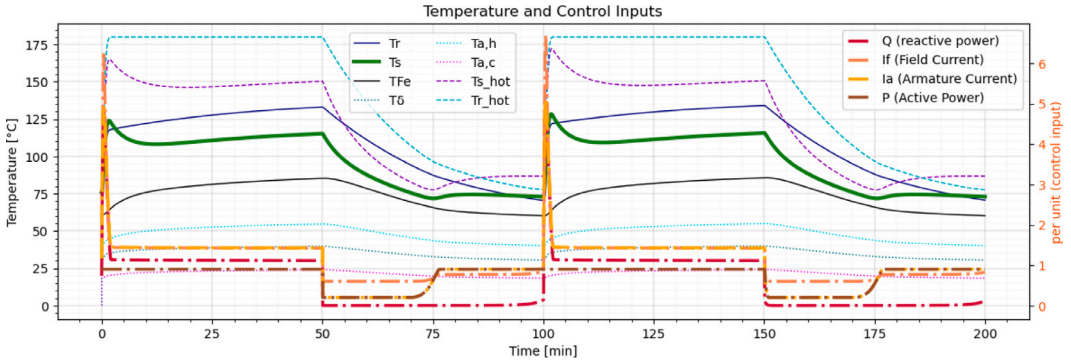


Fig. 21. Hydrogenerator's Hotspot Management using Dual-MPC with Insulation Class H (180 °C) for Emergency (< 5 min), Boosting (45<t< 50 min), and Recovery (50<t< 100 min) modes, i.e., 50 min thermal cycling. Here, it is depicted as an active and reactive power (P,Q) control while ensuring safe temperature limits for the different modes considering P_{min} = 0.2 pu and Q_{min} = 0 pu. The model (under development) is an enhanced Lumped-Parameter Thermal Network from Øyvang et al. (2019).

Table 26
Large SC reactive power optimization.

Parameter	Optimization	Outcome
X_d	Fixed or minimize	Keep the same SCR or enhanced it
X'_d	Minimize	Enlarge transient dynamics.[a]
X''_d	Minimize	Enlarge sub-transient dynamics
T'_d	Minimize	Achieve rapid response of excitation system
T''_d	Maximize	Extend sub-transient dynamics.

[a] The transient reactance may be minimized. However, the transient dynamics are reduced since the T'_d is minimized to achieve rapid response of the AVR.

In contrast, Hydropower is the seventh place out of 9 (Association et al., 2021).

Even though the G-PST consortium is searching for general guidelines for high penetration of Inverter-Based Resources (IBRs), hydropower technologies play a vital role in the energy mix in some countries, mainly if new services can be provided. Jacobson et al. in 2011 (Jacobson and Delucchi, 2011; Delucchi and Jacobson, 2011) introduce the concept of Wind, Water, and Solar (WWS) to create a fully renewable energy mix worldwide by 2050, as depicted in Table 28.

A study (Jacobson et al., 2018) made with 100 % WWS in 139 countries among 20 world regions, where different mixes of generation, storage, transmission, and demand response might reach different local optimal. Three scenarios were discussed, but only one included additional hydropower turbines or refurbishments, which increase 0.28 % of total system cost over all the grid regions. The extra cost is because new hydropower is not PSH; hence, it cannot store excess electricity.

Table 27
Hydropower operation: Power producer perspective.

Areas	Challenges	Opportunities	Citations
Transient thermal dynamics	- Thermal measurements - Thermal time constants vs Machine size - Extreme thermal cycles, e.g., starts and stops - Accurate models	- Adaptive cooling and extra cooling systems - Thermal parameter identification - Wireless communication devices for thermal monitoring - Thermal available capacity for enhanced operation - Model Predictive Control with Peltier Modules Cooling - Multiphase winding topologies	Znidarich (2008a), Øyvang et al. (2019) Dang et al. (2018, 2020) Zeaiter (2020), Jichao et al. (2021a,b) Carunaiselvane et al. (2018), Carounagarane et al. (2020), Armando et al. (2021) Lomei et al. (2017), Boglietti et al. (2016, 2018, 2019), Øyvang et al. (2020), Boglietti et al. (2020), Savin et al. (2021) Stoop et al. (2022), Kellermann et al. (2021), Gou et al. (2022), Aleikish et al. (2021), Pandey et al. (2020), Lee and Hwang (2022), Phuc et al. (2021), Zhao et al. (2021)
Super-saturated electromagnetic dynamics	- Cross-saturation between d- and q-axes - Self-saturation of d- or q-axis - Interplane cross-saturation, i.e., multiphase machines - High short circuit current and torque	- FSCW or high reactance philosophy of design - New iron core material for higher saturation - Grain-oriented teeth and rotor poles for higher saturation - Superconducting slotless machines - Accurate super-saturated electromagnetic models - Reactance optimization	Kundur et al. (1994), cyiza Karekezi et al. (2022) Penney (1929), Calamaro et al. (2015), Castilla et al. (2009), van der Sluis (2001), Machowski et al. (2020), Levi and Levi (2000), Jecmenica et al. (2019), Ma et al. (2021), Chen et al. (2019), Ma et al. (2019), Zhi-peng et al. (2022), Duan et al. (2022)
Refurbishments	- Boosting requirements - Grid losses reduction - System optimization - Hybridization of Power Plants	- Rotor wireless measurements - Adaptive cooling plus Power Electronic cooling (Peltier) - New control algorithm for system optimization - Digital Twin for online monitoring and control - Digital Sibling for extreme scenarios analysis - Rewinding: same winding layout or a new one - Hybrid Hydropower Plants with Storage for Fast Frequency Reserve - Hybrid Hydropower Plants with VREs for increased energy production	Electric Machinery Committee of IEEE (2010), Znidarich (2008a) Boscaglia et al. (2021) Rahi and Chandel (2015), Lehoczky (1988), Moore (2001), Akiror et al. (2016), Behravesh et al. (2021) Rodič (2017), Jones et al. (2020), Rasheed et al. (2020), San et al. (2021), Pawar et al. (2021), Bárkányi et al. (2021), Tahkola et al. (2020), Guo (2020)
Operation	- Lower efficiency for boosting modes - Higher costs of operation for boosting modes - Underexcited operation: extra heating and stability issues	- Reactive power markets - Boosting VAr markets - Hybrid excited rotors - Q at night markets	Holzer et al. (2020), cyiza Karekezi et al. (2022) Kougias et al. (2019), Kim et al. (2019), Piancó et al. (2022), Lee et al. (2020) Adams and McClure (1948), Kekela and Firestone (1964), Phillips and Rubenstein (1956), Estcourt et al. (1953) Coelho et al. (2015), Morais et al. (2016), Baracho et al. (2018), Park et al. (2018), Tian et al. (2019), Jichao et al. (2019, 2020), Kamiev et al. (2016), Ammar et al. (2012), Ployard et al. (2014, 2016)

Table 28
Challenges to achieve fully Wind, Water, and Solar (WWS) energy infrastructure.

Aspects	Description	Challenges
Resources	- By 2030 the world will require 17 TW. The available world capacity is up to 85 TW for wind power, 1.6 TW for hydropower, and 340 TW for solar. - WWS resources equivalent footprint is 0.74 % to power the world.	- Abrupt climate changes, i.e., high variability of resources. - Energy models to account for actual energy produce. - Climate justice and right-of-way to construct any project. - Increase social welfare and resiliency to climate change.
Economic	Complementary and gap-filling WWS resources such as hydropower, smart demand-response management, and better forecasting have little or no additional cost.	- Development of models, schemes, technologies, policies, markets, and operation tools require plenty of economic and human resources. - Additional challenges applies to countries, institutions, and regions with insufficient scientific and financial resources. - R&D inequality to achieve a planetary grid with 100 % WWS.
Technological	Electrification may reduce 30 % the world power demand by 2030.	- Transmission infrastructure - Mass storage deployment - Hydrogen systems and markets

In contrast, relatively expensive batteries can store excess electricity and reduce energy costs. Furthermore, interconnecting countries in a 100 % WWS scenario benefits the grid's reliability and reduces annual costs, shedding, and land requirements further by reducing storage requirements and excess generation capacity installed (Jacobson, 2021).

Recently, Jacobson et al. (2022b,a) made a complete study for 100 % WWS, heat plus storage, and demand response. No batteries with over 4 h storage are needed. The study finds that all regions can avoid blackouts despite variable and extreme weather. The most significant premises are the perfect interconnection between areas, i.e., lack of power flow, and the 30 s time resolution, i.e., lack of stability criteria. Moreover, requirements to achieve 100 % WWS are shown in Table 29.

To improve interconnectedness, a study by Acevedo et al. (2021) of an HVDC macrogrid transmission for the continental U.S. with a high share of VREs shows economic savings over a 35 year period that exceeds its investment cost. These savings result from access to cost-effective renewables, decreased energy costs, and less expensive flexibility services, i.e., regulating, contingency, and planning reserves. Achieving a benefit-to-cost ratio ranging from 1.2 to 2.5 over different HVDC interconnection designs for the continental U.S. Bloom et al. (2022). Further improvement can be achieved with a proper PMU-based centralized controller in the frequency nadir of the HVDC macrogrid under contingency conditions by 25.9 %. As a result, the primary frequency response is increased by 33.4 % (Mohan et al., 2021). In the case of China, the Ultra-High Voltage (UHV) transmission lines contribute to interprovincial consumption, reducing the wind and solar curtailment rate (Zhang and Wu, 2022).

To face the challenges imposed by a high share of VREs or IBRs, i.e., flow, balance, quality areas, dynamic stability, lack of inertia, missing capability to provide reactive power, and reduced short circuit capacity, especially when connected to weaker power grids (Gu and Green, 2023; Sinsel et al., 2020). Grid-strengthening solutions have been proposed, such as flexible AC transmission system (FACTS) devices, grid-forming inverter controls, and synchronous condensers (Jayasinghe and Bahrani, 2021). These technologies can be introduced and evaluated using a voltage sensitivity analysis method to mitigate changes in the power system strength (Boričić et al., 2022).

As discussed in the previous sections, new or modified large hydrogenerators may improve all balance issues and specific stability challenges, such as frequency control reserves, coordination of frequency trip limits, and short-circuit power levels, and provide extended reactive power capabilities and complimentary services, e.g., inertia. For instance, according to the future system inertia report from ENTSOE (Orum et al., 2018), in the Nordic Power System (NPS), the volume needed to affect the minimum frequency by 0.1 Hz in an 80 GWs system is 20 GWs. Thus, the use of the existing synchronous condensers, i.e., 2.8 GWs, and the synchronous generators (without

Table 29
Requirements to achieve VREs integration and grid's reliability and resiliency.

Requirements	Challenges	Goals
Transmission	- Investment costs - Design and planning - Right-of-way - DC or AC	- Cross-regional renewable energy consumption - Reduces wind and PV generation curtailment
Spatiotemporal resolution	- Energy models - Stability studies - Big data (load-supply) - Confidentiality	- Ensure grid's reliability and resiliency - Deal with intermittent behavior of VREs
Storage	- Investment costs - Control and coordination - Size and mass-production	- Storage energy to reduce wind and PV curtailment - Optimize resources by shifting seasonally and daily
Demand Response	- Active demand-side. - Incentives - Participation in the grid and the electricity market	- Reduce consumption - Optimize use of Renewable Energies - Integrate distributed energy sources and microgrids

Table 30
Constraints due to a high share of VREs.

Aspects	Challenges	Countermeasures
Local: Voltage Stability (Munkhchuluun and Meegahapola, 2017; Flynn et al., 2017) (Kawabe and Tanaka, 2015; Remon et al., 2017; Jongudomkarn et al., 2020; Pereira et al., 2014; Awal and Husain, 2021)	High share of VREs imperils the operational security of the grid at a local scale due to lack of grid forming requirements.	Grid forming features: Dynamic voltage support, Synchronous Power Controller, and Virtual Synchronous Generator Control Schemes with Fault Ride Through capabilities. - Flexible AC Transmission Systems (FACTS) Hybrid-PSH systems may incorporate previous PE features for voltage support. Hydrogenerator may have additional available thermal capacity to boost reactive power delivery.
Global: Grid Operation (Badrzadeh et al., 2021; Anon, 2021b)	Zero marginal costs of VREs reduces minimum synchronous unit commitment to provide inherent or controlled characteristics. Such as voltage and angle support, inertia, and frequency control. New stability resonance and converter-driven stability issues appear.	- Australian Energy Market Operator developed a wide-area electromagnetic transient model to obtain minimum unit commitment, long-term planning, operational decision-making, and event analysis. - Strategic location to install synchronous condensers - System-level inverter-based resources parameter tuning. - Coordination between TSOs and DSOs to allow the minimum number of large hydrogenerators to operate in condenser mode and far below their minimum stable load points.

considering hydropower) operated as condensers, i.e., 4.7 GWs, is not a sufficient measure for low-inertia situations. Moreover, hydropower with Pelton units may represent 25 GWs. However, the most cost-efficient represents little less than half the total potential. In the case of gas turbines, the prospect may be 6.2 GWs, and the pump-storage hydro is around 3 GWs. Therefore, the NPS may require proper optimization to support the system during low inertia, e.g., when large wind power plants are deployed. Moreover, flow-based market coupling may present higher precision, better resource utilization, increased total welfare, and smaller prices for a high share of Wind Power (Cui et al., 2021; Bo et al., 2020).

5.3.1. *New constraints for high share of VREs*

Detailed modeling of grid operation is crucial for power system operators and planners to ensure system security at a high share of VREs. For instance, in cross-border congestion management, parametrizing can have enormous impacts with a thoroughly calibrated model (Hobbie et al., 2022). The new constraint for Power Systems can be divided into local and global aspects, as presented in Table 30.

EMT-based simulators are universally used to simulate complex power systems with non-linearities and Power Electronic Devices. However, EMT is computationally intensive; thus, time-varying Dynamical Phasor (DP) based methods are used to reduce the computational burden. Moreover, co-simulation techniques for EMT simulators have recently become a trend due to their reduced computational intensity and high accuracy. Co-simulation utilizes detailed EMT models to capture detailed dynamics and phasor-based simulations. Subedi et al. (2021). EMT and co-simulation techniques are advised to define needs for future scenarios under normal and defined contingency conditions (Matevosyan et al., 2021). Furthermore, the development of a synthetic power system digital twin, or Reduction Order Model (ROM) (Grdenic et al., 2022), may represent the complexity and size of real systems while avoiding the release of confidential information, thus providing real-time models and what-if scenarios for risk assessment (Arrano-Vargas and Konstantinou, 2022).

Moreover, Grid-Forming Technology is a necessary enabler of the high penetration of IBRs but insufficient to resolve all issues (Matevosyan et al., 2021). For instance, in Great Britain, a minimum non-mandatory specification for grid forming capabilities has been recently developed, where the plants wishing to offer a Grid Forming ability will support the grid during unplanned events/faults, as is shown in Table 31.

Lastly, Table 32 presents the challenges of unit commitment, transmission grid interconnectedness, and dunkelflaute challenges to ensure system security. Appropriate transmission, distribution, and generation planning must capture the benefits and dynamics of the different hydropower technologies. Hence, adequate energy and grid models considering each country and region's particularities may result in multiple solutions to achieve security standards.

Finally, Table 33 summarizes the hydropower challenges from a Grid Owner Perspective. Five areas are presented where challenges and opportunities are depicted. The latter incorporates the hydropower plant operation perspective and the entire grid operation. Thus, the opportunities go beyond hydropower technologies. Nonetheless, this work advocates for a holistic view of the Power System to achieve the Climate Change Goals 2050+ Agenda.

6. Hydropower modeling

In a system-wide optimization, the opportunity cost of the water must be estimated accurately. Therefore, hydrothermal operation

Table 31
Grid forming capability minimum specification in UK (National Grid ESO, 2021).

Stability	Support type	New definitions
Frequency	- Limit rate of change of frequency (ROCOF) - Instantaneous active power and fast fault current injection after a phase change	- Active ROCOF response - Active Phase Jump Power - Active Control Based Power - Fast Fault Current Injection
Voltage	- Improve voltage profile during a fault, e.g., a pre-requisite fault ride through.	- Control Based Reactive Power - Voltage Jump Reactive Power
Rotor	- Damping power - Limiting vector shift - Synchronizing torque	- Active Damping

Table 32
Opportunities for hydropower technologies.

Aspects	Challenges	Opportunities
Unit Commitment (Hydro Tasmania, 2018; AEMO, 2019; Aziz et al., 2018)	Low inertia due to synchronous generator are reduce.	- Hydrogenerators acting as synchronous condensers (SC). - Retiring hydropower plants retailored into SC.
Transmission Grid (Rivera et al., 2020)	Radial grids may experience low voltage due to reactive power shortages.	- Hydrogenerator's extended thermal limits to provide extra reactive power services.
Dunkelflaute (dark doldrums) and Storage (Boston et al., 2022; Koritarov et al., 2022; Ruhnau and Qvist, 2022)	- An extended period based on historical weather or simulation of several years should be modeled to account for system security. - Optimum storage is achieve at less-than-a-day scenarios. - Firm generation is more cost-effective for once-a-year wind and solar reductions.	- Long-term storage can ensure system security during winter. - Diverse energy mix portfolio to reduce expensive storage. - Large-deployment of Hydrogen may cope for energy deficiency. - PSH technologies capture: (1) Intra-hour variability and may provide balance to the system. (2) Long-duration storage to overcome extreme weather events or prolonged drought periods.

Table 33
Hydropower challenges: Grid owner perspective.

Areas	Challenges	Opportunities	Citations
High share of VREs constraints	- Low inertia - Low SCR - Intermittency	- Minimum Super-flexible Hydro Unit Commitment with EMT grid models - Fast Frequency Reserve for PE-interfaced hydropower plants - Enhanced Reactive Power Services for grid resilience and losses minimization	Koritarov et al. (2022), Hobbie et al. (2022) Badrzadeh et al. (2021), Subedi et al. (2021), Matevosyan et al. (2021), Grdenic et al. (2022), Arrano-Vargas and Konstantinou (2022), National Grid ESO (2021), Hydro Tasmania (2018), AEMO (2019), Aziz et al. (2018) Boston et al. (2022), Ruhnau and Qvist (2022)
Energy Mix Optimization and Coordination	- Achieve the highest VRE share - Minimum system costs - Maximum generation revenue - Complex monitoring and forecast errors - Cost of the water	- Pumped-Storage Hydro optimization and coordination for VREs integration - Multi-time scale hydrological horizons optimization and coordination - Wider/Finer Spatio-temporal granularity for multi-basin optimization and coordination - Transdisciplinary working group for holistic approaches (Energy Models)	Spodniak et al. (2021), Xia et al. (2020), Rahman et al. (2021) Xie et al. (2021), Zhou et al. (2021), Imcharoenkul and Chaitusaney (2022) Shafiullah et al. (2022), Yang et al. (2022), Cousse (2021), Yin et al. (2021), Huxley et al. (2022), Rosemberg et al. (2022) Huang et al. (2022), Wang et al. (2020), van Meerwijk et al. (2016), Craig et al. (2022), Kies et al. (2021)
Electricity Markets	- Computational burden for real-time markets - ESS subsidies and incentives - Power and capacity forecasting - Scarcity and flexibility pricing - Capacity and forward electricity markets - Missing Money Problem	- Minimization of short-term market distortion - Pumped-Storage Hydro Investment - Mixed scarcity and capacity market mechanisms for short- and long-term conditions - An accurate formulation for scarcity pricing - Reactive Power Markets to incentivize enhanced reactive power services - Flow-based Market Coupling - Transmission Grid Expansion Investment - Hydropower's power adequacy and resilience incorporation into capacity markets - Minimizing the load lowest margin of operation for hydropower technologies - Remuneration for hydropower fast ramping services	Koritarov et al. (2022) Bistline (2019) Cui et al. (2021), Bo et al. (2020) Bichler et al. (2022), Zakeri et al. (2019), Tan et al. (2022) Liu et al. (2018), Thomaßen et al. (2022), Hogan (2022), Bajo-Buenestado (2021), Papavasiliou (2020), Papavasiliou and Bertrand (2021), Subramanyam and Zhang (2020), Duggan (2020), Prakash et al. (2022) Li et al. (2022) Sanjab et al. (2022), Cochran et al. (2014) Sreekumar et al. (2022), Naversen et al. (2022), Batlle et al. (2021), Ela et al. (2021), Mohandes et al. (2019)
Power Grid Control	- More participants and higher complexity - Intertwined risk analysis and decision-making - Complex real-time decision-making	- Power System Digital Twins concatenated with other modular cyber–physical models - Artificial intelligence in decision-making and advanced control systems - High-quality open-source datasets to improve AI models for complex real-time decision-making - Big data, IoT, and Cyber-security investment - User friendly AI applications	Marot et al. (2022), Kanabar et al. (2022), Xie et al. (2022)
Environmental Impact Large Hydro	- Decrease annual hydropower - Resilient hydrothermal dispatch - Pest species, e.g., water hyacinth - Greenhouse Gas (GHG) emissions	- Reduce GHG emissions with deeper dams - Multi-basin maximization for elevation of hydropower plants and reduction of GHG - Meet 2030 Climate Change Agenda for Hydropower Generation with appropriate policies	Craig et al. (2018), Wasti et al. (2022), van Vliet et al. (2016), Paltán et al. (2021), Obahoundje and Diedhiou (2022) Lu et al. (2020), Li and He (2022), Almeida et al. (2019), Jager et al. (2022)

planning problems must be assessed with better formulation to have fewer inconsistencies of dispatch costs, spot prices, and thermoelectric dispatches (Rosemberg et al., 2022). Moreover, hydro uncertainty, climate change effect, and the high share of VREs introduce intra-hourly variability that challenges the flexibility of hydro scheduling, i.e., multistage stochastic optimization as a modeling framework and stochastic dual dynamic programming as an algorithm (Lorca et al., 2020).

While historical hydrological records remain the foundation for hydropower scheduling, their utility is becoming increasingly limited by non-stationary hydroclimatic regimes and competing anthropogenic water demands across basins. In that sense, multi-basin hydrological

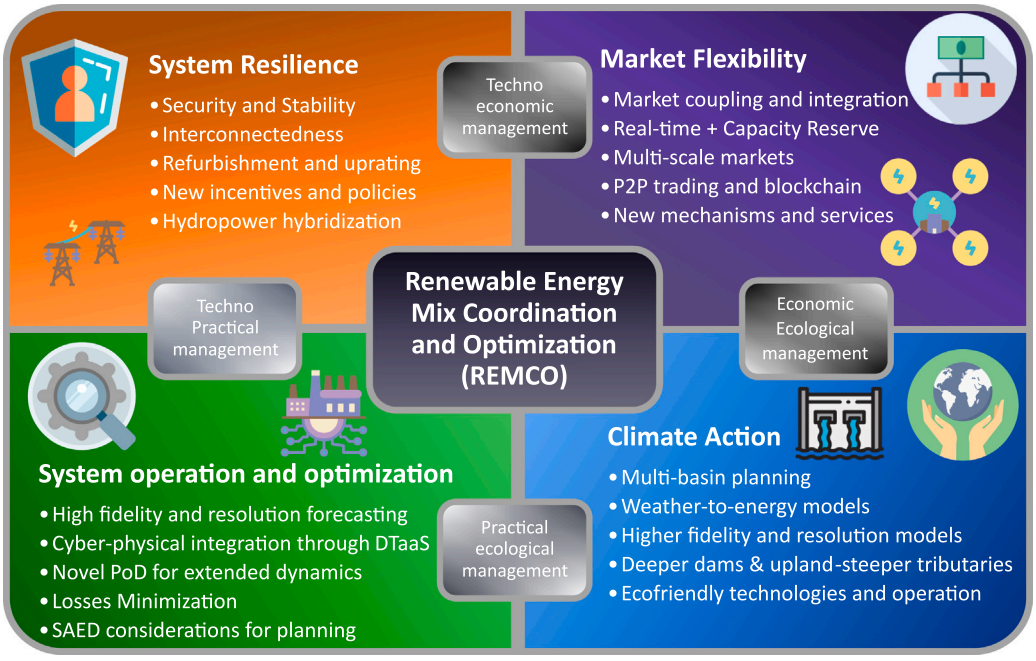


Fig. 22. Renewable Energy Mix Coordination and Optimization (REMCO) strategy. Here, is presented the epitome of various concepts and their interaction areas due to climate change, earth sciences, and power systems as a holistic approach with emphasis in hydropower technologies.

modeling and dynamic reservoir operation integrate regulated river flows, climate-driven inflows, and coordinated reservoir management across interconnected basins. This framework for hydropower generation enables finer temporal resolution to better capture responses to hydrological extremes. Moreover, it is possible to incorporate operational constraints critical at subcontinental scales, including environmental spill requirements, hydropeaking ramping rates, and coordination with adjacent projects, while aligning releases with wholesale electricity prices and local load patterns (Turner and Voisin, 2022).

Lastly, integrating water systems at a basin-scale and energy models for hydropower management may result in an increase of accuracy of large reservoir inflow for better representation of the hydropower system that handle with all water dynamics, i.e., downstream–upstream coordination, urban, agricultural, environmental, and recreational, as well as electricity generation considering micro-macro economics linkage, and Climate and Global Change conditions (Rheinheimer et al., 2023; Expósito et al., 2020).

Fig. 22 outlines the strategy coined as Renewable Energy Mix Coordination and Optimization (REMCO) to achieve a high share of VREs. It aims to ensure grid resiliency and stability by leveraging services provided by hydropower and other synchronous generation facilities. These services include reactive power provision, voltage support, fast ramping, inertia provision, and enhanced short circuit capabilities for extended thermal and electromagnetic dynamics. Four main areas are presented to be more relevant. These are system resilience, market flexibility, system operation and optimization, and climate action. Each area epitomizes the various concepts reviewed throughout the entire work.

System resilience can be summarized as ensuring security and stability through FACTS, Grid-forming inverter controls, and Synchronous Condenser mode; promoting interconnectedness, that is, local- and regional-wise meshing and incorporating multi-microgrids; refurbishment and uprating to keep the fleet of hydropower plants with higher capabilities; new incentives and policies to allow investment and operation of new and uprated hydrogenerators for enhanced power dynamics; hydropower hybridization with other renewable energies to improve energy mix.

Market flexibility covers flow-based market coupling and integration of different regional-wise markets; real-time market and capacity reserve market integration for short- and long-term operation, mainly to avoid instability during scarcity periods; multi-scale market to include microgrids and DSOs; P2P trading and blockchain to allow a more decentralized and transparent market; new market mechanism and services to allow the entry of hydrogenerator and other synchronous generator additional services and capabilities for broader short- and long-term thermal dynamics to face outages, voltage instabilities, or constant *dunkelflaute*.

System operation and optimization include high fidelity and resolution for forecasting to cope with VREs' stochastic behavior; cyber–physical integration through Digital Twin as a Service (DTaaS) for seamless decision-making using the Hypervision control approach; Novel Philosophy of Design (PoD) in the infrastructure for extended thermal and electromagnetic dynamics; losses minimization using DTaaS and the extended capabilities of hydrogenerators and other synchronous machines; and Spatial Annually Energy Density (SAED) considerations for planning.

Last, the **climate action** area constitutes multi-basing planning to construct efficient and environmentally friendly hydropower plants; weather-to-energy model as interdisciplinary study of climate change, earth science, and power system engineers to achieve optimal hydrological dynamics into design and planning; higher fidelity and spatiotemporal resolution for optimal design, planning, operation, and forecasting of hydropower plants; deeper dams and upland-steeper tributaries to achieve reduced GHG emissions; and eco-friendly technologies that do not contaminate the water quality, and operation to maintain a sustainable water flow.

7. Emerging trends

7.1. Digital twins

The digitalization of hydropower installed worldwide capacity could increase by 42 TWh the annual production of energy, which is equal to USD 5 billion in annual operational savings and a significant reduction of GHG emissions (Kougias et al., 2019). A hydropower Digital Twin

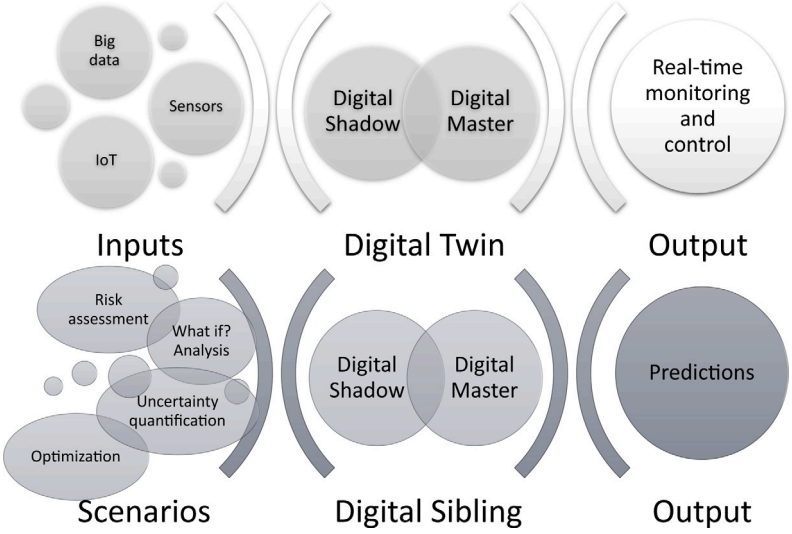


Fig. 23. Digital twin and digital sibling.

Table 34
Approaches for Capacity Twin Model (Rasheed et al., 2020; San et al., 2021; Pawar et al., 2021; Bárkányi et al., 2021).

Model	Description
Physic-based	- Mathematical equation - Other physical phenomena is ignored to reduce complexity
Data-Driven	- Extensive data to account for all the (un)known physics. - Not reliable when new physical phenomena appears
Hybrid Analysis Modeling (HAM)	- Combines previous two. - Suitable for Surrogate Models

(DT) paves the way for minimizing penstock fatigue, increasing service lifetime, and increasing efficiency and energy production (Quaranta et al., 2020). Moreover, it is calculated that in Europe the digitalization in improving better load share on turbines, better blade-opening, draft tube operation, better inflow forecast, spill reduction, and prevention of unplanned shutdowns, the benefits could reach up to EUR 1.7 billion (Quaranta et al., 2023).

A DT is a cyber–physical system (CPS) for real-time monitoring and control, whereas a digital sibling is used for hypothetical scenarios, e.g., what if? Analysis, risk assessment, optimization, and uncertainty quantification, as shown in Fig. 23. However, both CPSs consist of a digital shadow and a digital master. The first represents the structured data collection generated by operational, conditional, and process data. The latter is a universal model of the assets (e.g., machines) and their relations (Rodič, 2017).

A DT is a synchronized system of convenient bidirectional information and management between physical and virtual entities, enabling convergence between states. The DT can exist at any life-cycle stage and aims to leverage aspects of the virtual and physical environments over the life-cycle (Jones et al., 2020).

For hydrogenerators, a Capacity Twin Model (CTM) can be created for sufficient predictive quality of control to optimize enhanced reactive power support ensuring operational security with a reliable model. Table 34 shows the different approaches for a CTM. In addition, a CTM may be constructed as a 3D Finite Element Model (3D-FEM) that calculates and predicts the available thermal capability without compromising the components' health. An example is a Smart Power Plant Supervisor Project, a digital tool that optimizes hydroelectric unit operation and maintenance to improve ancillary services provision to the power system (Vagnoni et al., 2021).

The 3D-FEM offers a virtual space for examining various physical phenomena, but its computational demands make it unsuitable for

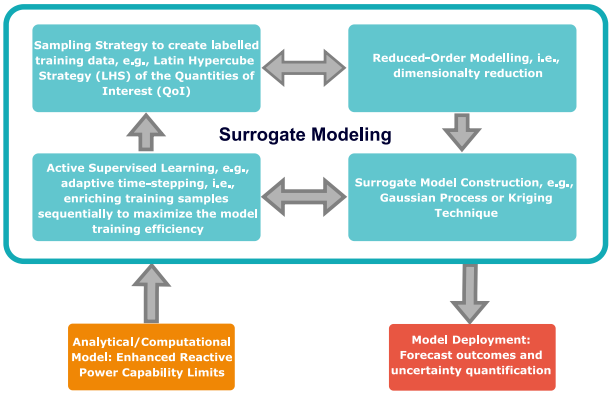


Fig. 24. Surrogate Modeling Approach Example.

real-time control and operation. Instead, machine learning models are recommended to create Multiphysical-Informed Machine Learning, resulting in faster and more precise calculations.

In that sense, the surrogate model (SM) delivers accurate, fast simulation output. This trained model replaces the original computer simulation as a DT, and the training data is obtained by conducting complete simulation outputs at several intelligently selected locations in the design parameter space—the procedure is shown in Fig. 24 (Tahkola et al., 2020; Boscaglia et al., 2021). An SM is a case of supervised machine learning applied in engineering design in which polynomial regressions, support vector machines, Gaussian Processes, and neural networks are widely adopted to accelerate the product design and analysis processes (Guo, 2020).

7.2. Power grid control and monitoring

The path toward 2050 unequivocally changes the control side due to the real-time market, more participants in the market, increased interconnected grids, intermittent power generation, and CPS for TSOs, DSOs, and power producers. The trend for future control centers is to increase the automation processes that are time-consuming, tedious, and repetitive tasks. Due to increasingly complex risk analysis and decision-making, operator methods must shift from real-time to Planning- and navigation-based methods. These new methods entail operators relying on the system's real-time automation, forecast accuracy,

and study tool optimization ahead of real-time, such as artificial intelligence and DTs. Moreover, achieving coordination between several CPS models would come as a hierarchical modular and configurable cyber layer, in which the top one is an *system-optimization* layer. All that, as mentioned earlier, can be integrated into a **Hypervision** framework, a simplified and unified interface for all operators that instantiates structured decision-making (Marot et al., 2022).

Hydrogenerator and transformers (Mansour et al., 2023) can be deployed as a DT. Hence, entire power plants can be modeled as DTs for real-time monitoring and control, assessing the life-cycle, and predicting risk assessment, what-if analysis, optimization, and uncertainty quantification scenarios. Similarly, power substations may have a CPS to track electrical, chemical, and thermal characteristics to support informed decision-making, improve grid reliability, and optimize operation and maintenance costs (Kanabar et al., 2022).

DTs can be categorized according to their timing and model depth, which, faster than real-time and high fidelity, may account for control and forecasting. The features of a DT can be distinguished as model phenomena fidelity, geographical coverage, execution speed, and model accuracy. Constructing DTs can be white, gray, or black box models. Furthermore, using Distributed Co-Simulation Protocol (DCP) and Functional Mock-Up Interface (FMI) standard enables multi-domain models through smaller subsystems, state-of-the-art high fidelity models, and embedded black-box models to retain sensitive data (Palensky et al., 2021).

A new paradigm of Digital Twin as a Service (DTaaS) is emerging through the synthesis of Internet-of-Things and Big Data for Information gathering, cloud computing for scalability and industrialization, Machine Learning and AI technologies for knowledge acquisition, predictive and prospective analytics for Wisdom procurement, and blockchain technologies to ensure transparency, accessibility, trusted traceability, and immutability of transactions. Aheleroff et al. (2021), Hemdan et al. (2023), Ardolino et al. (2018). Fig. 25 presents a schematic of a Power System Digital Twin integration (Song et al., 2023) with DTaaS and Cyber Layer.

Finally, three challenges must be faced to achieve a digitized power grid. (1) high-quality open-source datasets that AI communities may use to train, calibrate, test, and benchmark machine learning models; (2) advanced algorithms, cyber-security, massive streaming data, hybrid edge, and cloud computing to account for Big Data coming from all the sensors for data acquisition; and (3) user-friendly AI applications for interpretability of the physics behind, robustness to perturbances, and scalability for large-scale real-world systems (Xie et al., 2022).

7.3. Holistic hydropower approach

A three-pillar interplay is schematized in Fig. 26, which elucidates the need for a cooperative plan for Hydropower in the transition toward achieving the Climate Change Goals of the 2050 Agenda, chiefly to grow the same capacity over the next 25 years as it did over the 140 years prior. In that sense, outdated hydropower plants with decommissioning plans should be reassessed for modernizing, uprating, or refurbishing.

Complex power systems demand a higher temporal granularity for modeling, analysis, and planning to consider market integration and stability concerns. Hydropower's viability and efficiency, like that of other VREs, are strongly dependent on geographic location. Therefore, a finer-wider spatiotemporal granularity will be required to integrate renewable technologies and coordinate multi-basin approaches.

Deeper dams, multi-basin optimization, and policies may be used to address environmental issues, habitat loss, and emissions. In addition, the effects of climate change on water stream flows — which vary depending on the basins — for the potential for exploitable hydropower imposed complex, significant spatial scale uncertainties for system modeling, planning, optimization, and operation.

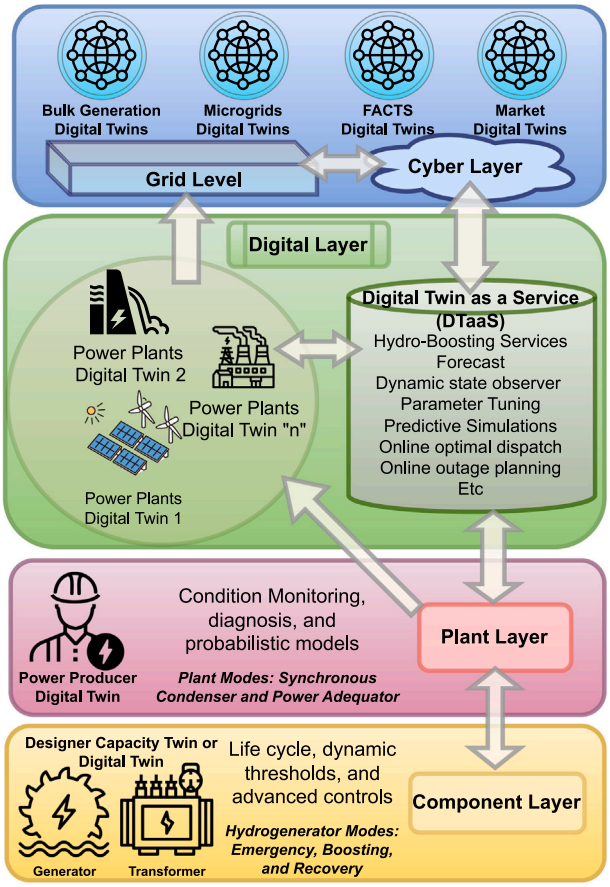


Fig. 25. Power System Digital Twin as a Service Approach. The hierarchical structure starts with the component layer, where detailed DTs of unique power system components are created. These are integrated into the plant layer for holistic operational insights. The digital layer employs the FMI to harmonize and aggregate these insights, forming the DTaaS capabilities. At the top, the grid-level operator accesses the cyber layer, utilizing co-simulation within a scalable and distributed computing environment for real-time optimization, enhancing the resiliency and security of the power system.

Hydropower is an economical energy source, but it requires significant upfront investments and long-term financial planning, which can be difficult in a volatile economy. Natural market operations also decrease hydropower dispatch with a high proportion of VREs. Therefore, the hydrogenerators have to run as synchronous condensers in order to maintain a minimal level of system strength. Furthermore, because energy markets need a suitable mechanism to compensate for these services, reactive power and voltage support do not yield considerable economic benefits. Because there are no profits, land and water usage is squandered.

Finally, hydropower's energy density might differ and sometimes be even lower than that of fossil fuels, as in the case of gas power plants. Still, it has the highest energy density out of all the renewable energies. It is, therefore, imperative that its scalability be improved. Its integration with other VREs needs to be handled as well in order to optimize environmental impact reduction and efficiency gains. These difficulties could be addressed by developing new hydrogenerator Philosophies of Design (PODs) and hydropower plant operating modes that provide more adaptable envelopes for improving reactive power while accounting for hotspot thermal dynamics.

The limitations of the current work are (1) Lack of multi-sector study, including fuels, biofuels, EVs, transportation, heating, and other sources of storage technology. (2) The analysis based on hydropower accounts for potential ways to achieve a high percentage of VREs in specific geographies. (3) Not an analysis based on weather-to-energy.

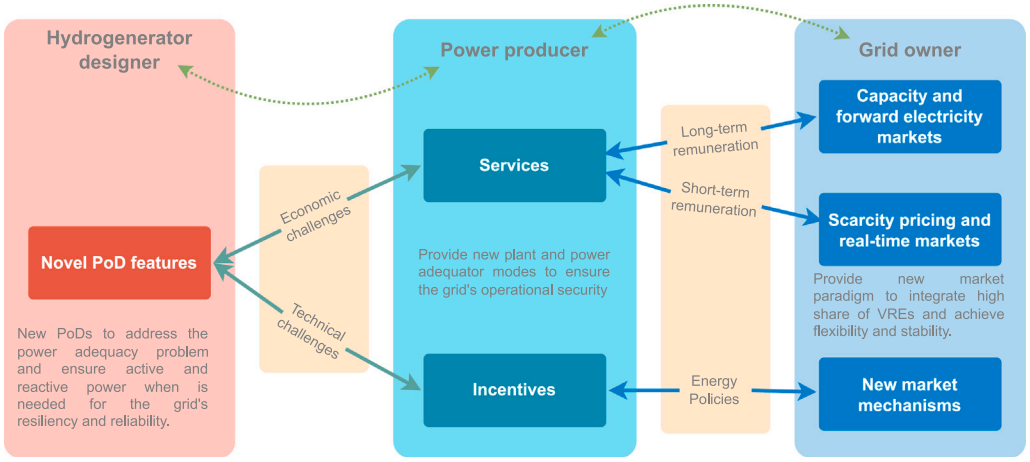
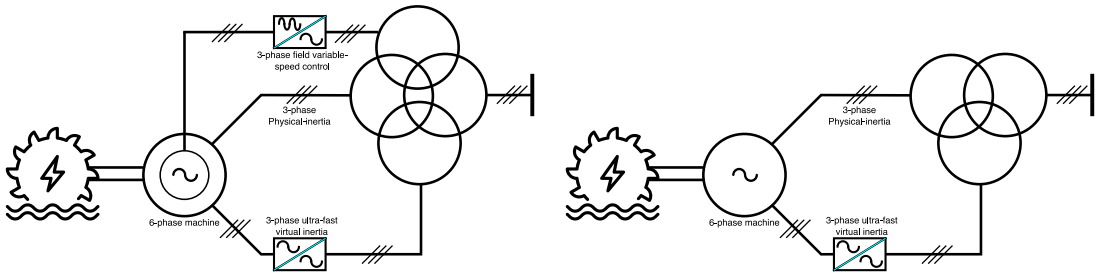


Fig. 26. Pillars interaction. The interrelation between designers and power producers is given by the economic and technical challenges driven by services and incentives. Economic challenges are linked to services because of the investment in novel PoDs. In the case of technical challenges, incentives are required to achieve technological disruption. Furthermore, the interrelation between power producers and grid owners is done between the remuneration period for the new services and the energy policies to give incentives. The energy policies impose climate- and social-driven constraints.



(a) Doubly-fed Machine Variable Speed (b) Synchronous Machine Fixed Speed

Fig. 27. Proposal of Long-term system: A dual three-phase machine with hybrid inertia.

It covers only the power system participants shown in Fig. 1. (4) Upgrading or refurbishing solely the hydrogenerators and not the plant as a whole or other areas related to the environment and civil society.

7.3.1. Machine designer challenges

Based on the prognosis timeline, two stages for hydropower technologies are essential in achieving the 2050+ Agenda of Climate Change goals. Stage 1: the transitioning systems up to 2050 that may obey power grids with low- and mid-level penetration of VREs, e.g., less than 50 %, and fewer meshed transmission grids. Stage 2: long-term systems up to 2100, which accounts for high-level penetration of VREs and highly meshed transmission grids.

For the transitioning systems, improved operation for extended and flexible dynamics can be achieved by novel cooling features, hybrid solutions, and hybridization with VREs while having accurate efficiency and thermal hotspot management with advanced control systems to track optimal points of an operation with a lifetime assessment. Hence, digital twin as a service (DTaaS) is vital for hydrogenerators. For instance, the hydrogenerator digital twin can give the actual efficiency of the machine to assess the opportunities for refurbishment and updating to comply with codes, improve profits, and provide new services to integrate intermittent renewable resources into the grid. Moreover, PSHs must be further developed and implemented with new services to ensure the grid's resilience and reliability.

In the case of non-PE-interfaced hydropower plants, the most significant advantages are the physical inertia and reactive power enhanced support capabilities because of the broader thermal dynamics. Additionally, do not introduce harmonic pollution into the grid, deteriorating the overall power quality to the consumers. Moreover, it avoids complex calculations such as Geometric Algebra (Montoya et al., 2022; Eid and Montoya, 2022) or Conservative Power Theories (Ding et al., 2021) to calculate appropriate harmonic filters.

New PoDs must be designed for long-term systems to account for ultra-fast services, real-inertia support, extended dynamics, and variable-speed pumped-storage hydro. Therefore, proposals are presented in Figs. 27 and 28. The former are multiphase hybrid-inertia hydrogenerators (MHIH) in which a bundle of three-phase systems is connected directly to the grid, and the other bundle systems are connected via power electronics. This hybrid configuration, i.e., physical and PE-connected MHIH, can provide fast frequency reserve, frequency control ancillary services, network support and control ancillary services, system restart ancillary services, real and synthetic inertia, regulation services, allocation of excess energy, storage optimization, extended reactive power for long-term voltage stability with extended thermal availability, and with various three-phase inverters, each providing different services, such as various control algorithms for Grid-forming simultaneously. Even mitigate harmonic distortion due to other IBRs. Thus, MHIH may provide broader services for power quality and optimization schemes. The latter proposal in Fig. 28 is a direct-cooled rotor-turbine to improve field winding cooling and enhanced reactive power boosting.

7.3.2. Power producer challenges

The challenges for the power producers are the operation modes in which the profits are maximized. Moreover, the operation has to face the new conditions of peaking function imposed by an electricity market with a high share of VREs. Hence, the operation modes are changed from a constant power production to a power adequacy services.

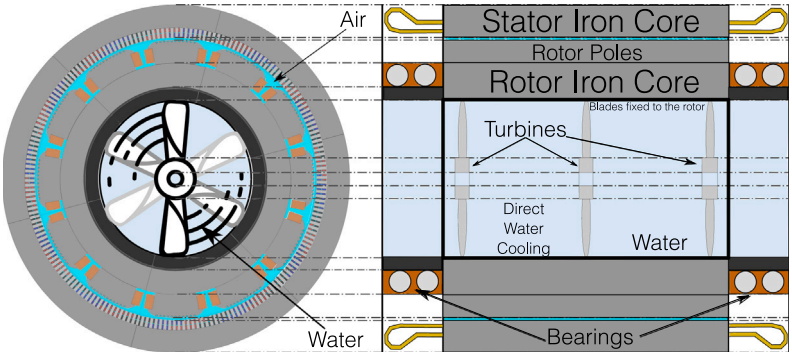


Fig. 28. Rotor-turbine Conceptualization. This proposal shows a self-incorporated hydro-turbine to reduce mechanical losses and utilize direct water cooling to the rotor. Hence, enhanced field current operation for power reactive boosting can be achieved. The technology has been under development with shrouded tidal turbines (Wani et al., 2020; Ebrahimi, 2020; Qin et al., 2022).

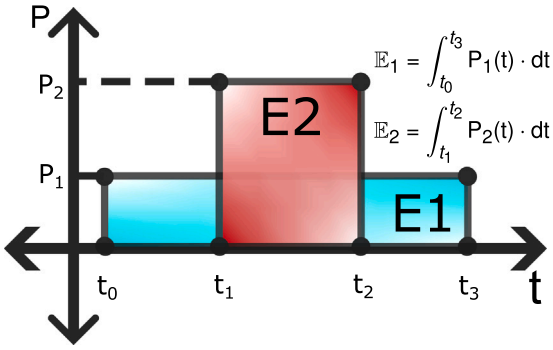


Fig. 29. Energy-to-Power transition. Here, the energy usage due to operational transition of hydropower generation is depicted.

Hydropower technologies' fast ramping capabilities have to be further exploited to provide new services for capacity markets and wind and solar energy scarcity periods. In that sense, each power plant must optimize grid security and loss reduction to maximize profits. Therefore, plant modes and hydrogenerator modes may use an optimization algorithm.

Furthermore, hydropower production is facing several challenges. Namely, the capacity factor is diminished because the baseload operation is changing to a peaking operation. Thus, there can be three main scenarios of hydropower peaking operation:

1. Reducing annual energy for the same installed capacity because of the diminishing water flows in rivers and the natural market operation with a high share of VREs,
2. Increasing the installed capacity to produce the same annual energy to face the scarcity periods of wind and solar energy,
3. Increasing installed capacity and annual energy, where is possible.

The former is straightforward to see that the capacity factor is reduced. In this case scenario, represents the hydropower facilities where there is no action for uprating the capabilities. The second case obey strictly to comply with the grid's reliability and resiliency; therefore, it is a consequence of the need for more capacity installed during the scarcity periods of wind and solar energy.

For instance, assuming the energy for the red and blue rectangles in Fig. 29 have equal areas, i.e., the same water flows. The new capacity factor C_{p_2} of the hydropower energy production of the red rectangle is written as follows.

$$C_{p_2} = C_{p_1} \cdot \frac{P_1}{P_2} \tag{18}$$

Thus, when the P_2 is increased there is a reduction of the capacity factor. In order to keep the same energy, it requires different scheduling and optimization algorithms to couple near-real-time, short-, mid-, and long-term periods. However, this is true if the annual energy production is matched with the power production over the scarcity periods. In other words, solar and wind energy production, energy storage, and the well-interconnected vast regions could convey to push away hydropower power production. Therefore, the power production at the end of the year will not meet the annual energy expected.

Finally, increasing installed capacity and energy may lead to keep the same capacity factor. However, it will depend on the annual energy production and the market mechanism to integrate hydropower production into the scarcity periods, or other mechanism such maintaining stable electricity prices, optimize the grid's losses, and ensure system reliability and resiliency.

7.3.3. Grid owner challenges

The grid owners face the challenges of developing new market mechanisms to achieve grid reliability within a high share of VREs. An inherent contradiction exists between ensuring supply security, the electricity market's functioning, and achieving climate change goals.

Moreover, due to the increment of IBRs, the grid owner may need new approaches and technologies for grid planning, operation, and optimization. For instance, transmission investment accompanied by generation expansion should be planned to address long-term supply security. Furthermore, the flow-based market coupling may contribute to facing scarcity events in interconnected bidding zone borders with meshed transmission grids.

In that sense, hydropower technologies may provide power adequacy and storage capacity services to integrate a high share of VREs, chiefly for an Energy-to-Power transition, where the capacity factors are reduced because of the refurbishment and uprating of hydropower plants. Here, new PoDs for hydrogenerators will meet the new stressed and broader range of operating points and the extended dynamics for reactive power boosting and power adequacy services, resulting in suitable for emerging Electricity Markets.

7.4. Future outlooks and recommendations

Building on the three-pillar framework and scientific contributions of this work, the following priorities emerge for advancing hydropower's role in high-VRE grids:

1. Hydrogenerator Design Innovations

   - **Priority:** Develop variable-speed hydrogenerators with hybrid excitation systems to extend reactive power capabilities under erratic VRE conditions.

- **Action:** Establish testbeds for climate-resilient designs to validate extended thermal/electromagnetic dynamics.
- **Gap Addressed:** Current PoDs lack adaptability to climate-driven hydrological extremes in combination with wind and solar generation variability.

2. Power Producer Operation Enhancements

- **Priority:** Implement AI-driven digital twins to optimize real-time hydropower cycling (e.g., hydropeaking mitigation) while minimizing wear and tear.
- **Action:** Pilot hybrid hydropower-VRE systems (e.g., floating PV on Norwegian reservoirs) to validate multi-energy coupling benefits.
- **Gap Addressed:** Operational modes neglect VRE-induced thermal cycling and revenue trade-offs.

3. Grid Owner Integration Strategies

- **Priority:** Design capacity markets that reward hydropower's inertia, voltage support (i.e., reactive power support), ramping, and seasonal storage.
- **Action:** Adopt transboundary frameworks (e.g., EU-wide REMCO strategy) to coordinate multi-basin hydropower and VRE integration.
- **Gap Addressed:** Siloed market mechanisms undervalue hydropower's system services.

4. Cross-Pillar Synergies

- **Priority:** Create policy-economic-technical co-design task forces to align IPCC targets (e.g., 1.5 °C pathways) with hydropower expansion.
- **Action:** Launch Nordic-EU partnerships to harmonize grid codes, environmental regulations, and turbine modernization.
- **Gap Addressed:** Regulatory frameworks lag behind technical innovations opportunities.

8. Conclusions

This work presents a comprehensive overview of planetary power systems and the interplay of hydropower technologies and solutions to achieve resilience and reliable grids toward the 2050+ Agenda of Climate Change goals.

A bottom-up survey is done by implementing a three-pillar scheme of intertwined actors, i.e., (1) grid owners, (2) power producers, and (3) designers. Furthermore, these actors impose conditions on other layers to achieve hydropower's flexibility compatibility for the grid's resilience and robustness for VRE integration. Hence, these conditions become challenges because of their inherent holistic and complex approach. For instance, designers cannot create and suggest innovative solutions if there is no prior arrangement from the power producer to offer new services and if there is no prior agreement between the electricity market and the power grid security. Therefore, any new proposal of PoD, service, or market mechanism is a nested problem that entangles the three pillars.

The main limitation of enhanced hydropower flexible services is the economic constraints of limited market mechanisms to offer incentives and remuneration. Thus, hydropower technologies must have specific services, tariffs, and incentives for their products to achieve VREs integration and support and ensure the grid's resilience and reliability.

Custom-made hydropower services and market mechanisms will incentivize investment in new installed capacity. Moreover, investing in hydropower technologies may expand the system's flexibility to achieve VRE integration through resilience and reliability. However, further development must be done for transdisciplinary approaches for Weather-to-Energy models to account for more accurate and robust analysis, design, planning, operation, and optimization. Specifically for hydropower technologies, a wider-finer Spatiotemporal granularity for multi-basin modeling must account for the hydrological behavior due to climate change and the interplay of well-interconnected power grids. It is important to remark that many articles assume almost perfect interconnected transmission grids for 100 % VRE integration; thus, transmission planning is of utmost importance.

Furthermore, the effort to meet hydropower global installed capacity goals is divided into new plant construction and old plant refurbishment. The latter may present the most significant challenge because the decommissioning of large hydropower plants may be done for several reasons, such as degradation, environmental impact, out-of-service, and politics. However, with refurbishment, decommissioning must be avoided as much as possible to achieve the goals of hydropower global installed capacity.

Finally, some noteworthy points epitomize the possible next steps in hydropower technologies and power systems: (1) New philosophy of design (PoD) for hydropower and pumped-storage hydro that incorporates synchronous condenser and Power Adequacy modes. (2) Market-based initiatives to incentivize reactive power services, flexible power producers, storage technologies like pumped hydro, investments in new capacity, and new and interconnected transmission capacity. (3) Enhance accuracy in scarcity pricing mixed with capacity and forward electricity markets to settle the short- and long-term markets. (4) A multi-basin approach and transdisciplinary weather-to-energy models to achieve broader and finer spatio-temporal granularity. This approach will aid in better forecasting and ultimately lead to accurate and robust design, planning, operation, and optimization of power systems and electricity markets.

CRediT authorship contribution statement

Dany Josue Tome Robles: Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Frédéric Maurer:** Supervision, Resources, Methodology, Conceptualization. **Jonas Kristiansen Nøland:** Supervision, Resources, Methodology, Conceptualization. **Mayank Panwar:** Supervision, Resources, Conceptualization. **Sambeet Mishra:** Supervision, Resources, Conceptualization. **Thomas Øyvang:** Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dany Josue Tome Robles reports financial support was provided by SysOpt Grant 326673. Thomas Oyvang reports financial support was provided by SysOpt Grant 326673. Jonas Kristiansen Noland reports financial support was provided by SysOpt Grant 326673. Frederic Maurer reports financial support was provided by SysOpt Grant 326673. Sambeet Mishra reports financial support was provided by SysOpt Grant 326673. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Extended version of the working review paper can be found in Tome-Robles et al. (2024).

Data availability

Data will be made available on request.

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